

SEMESTER 3

**ELECTRONICS & BIOMEDICAL
ENGINEERING**

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE-3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9

4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

ELECTRONIC DEVICES & CIRCUITS

Course Code	PCEBT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives

1. This course aims to develop the skill of analyze and design of different types of analog circuits like amplifiers, power amplifiers, differential amplifiers, oscillators and pulse circuits using discrete electronic components.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	DC Biasing in BJTs: Operating point and DC load line. Fixed, emitter stabilized and voltage divider biasing. Bias stabilization. BJT Amplifiers: RC coupled amplifier (CE configuration) – need of various components and design. Small signal analysis: re and hybrid equivalent models for CE configuration-small signal voltage gain, input and output impedance of CE configuration	12
2	Junction Field Effect Transistors and MOSFETs: Structure, principle of operation and characteristics of JFETs. Structure and physical operation & characteristics of Enhancement and depletion type MOSFETs. MOSFET amplifier circuits: Biasing in MOSFET amplifier-Single stage MOSFET amplifier Circuit (CS configuration)-Design.	12
3	Power amplifiers: classification, class A and Class B power amplifier, Class AB, Class C and Class D Power Amplifiers efficiency and distortion (no analysis is required). Feed-back amplifiers: General feedback structure, properties of negative feedback, and four basic feedback topologies. Sinusoidal oscillators: criterion for oscillation, basic principles of RC and crystal oscillators.	10

4	Pulse Circuits: Pulse shaping using RC circuits- Differentiating and integrating circuits–applications. Multivibrators: astable, monostable and bistable circuits – applications. Differential amplifiers: BJT differential pair, operation with Common-Mode and differential mode input voltages, CMRR	10
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze basic amplifiers using BJT and MOSFET	K4
CO2	Analyze power amplifiers and feedback amplifiers	K4
CO3	Design sinusoidal and non-sinusoidal signal generators.	K3
CO4	Design and implement wave shaping circuits.	K3
CO5	Analyze the operation of differential amplifiers.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	2	2	2				
CO2	3	3	3	3	2	2	2	2				
CO3	3	2	3	2	3	2	1	2				
CO4	3	2	3	2	3	2	1	2				
CO5	3	3	3	3	2	2	2	2				

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Devices and Circuit Theory	Robert L Boylestad and L Nashelsky:	Pearson	11 th Edition, 2015
2	Microelectronic circuits	Sedra A. S. and K. C. Smith	Oxford University Press	8 th Edition, 2019
3	Electronic Devices & Circuits	Allan Mottershead	Prentice Hall of India,	2003
4	Pulse, digital and Switching Waveforms	Millman and Taub	Tata McGraw Hill	3 rd edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Microelectronics	Razavi B	Wiley Education	3 rd edition, 2021
2	Integrated Electronics	Millman & Halkias	Tata McGraw Hill	2 nd Edition, 2017
3	Analysis and Design of Analog Integrated Circuits	Gray & Meyer	John Wiley & Sons	5 th Edition, 2009
4	Electronic Devices and Circuits	David A Bell	Oxford University Press	5 th Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/102/108102095/ , https://archive.nptel.ac.in/courses/108/102/108102112/ https://nptel.ac.in/courses/115102014
2	https://archive.nptel.ac.in/courses/108/102/108102095/ , https://archive.nptel.ac.in/courses/108/102/108102112/ https://nptel.ac.in/courses/115102014
3	https://archive.nptel.ac.in/courses/108/102/108102095/ , https://nptel.ac.in/courses/117101106 https://nptel.ac.in/courses/115102014
4	https://archive.nptel.ac.in/courses/108/102/108102095/ https://archive.nptel.ac.in/courses/108/102/108102112/ https://nptel.ac.in/courses/115102014

SEMESTER S3

LOGIC CIRCUITS AND DESIGN

Course Code	PCEBT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives:

1. To get familiarized with the basics of digital circuits and logic families
2. To design and implement digital circuits

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Digital logic: Binary, Octal, and Hexadecimal number systems – Representation of negative numbers in binary – Binary arithmetic. Binary codes: BCD & BCD addition, XS-3 & Gray Codes, Error detection and correction codes – Parity & Hamming codes. Boolean algebra: Operations, Laws & Theorems, De Morgan's theorems – SOP & POS Boolean expressions and truth tables - Formation of switching functions from word statements, Minterm and Maxterm expansions, Incompletely specified functions. Minimization of Boolean Expressions: Algebraic, Karnaugh map (up to 5 variables) - Realization using basic gates and universal gates.	11
2	Combinational Logic Circuits & Design: Adders & Subtractors – Types, Ripple carry & Carry look ahead adders, BCD adder. Code converters – examples & Comparators. Multiplexers, Demultiplexers, Decoders & Encoders. Basics of Verilog programming: Basic language elements - identifiers, data objects, scalar data types, operators. Modelling and Implementation of logic gates using Verilog codes. Modelling and Implementation of simple combinational circuits at the gate level using Verilog codes. Sequential Logic circuits & Design: Latches – SR Latch. Flip-Flops – SR, JK, D & T Flip Flops – Level & Edge triggered flip flops – Synchronous & Asynchronous inputs - Conversion between flip flops. Master slave flip flops.	14

3	Shift Registers: SISO, SIPO, PISO, PIPO shift registers, Right & Left shifts, Bidirectional & Universal shift registers. Applications: Serial binary adder Counters: Asynchronous counters - Up, Down and Up/ Down counter, Mod n counters. Shift register counters: Ring & Johnson counters. Design of Synchronous counters Introduction to design of synchronous sequential circuits using Finite State Machines: Mealy & Moore types – Flip flops.	11
4	Logic families: Introduction to different logic families, Standard logic levels – Current and voltage parameters – fan in and fan out – Propagation delay, noise consideration. TTL: Basic working principle of a TTL NAND gate – Totem pole and Open collector gate output configurations – Tri-state logic – characteristics of a TTL NAND gate. CMOS: Basic working principle of a CMOS inverter, Comparison of TTL & CMOS, Interfacing TTL & CMOS ICs.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of logic circuits, number systems and digital codes and Boolean algebra.	K2
CO2	Design and implement combinational logic circuits.	K3
CO3	Design and implement sequential logic circuits.	K3
CO4	Familiarize with the concepts of different logic families.	K2
CO5	Simulate gate level logic circuits using Verilog HDL	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3	3									
CO3	3	3	3									
CO4	3	3										
CO5	3	3	3		3				3			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Logic Design	Charles Roth	Cengage	7e/ 2021
2	Fundamentals of Digital Circuits	A. Ananthakumar	PHI	4e/ 2016
3	VHDL Basics to Programming	Gaganpreet Kaur	Dorlin Kindersley	1e/ 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern digital Electronics	R.P. Jain	McGraw Hill	5e/ 2022
2	Digital Design	Mano M.M., Ciletti M.D.	Pearson	4e/ 2008
3	Digital Fundamentals	Thomas L. Floyd	Pearson	11e/ 2017
4	Verilog HDL: A Guide to Digital Design and Synthesis	Samir Palnikar	McGraw Hill	2e/ 2003
5	VHDL: A Design Oriented Approach	S. S. Limaye	McGraw Hill	2e/ 2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/106/117106086/
2	https://archive.nptel.ac.in/courses/106/105/106105185/

SEMESTER S3

BIOSENSORS AND TRANSDUCERS

Course Code	PBEBT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Learn the basic principles of biosensors, transducers & their applications.
2. Provide students an insight into biopotentials and electrodes.
3. Introduce Nano biosensors & their applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Transducers: Classification of transducers, characteristic of transducers. Temperature transducers: Resistance temperature detector (RTD), Thermistor, Thermocouple, chemical thermometry. Displacement transducers: potentiometer, resistive strain gauges, inductive displacement, capacitive displacement transducer. Pressure transducer: variable capacitance pressure transducers, LVDT transducers, strain gauge transducers, catheter tip transducers, Piezoelectric transducer. Photoelectric transducers: photo-emissive tubes, photovoltaic cell, photoconductive cell, photodiodes.	9
2	Bio-Potential Electrodes: Electrode theory, Skin Contact Impedance. Types of Electrodes: Microelectrodes- Metal microelectrodes, micropipette (metal filled glass and glass micropipette electrodes). Depth and needle electrodes- Depth electrode, Needle electrode. Surface electrodes- Metal plate surface electrodes, Suction Cup electrode, Adhesive tape electrode, Multipoint electrode. Bio-potentials: ECG - Generation of cardiac action potentials- Characteristics of ECG Signal, Einthoven's triangle -Lead Systems. Electrodes for ECG acquisition. EEG - Electroencephalogram - Brain action potentials - Characteristics of signal-Electrodes and 10-20 system for EEG signal acquisition. Evoked Potentials. EMG - Electrical activity of muscles – Characteristics of EMG signal. Electrodes for EMG acquisition. (Instrumentation not needed for ECG, EEG & EMG)	9

3	Fundamentals of Biosensors -Basic Principle, Characteristics of a Biosensor. Types - Based on Transducer: Electrochemical- Amperometric, Potentiometric; optical-Surface plasmon resonance, Evanescent Wave Fluorescence Biosensors; piezo electric; calorimetric biosensors. Based on Bio recognition elements in biosensors- natural-semisynthetic-synthetic. immobilization methods ISFET, Enzyme electrodes. Principles of Fiber-Optic Sensing - Definitions, Classification, Working Principles. Applications of biosensors -Biosensors for clinical diagnostics-Glucose detection- cancer detection.	9
4	Nano biosensors: Significance, Types of Nanomaterials - Gold Nano- particles, Quantum Dots, Carbon Nanotubes & Graphene (Key features). Synthesis of Nanomaterials-Top-down Approach & Bottom down Approach. Types of nano biosensors: nanotubes-based sensors, nanoparticles-based biosensors, nanowire-based sensors, quantum dots-based sensors (Basics of sensing only). Application of Nanomaterial based biosensors : Metal nanoparticle-based biosensors, carbon nanotube-based biosensors.	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the principle of transduction, classifications and the characteristics of different transducers and study its biomedical applications.	K3
CO2	Understand the concepts, types, working and practical applications of electrodes for acquiring and recording various physiological parameters.	K2
CO3	Understand the fundamentals of biosensors and its applications	K3
CO4	Identify and Evaluate Various Types of Nanomaterials and its applications.	K3
CO5	Design and implement simple projects for data acquisition and display the output	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2			1						
CO2	3	2	2			1						
CO3	3	2	2			1						
CO4	3	2	2			1						
CO5	3	3	3		3	3			3	3	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of Medical Physiology	Guyton and Hall	Saunders, an imprint of Elsevier Inc	12e/ 2011
2	Biosensors Fundamentals and Applications	Bansi Dhar Malhotra, Chandra Mouleypany	Smithers Rapra	1e/ 2017
3	Introduction to Biosensors – From Electric Circuits to Immunosensors	Jeong-Yeol Yoon (auth.)	Springer International Publishing	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Applied Biomedical Instrumentation	Geddes and Baker	Wiley Inter science	1989
2	Biomedical Transducers and Instruments	Tatsuo Togawa, Toshiyo Tamura and P. Ake Öberg	CRC Press	2000
3	Fiber-Optic Sensors for Biomedical Applications	Daniele Tosi Guido Perrone	Artech House Applied Photonics Series	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLp6ek2hDcoNBrYuh8TYc3YNQUvKanqiRa&si=s0dTnwc4f63NzELM
2	https://www.youtube.com/watch?v=0EDwazhDRQk https://drmcet.digimat.in/nptel/courses/video/108108167/L10.html
3	https://youtu.be/kQ6CY1qpGjY?si=Hh-S5INJwkw2jUbT
4	https://archive.nptel.ac.in/courses/127/105/127105235/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11

3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome- extension://efaidnbmn nnibpcajpcglclefindm kaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	6

2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world	6

	case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*
CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.

- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

ELECTRONIC DEVICES AND CIRCUITS LAB

Course Code	PCEBL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. This course aims to equip students with the skills to design and analyse electronic circuits including rectifiers, filters, voltage regulators, amplifiers, and waveform generators, emphasizing practical implementation and simulation software tools. Students will learn to create and assess various wave shaping and filtering circuits, and understand the frequency response of amplifiers. Additionally, the course fosters teamwork, effective documentation, and communication skills for professional technical reporting and collaboration.

Exp t. No .	Experiments (Any 10 from the list is mandatory)
1	Characteristics of diodes (Si, Zener and Light Emitting diodes)
2	Rectifier circuits
3	Filter circuits
4	Clipping and clamping circuits
5	Characteristics of transistors
6	Biasing of BJT – Fixed and voltage divider biasing
7	Zener voltage regulator
8	Design of single and dual power supplies
9	Frequency responses of RC low pass & high pass filters
10	RC differentiating and integrating circuits
11	Switch circuits using BJTs.
12	RC coupled amplifiers using BJT
13	Astable Multivibrator using BJTs
14	Monostable Multivibrator using BJTs
15	Simulate Clipping and clamping Circuits
16	Simulate sweep circuit
17	Simulation of Zener voltage regulator
18	Simulation Rectifier Circuits

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design circuit of rectifiers, filters, voltage regulators, and complete power supplies.	K3
CO2	Implement Wave Shaping, Multi-Vibrator, switching & sweep Circuits	K3
CO3	Analyze RC Coupled Amplifiers and its frequency response	K4
CO4	Simulate the electronic Circuits Using simulation software	K2
CO5	Develop skills to document their work comprehensively, collaborate within a team, and communicate technical information clearly and professionally	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		3	2		2	2		3	
CO2	3	3	2		3	2		2	2		3	
CO3	3	3	2		2	2		2	2		3	
CO4	3	3	3		3	2		2	2		3	
CO5	2	2	2		2	3		3	3		3	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Devices and Circuit Theory	Robert L. Boylestad and Louis Nashelsky	Pearson Education	2007
2	Electronic Devices & Circuits	Allan Mottershead	Prentice Hall of India	2003
3	Pulse, Digital and Switching Waveforms	Jacob Millman and Herbert Taub	Tata McGrawHill	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microelectronic Circuits	Microelectronic Circuits	Oxford University Press	7th edition (March 1, 2014)
2	Electronic Devices	Thomas L. Floyd, David M. Buchla	Pearson	10th edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=e2qCGpZ8wQ0
2	http://vlabs.iitkgp.ernet.in/be/exp7/index.html
3	https://www.vlab.co.in/participating-institute-iit-delhi
4	http://vlabs.iitkgp.ac.in/psac/newlabs2020/ssds/newexp1/index.html
5	https://www.vlab.co.in/participating-institute-iit-delhi
6	http://vlabs.iitkgp.ernet.in/be/exp10/index.html
7	https://www.youtube.com/watch?v=6Otr1I0OR18
8	https://be-iitkgp.vlabs.ac.in/exp/frequency-response/
9	http://vlabs.iitkgp.ernet.in/be/exp15/index.html

10	http://vlabs.iitkgp.ernet.in/be/exp13/index.html
11	http://vlabs.iitkgp.ernet.in/be/exp13/index.html
12	https://www.youtube.com/watch?v=PTQI8E49vOI

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

LOGIC CIRCUITS LAB

Course Code	PCEBL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To provide experience on design, implementation, simulation and testing of digital electronic circuit.
2. To introduce and familiarize with Verilog programming for the design of digital circuits.

Exp t. No.	Experiments (Any FOUR from combinational circuits, Any FOUR from sequential circuits and Any TWO Verilog HDL coding of simple combinational circuits at gate level mandatory)
1	Familiarization of logic gates, and the realization of logic gates using universal gates.
2	Design and implementation of Adder and Subtractor circuits using basic gates and Universal gates.
3	Design and implementation of Binary Adder cum Subtractor circuit with mode control using IC 7483.
4	Design and implementation of BCD Adder using IC 7483.
5	Design and implementation of 2- bit comparators and code converters using basic logic gates.
6	Design and implementation of 4:1 Multiplexer and 1:4 Demultiplexer circuits using basic gates and study of mux/De-mux ICs.
7	Study of Flip-Flop circuits and Flip-Flop ICs- Implementation and analysis of different types of Flip-Flops.
8	Design and implementation of following asynchronous counters 1. up and down asynchronous counter 2. up/ down asynchronous counters using mode control.
9	Study of counter ICs 7490, 7493 and 74193. Implementation of mod-N asynchronous counters IC 7493.

10	Design and implementation of following synchronous counters 1. up and down synchronous counter 2. up/ down synchronous counters using mode control 3. mod-N synchronous counters
11	Design and implementation of Ring and Johnson Counters using Flip Flops.
12	Design and implementation of following shift registers. (any two) 1. Serial In- Serial Out (SISO) shift registers 2. Serial In- Parallel Out (SIPO) shift registers 3. Parallel In- Serial Out (PISO) shift registers 4. Parallel In- Parallel Out (PIPO) shift registers 5. Serial/Parallel Shift register using mode control

Note*- Familiarization with Verilog HDL coding: Modelling and simulation of basic gates, simple Boolean functions (any two) and simple combinational circuits (any two) using Verilog (gate level) shall be done.

**Course Assessment
Method (CIE: 50 marks,
ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarize the standard logic gates and universal gates.	K2
CO2	Design and implement combinational circuits using basic gates and digital ICs.	K4
CO3	Design and implement flip flops and counters using basic gates and digital ICs.	K4
CO4	Develop proficiency in Verilog HDL coding for modeling basic gates, Boolean functions, and simple combinational circuits.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3		1	2	3	2	1	3
CO2	3	3	3	3	3		1	2	3	2	1	3
CO3	3	3	3	3	3		1	2	3	2	1	3
CO4	3	3	3	3	3		1	2	3	2	2	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Logic Design	Charles H. Roth Jr.	Thomson Books/Cole	5 th edition, 2004
2	Fundamentals of Digital Circuits	A. Anand Kumar	PHI learning	2 nd edition, 2010
3	Digital Design with an Introduction to the Verilog HDL	M Morris, M. and Michael, D	Pearson	5 th edition, 2013
4	Fundamentals of digital logic with Verilog design	Brown, S.D. and Vranesic Z.G.	McGraw-Hill	3 rd edition, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Fundamentals	Thomas L Floyd	Pearson	10 th edition, 2011
2	Digital Design- Principles and Practices	John F Wakerly	Pearson	3 rd edition, 2000
3	Digital Design	Mano M M	PHI	3 rd edition, 2002
4	Modern Digital Electronics	R P Jain	Tata Mc Graw Hill	4 th edition, 2009
5	Digital System Design using VHDL	Charles H Roth Jr	Thomson learning	2 nd edition, 2004
6	Advanced digital design with the Verilog HDL	Michel D. Ciletti	Pearson	2 nd edition, 2010

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://archive.nptel.ac.in/courses/108/105/108105113/ https://archive.nptel.ac.in/courses/108/105/108105132/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

ELECTRONICS AND BIOMEDICAL ENGINEERING

SEMESTER S4
MATHEMATICS FOR ELECTRICAL SCIENCE– 4
(B Group)

Course Code	GBMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To expose the students to the basics of random processes essential for their subsequent study of analog and digital communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and <i>t</i> -distribution), Hypotheses and Test Procedures, Type I and Type II error, <i>z</i> Tests for Hypotheses about a Population Mean (for large sample), <i>t</i> Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Random process concept, classification of process, Methods of Description of Random process, Special classes, Average Values of Random Process, Stationarity- SSS, WSS, Autocorrelation functions and its properties, Ergodicity, Mean-Ergodic Process, Mean-Ergodic Theorem, Correlation Ergodic Process, Distribution Ergodic Process. [Text 2: Relevant topics from Chapter 6]	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z-tests and the one-sample <i>t</i> -test.	K3
CO4	Analyze random processes by classifying them, describing their properties, utilizing autocorrelation functions, and understanding their applications in areas like signal processing and communication systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Probability, Statistics and Random Processes	T Veerarajan	The McGraw-Hill	3 rd edition, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Probability and Random Processes	Palaniammal, S.	PHI Learning Private Limited	3 rd edition, 2015
4	Introduction to Probability	David F. Anderson, Timo, Benedek	Cambridge	1 st edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	
4	https://archive.nptel.ac.in/courses/117/105/117105085/

SEMESTER S4
LINEAR INTEGRATED CIRCUITS

Course Code	PCEBT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCEBT302	Course Type	Theory

Course Objectives:

1. To impart extensive knowledge to students of Integrated Circuit fabrication technology and the concepts of operational amplifiers.
2. To enable students to acquire the design and analysis skills for circuits using op-amps and other specialized ICs.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Microelectronic fabrication technology: Brief overview of microelectronic fabrication technology - Epitaxial Growth, Diffusion, Ion Implantation Processes -Description - Fabrication of BJT, MOSFETs. Operational amplifiers: Introduction to operational amplifiers, Internal block schematic of op amp- equivalent circuit - Voltage transfer curve— input and output impedance –Slew rate, Input bias current –Input offset current- Offset voltage- CMRR, SVRR. Open loop op- amp configurations- open loop gain – Frequency response- finite gain bandwidth –Ideal op amp parameters. Offset voltage compensating networks-offset minimizing resistor.	11
2	Closed loop op amp configurations: Feedback topologies- Voltage series, voltage shunt, current series and current shunt feedbacks – Frequency response, concept of virtual ground - Closed loop gain, input resistance, output resistance, band width and output offset voltage of feedback configurations, voltage follower. Summing amplifier, Difference amplifiers with one op amp and 3 op amps- Instrumentation amplifier, Log and Antilog amplifiers. Voltage to	11

	current and current to voltage converter - Integrator and differentiator. Peak detector circuit, Precision rectifiers. Comparators and Schmitt trigger circuits	
3	Active Filters: Transfer functions – LPF, HPF, BPF, BRF. First and second order filters. Approximation methods–Butterworth – Chebyshev. Sample and hold circuit, ADC -successive approximation, flash, integrating types. DAC – weighted, R-2R types, Multi vibrators - Astable and mono stable – Design and working. Wave generators- Triangular and saw tooth - RC phase shift and Wien bridge oscillators.	11
4	Specialized ICs and applications: IC regulators - 723 -block diagram, typical low voltage regulator circuit, 78XX and 79XX IC regulators, 317 - applications. Timers - 555 – Functional block diagram- Astable and monostable multi vibrators using 555. VCO – 566. PLL-IC 565- Block diagram, working and applications.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the steps of micro-fabrication technology and the constructional details of op amp.	K2
CO2	Apply the fundamental concepts of op amp to design analog circuits for different applications.	K3
CO3	Analyze the important circuits using op-amps.	K4
CO4	Select a suitable circuit for doing a specified analog signal processing operation.	K4
CO5	Choose specialized ICs to design and develop application circuits.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				1							1
CO2	3	2			1							1
CO3	3	2	2	1	1							1
CO4	3	2	2	1	1							
CO5	3	2	2	1	1							1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op-Amps and Linear Integrated Circuits	Ramakant A. Gayakwad	Pearson Education Asia	Fourth edition,. 2015
2	Operational Amplifiers Theory and Design	Johan Huijsing	Springer International Publishing	Third illustrated edition, 2016
3	Op Amps for Everyone	Bruce Carter, Ron Mancini	Elsevier Science	Fifth illustrated edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op amps and Linear Integrated circuits	Coughlin & Driscoll	Pearson Education Asia.	Sixth Edition. 2000
2	Microelectronics	Millman & Grabel	McGraw Hill International	Second edition. 2017
3	Microelectronic circuits	Sedra & Smith	Oxford University Press	Seventh Edition, 2014
4	Op amps and Linear integrated Circuits	D A Bel	Prentice Hall of India	1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/lpXNCwsnxjM
2	https://youtu.be/Iv4Cj2A3ldw?si=m0jNXOfP9-9m1uwH , https://youtu.be/4HV1ZZjvDn0?si=oJ7E5DF_k7NIplMO
3	https://youtu.be/M3yI0byaqKc?si=SHzcyMQToWpMpUjK
4	https://youtu.be/WFsPI8_ZKbc

SEMESTER S4
ANALYTICAL & DIAGNOSTIC EQUIPMENTS

Course Code	PCEBT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCEBT205 & PBEBT304 desirable	Course Type	Theory

Course Objectives:

1. Students will acquire an understanding of the basic principles behind analytical and diagnostic equipments used in clinical environments.
2. Students will be able to analyze the instrumentation of analytical and diagnostic equipments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Analytical equipment: Beer-Lambert's Law- Colorimeters - Spectrophotometry (UV/ Visible, IR): Instrumentation - Filters - Monochromators - Detectors - Automated clinical Biochemistry analysers – Applications. Electroanalytical Instruments: Methods - Ion selective electrode (ISE) - Blood Gas Analysers - Electrolyte Analysers - Blood cell counters. Sample preparation equipment - centrifuge, sonicator	11
2	Analytical techniques in Biochemistry and Molecular biology: Radioimmunoassay (RIA) - Enzyme Linked Immunosorbent Assay (ELISA/CLIA) - Applications; Polymerase Chain Reaction- RT PCR; Principle of NMR spectroscopy and Mass spectrometry. Principle and applications of Electrophoresis (Gel, Capillary and IEF), Chromatography (Gas and HPLC), Flow Cytometry. <i>(Class Assignment/Seminar topics: Lab on a Chip (LOC) biochemical sensor, Miniaturized Systems for (Bio) Chemical analysis and synthesis - glucometer-Point of Care Test equipment (POCT))</i>	11

3	Bioelectric Devices: Electrocardiograph- Pre-amplifiers- Filters- Isolation circuits- RL driven circuit- protection circuits; Analog front end of ECG machine (Block diagrams only). Patient monitoring systems – Instrumentation of Pulse oximeter, Arrhythmia monitors, Ambulatory recorders - Holter monitors. Electroencephalograph - Instrumentation, Types of Evoked potentials, Apnoea monitors. BP measurement - Working principle of manual sphygmomanometer	11
4	Cardiac output measurement- Fick method -Dilution Methods- Doppler method, Impedance Cardiography. Spirometry – Basic system and applications-Pulmonary function measurements: Respiratory volumes, lung capacity -different volume measurement. Impedance Plethysmography. Blood flow meters - Electromagnetic – Ultrasonic Doppler blood flow meters. Audiometers – Pure tone - speech audiometers.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):-40 Marks, ESE: 60Marks

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarise the instrumentation of various analytical equipment used in clinical laboratory.	K2
CO2	Recognize the principles and applications of biochemistry equipment used in clinical environment.	K2
CO3	Analyse the instrumentation of bioelectric devices.	K4
CO4	Analyse the impedance measurement concepts and related measurements.	K4
CO5	Interpret the principles related to respiratory, blood flow and audiometric measurements.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1								1
CO2	2	2	1	1								1
CO3	3	2	2	1								1
CO4	3	2	2	1								1
CO5	3	2	2	1								1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mahin Basha	Analytical Techniques in Biochemistry	Springer US_Humana	(2020)
2	Rüdiger Kramme, Heike Kramme	Springer Handbook of Medical Technology	Springer-Verlag Berlin Heidelberg,	2011
3	Andrew G Webb	Principles of biomedical instrumentation.	Cambridge University Press	2018
4	Anthony YK Chan	Biomedical device technology: principles and design.	Charles C Thomas Publisher,	2016.
5	R S Khandpur	Handbook of Bio-Medical Instrumentation	Tata McGraw Hill, 2nd Ed.	2003
6	Skoog A Dogulas, F. James Holler, Stanley R Crouch	Principles of Instrumental Analysis	Cengage, 6th Edition.	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Joseph D. Bronzino	The Biomedical Engineering Handbook	CRC Press	1995
2	Geddes, Leslie Alexander, and Lee Edward Baker	Principles of applied biomedical instrumentation.	John Wiley & Sons,	1975.
3	John G. Webster	Medical Instrumentation - Application and Design	Houghton Mifflin Co., Boston	1992
4	John G. Webster	Encyclopaedia of Medical Devices and Instrumentation	Wiley Interscience, 2nd Ed.	2006
5	Richard Aston	Principles of Biomedical Instrumentation and Measurements	Merril Publishing Co.	1990
6	Myer Kutz	Standard Handbook of Biomedical Engineering and Design	McGraw Hill	1993

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/104105084
2	https://archive.nptel.ac.in/courses/104/105/104105102/
3	https://archive.nptel.ac.in/courses/127/106/127106232/
4	https://nptel.ac.in/courses/108106173

SEMESTER S4

MICROCONTROLLER PROGRAMMING & INTERFACING

Course Code	PBEBT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCEBT303	Course Type	Theory

Course Objectives:

1. Familiarize Microcontrollers and its programming
2. To familiarize the interfacing of microcontroller with external devices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Intel 8051: Architecture - ALU, internal RAM and ROM, Organization of ports. Registers of 8051: data pointer, program counter, CPU registers, register banks, stack and stack pointer, PSW, Special Function Registers. Addressing modes of 8051. Instruction set of 8051- Assembler directives, Data transfer instructions, Arithmetic, logical, compare and rotate instructions- Bit processing instructions- Program flow control instructions	9
2	Programming using Assembly language: Illustrative examples using of all types of instructions. Programming the ports of 8051. Programming in C: Illustrative examples of C programming for 8051- time delay and I/O operations, logic operations	9
3	Timers of 8051: All modes of operations, programming the timers in assembly language and C language, timer as an event counter. Serial Communication with 8051: Introduction to serial communication, RS232 protocol. Serial communication programming in assembly language. Interrupt system of 8051: introduction to interrupts- interrupts in 8051-priority of interrupts- Interrupt handling and	9

	execution for timer, external and serial interrupts. Programming to handle all types of interrupts in assembly language	
4	Interfacing of 8051: LCD, stepper motor, ADC, 8255. Introduction to AVR microcontroller: Overview of AVR family, ATmega32 – Architectural overview, CPU core, memory organization. Introduction to PIC microcontrollers: Architectural features, Memory organization, Core and peripheral features	9

Project based learning: 9 Hrs

- In the first class, students shall be given an awareness and/ or guidelines on how to implement learn this course through a project and form groups of maximum 5 members in the first week.
- Each group identify a project where microcontrollers can be used for simple biomedical applications and perform a literature survey.
- Choose a suitable microcontroller/ microcontroller- based development boards for your application and the input-output devices to be interfaced
- Develop the block diagram and present before the jury within first month after the commencement of classes
- Implement necessary hardware and software to develop the prototype
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarize with the architecture and instruction set of Intel 8051 microcontroller to program 8051 using assembly and C language.	K3
CO2	Demonstrate the use of timers, serial communication and interrupts of 8051.	K3
CO3	Interface 8051 with peripherals.	K3
CO4	Familiarize with the architecture of 8-bit AVR Microcontrollers and PIC microcontrollers	K3
CO5	Design and implement simple projects using microcontrollers	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	-	-	-	-	-	-	-	-
CO2	3	3	3	2	-	-	-	-	-	-	-	-
CO3	3	3	3	2	-	-	-	-	-	-	-	-
CO4	3	3	1	-	-	-	-	-	-	-	-	-
CO5	3	3	-	-	3	3	-	-	3	3	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller and Embedded Systems using Assembly and C	Muhammed Ali Mazidi and Janice GillispieMazidi	Pearson Education	4 th edition
2	The AVR Microcontroller and Embedded Systems Using Assembly and C	Muhammad Ali Mazidi, Sarmad Naimi and SepehrNaimi	Pearson Education	2013
3	Design with PIC microcontrollers	John B. Peatman	Pearson Education	ISBN: 978-81-7758-551-3

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller	Kenneth J Ayala	Cengage Learning	3rd edition
2	Programming and Customizing the AVR Microcontroller	Dhananjay Gadre	McGraw Hill Education	
3	The 8051 Microcontrollers: Architecture Programming and Applications	K Uma Rao & Andhe Pallavi	Pearson Education	1e/2011
4	PIC data sheets and manuals	PIC18F2455/2550/4455/4550 Data Sheet – Microchip		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=V9AzfvkWPW0
	https://www.youtube.com/watch?v=3hltHQXAQm8
	https://www.youtube.com/watch?v=5V3N1SYgVhQ
	https://www.youtube.com/watch?v=AdMxMBH393Q
2	https://www.youtube.com/watch?v=TphNPocrvQE
	https://www.youtube.com/watch?v=MKzLaZxnqFY
	https://www.youtube.com/watch?v=GHYaAUVKtDA
	https://www.youtube.com/watch?v=2AVOxLPKjeA
3	https://www.youtube.com/watch?v=E81Vz-0ErPA
	https://www.youtube.com/watch?v=K4ItndQiwec
	https://www.youtube.com/watch?v=0fMLDXwcT_w
	https://www.youtube.com/watch?v=MqhxeOi8R1Q
	https://www.youtube.com/watch?v=JBv4y-4_oaw

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER 4

ELECTRICAL TECHNOLOGY FOR BIOMEDICAL ENGINEERS

Course Code	PEEBT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives:

1. **Understand** the principles and applications of electrical machines and power supplies in biomedical engineering, including the operation of different types of motors, transformers, switched-mode power supplies (SMPS), and special motors.
2. **Apply** advanced power electronics knowledge and electrical safety standards in biomedical equipment design and implementation, encompassing the construction and characteristics of power semiconductor devices, the operation of SCR in controlled rectifiers, and compliance with international electrical safety standards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic principles of electrical machines: Concepts of motoring & generating action. DC generator: Main constructional features, principles of operation, types of generators, emf equation, applications. DC Motors: Back emf, need for starters, types, speed and torque equation, characteristics of shunt, series, and compound motors, speed control, applications.	9
2	Transformers: Working principle and elementary theory of an ideal transformer, constructional features, emf equation, turns ratio, efficiency, regulation, losses, open circuit and short circuit test, Auto transformer and Isolation transformer working principle, numerical problems. Switched-Mode Power Supplies (SMPS): Principles, types (buck, boost, buck-boost), advantages, and applications in medical devices.	9

3	Synchronous Motors: Basic principles, Losses and Efficiency, Torque equation - Starting methods. Single-phase Induction Motor: Double field revolving theory, making single-phase induction motor self-starting, split phase and capacitor start induction run. Special Motors: Construction, principle of operation, and applications of BLDC, Stepper Motor, Servomotor and PMSM.	9
4	Overview of electrical safety of low-voltage systems: Basic Definitions and Nomenclature -Basic Insulation, Class 0, II, III Equipment, Exposed-Conductive-Part (ECP), Extra-Low Voltage, Extraneous-Conductive-Part (EXCP), Grounding System -IT, TN,TN-C, TN-C-S, TN-S and T T, Electrical Codes and Standards, Personal Protective Equipment (PPE). Fundamentals of Electrical Safety: Introduction, Protection Against Direct Contact, Protection Against Indirect Contact.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand various motor types and apply them appropriately for different applications.	K2
CO2	Analyze the constructional details and performance characteristics of a single-phase transformers to synthesize solutions for different electrical needs.	K4
CO3	Explore the operational principles of synchronous motors, single-phase induction motors, and special electric motors to understand their applications	K2
CO4	Analyze electrical safety standards for medical equipment to evaluate compliance and propose measures for safe operation in healthcare settings.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2			2						2
CO2	3	2	2			2						2
CO3	3	2	2			2						2
CO4	3	2	2			2						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Machinery Fundamentals	Stephen J. Chapman	McGraw-Hill Education	5th Edition, 2011
2	Transformers and Inductors for Power Electronics:	W.G. Hurley, W.H. Wolfle	John Wiley & Sons	2nd Edition, 2013

	Theory, Design and Applications			
3	Electrical Safety of Low-Voltage Systems	Dr. Massimo A. G. Mitolo	McGraw-Hill Education	
4	Electrical Safety Handbook	John Cadick, Mary Capelli-Schellpfeffer, Dennis K. Neitzel	McGraw-Hill Education	4th Edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A textbook of electrical technology	Theraja, B. L	S. Chand Publishing	2008
2	Power electronics: circuits, devices, and applications	Rashid, Muhammad H	Pearson Education India	2009
3	Bioelectronics and Medical Devices: From Materials to Devices- Fabrication, Applications and Reliability	Pal, Kunal, Heinz-Bernhard Kraatz, Anwesha Khasnobish, Sandip Bag, Indranil Banerjee, and Usha Kuruganti, eds	Woodhead Publishing	2019.
4	Advanced electrical drives: analysis, modeling, control	De Doncker, Rik W., Duco WJ Pulle, and André Veltman	Springer Nature	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ee60/preview
2	https://nptel.ac.in/courses/108106071 https://archive.nptel.ac.in/courses/108/108/108108036/
3	https://archive.nptel.ac.in/courses/108/105/108105131/
4	https://onlinecourses.swayam2.ac.in/nou20_cs08/preview

SEMESTER S4
DESIGN OF LOGIC SYSTEMS

Course Code	PEEBT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	PCEBT303	Course Type	Theory

Course Objectives:

1. This course aims to make students able to understand the fundamental principles of logic systems, their analysis, design and implementation, as well as to gain proficiency in HDL languages.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to digital design methodology using Verilog HDL - Design flow - Modeling abstraction levels, gate level model, RTL model, behavioral model - Simulation and synthesis - Modeling for ASIC/FPGA - Language concepts - Data types and operators - Structural, data flow and behavioral models - Hierarchical structure - Combinational and sequential circuit description - Continuous and procedural assignments - Blocking and non-blocking assignments - Delay modeling - Parameterized reusable design- Compiler directives - Testbenches.	9
2	Combinational Modules, Modelling and simulation using Verilog: Adders Half adder, Full adder, ripple carry adder), comparators, Decoders, Encoders, Multiplexers, Demultiplexers. Sequential modules Modelling and simulation using Verilog: Flip flop (JK, D), Shift registers (SISO, SIPO, PISO, PIPO,	9

	Universal shift Register), synchronous Counters (up, down, Up/down using mode control, Mod-N).	
3	Timing characteristics and Impediments to synchronous design of sequential networks – setup time – hold time – propagation delay – maximum clock frequency- clock skew. Analysis of synchronous sequential circuits - Derivation of state graphs and state tables, General model of sequential networks. Design of synchronous sequential circuits - Finite state machines - Mealy and Moore models - State graphs - State assignment - State table - State reduction- Design of Sequence Detector	9
4	Programmable LSI techniques -PLA/PAL/PLDs - Building Logic Functions with PLA/PAL, Read-Only Memory (ROM)- Combinational Circuit Implementation with ROM- Sequential Circuit Implementation with ROM CPLD and FPGAs - Xilinx/Altera series FPGAs - Programmable System on Chip - Zynq So design overview.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	students will understand and interpret the constructs of Hardware Description Language.	K3
CO2	students will understand the design, modeling, and simulation of combinational and sequential modules using Verilog at different levels of abstraction.	K4
CO3	students will understand the design principles and challenges of synchronous sequential circuits, including timing characteristics, state modeling, and various types of sequential systems.	K4
CO4	students will understand the design and application of programmable logic devices (PLDs), including PLA, PAL, ROM, CPLDs, FPGAs, and the Zynq SoC architecture.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	3	3	3	3	-	-	-	1	1	3	3
CO2	1	3	3	3	3	-	-	-	1	1	3	3
CO3	1	3	3	3	3	-	-	-	1	1	3	3
CO4	1	3	3	3	3	-	-	-	1	1	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Digital Systems	Milos Ercegovac, Tomas Lang, Jaime H. Moreno	John Wiley & Sons, Inc	1 st edition, 1998
2	Digital Systems Design Using Verilog	Charles Roth, Lizy K. John, Byeong Kil Lee	Cengage Learning	1 st edition, 2015
3	Digital Design with an Introduction to the Verilog HDL	M Morris, M. and Michael, D	Pearson	5 th edition, 2013
4	Fundamentals of logic design	Roth Jr, C.H., Kinney, L.L. and John, E.B	Cengage Learning	6 th edition, 2020
5	Fundamentals of digital logic with Verilog design,	Brown, S.D. and Vranesic Z.G.	McGraw-Hill	3 rd edition, 2014
6	Advanced digital design with the Verilog HDL	Michel D. Ciletti	Pearson	2 nd edition, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Logic Design	Charles H. Roths Jr.	Thomson Books/Cole	5 th edition, 2004
2	Digital Design- Principles and Practices	John F Wakerly	Pearson	3 rd edition, 2000
3	Digital logic: Applications and design	Yarbrough, John M., and John M. Yarbrough	Eagan: West Publishing Company	1 st edition, 1997
4	Logic Design Theory, Prentice	Nripendra N Biswas	Hall of India	2001
5	Digital System Design using VHDL	Charles H Roth Jr	Thomson learning	2004
6	Verilog HDL Synthesis: A Practical Primer	J. Bhasker	B. S. Publications	2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/106/108106177/ https://archive.nptel.ac.in/courses/106/105/106105165/
2	https://archive.nptel.ac.in/courses/106/105/106105165/
3	http://www.digimat.in/nptel/courses/video/117106109/L28.html
4	https://archive.nptel.ac.in/courses/108/106/108106177/ https://archive.nptel.ac.in/courses/117/108/117108040/

SEMESTER S4

CONTROL SYSTEMS ENGINEERING

Course Code	PEEBT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Characterize control systems using time, space and frequency domain techniques and simplify them for analysis and design.
2. Employ time, space and frequency domain analysis methods to predict and evaluate performance parameters of control systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Control Systems - Definition, Examples (including physiological control systems), System configurations, Design considerations, Control system design process; Review of Laplace Transforms, Transfer function of LTI system, Model of simple electrical and mechanical systems; Concept of feedback, Reduction of Block diagrams, Signal flow graph- Use of Mason's gain formula.	9
2	Time domain analysis- Standard test signals, Time response specifications, First order system response, S-plane root location and transient response, impulse and step response of second order systems, performance and characteristics in the time domain, Steady state response - Steady state error and static error coefficients.	9
3	Frequency domain analysis - Frequency response techniques - frequency response specifications, bode plots, gain margin and phase margin via Bode plots; Routh - Hurwitz criterion, Root Locus techniques, Concepts of relative stability. Overview of compensators for design improvement.	9
4	State space models - Introduction to the state variable concept, State model of mechanical and electrical systems. Various forms of	9

	representing state space systems; Solution of continuous time state equations - (Homogeneous case only), State transition matrix - properties, Controllability, Observability.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Characterize a control system in Laplace domain and model it in simpler forms.	K3
CO2	Employ time domain analysis to predict and investigate transient and steady state performance parameters of the system for standard test signals.	K4
CO3	Formulate different types of analysis in frequency domain to explain the nature of stability of the system.	K4
CO4	Apply state space representations to evaluate the controllability and observability of control systems	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3			2				2		3
CO2	3	3	3			2				2		3
CO3	3	3	3			2				2		3
CO4	3	3	3			2				2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Control Engineering	Katsuhiko Ogata	Pearson	5th Edition, 2010
2	Modern Control Systems	Richard C. Dorf and Robert H. Bishop	Pearson	13th Edition, 2016
3	Control Systems Engineering	Norman S. Nise	Wiley	The 8th edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biological Control System Analysis	Robert B. Northrop	CRC Press	2 nd Edition, 2010
2	Biomedical Engineering Principles	Arthur B. Ritter and Joseph D. Bronzino	CRC Press	4 th Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/107/106/107106081/
2	https://nptel.ac.in/courses/108106098
3	https://nptel.ac.in/courses/108102043
4	https://nptel.ac.in/courses/101104330
5	https://nptel.ac.in/courses/108107115

SEMESTER S4
BIOMATERIALS & APPLICATIONS

Course Code	PEEBT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Recognise the suitability of various materials for biomedical applications by evaluating their inherent properties, such as biocompatibility, mechanical strength, and degradation kinetics.
2. Proficiency in utilizing design principles and fabrication techniques to produce biocomposite materials, including conventional composite processing methods and advanced 3D printing technologies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Metals - Definition of Biomaterials – Classification – Metals-Stainless steel-Co-Cr alloys-Ti based alloys- Dental Metals -Other metals –corrosion of metallic implants-manufacturing implants-mechanical properties-applications in medicine. Biocompatibility -host and tissue response- in-vitro & in-vivo tests for implants. Ceramics - Ceramic materials- Aluminium oxides, -Glass ceramics, carbons. Bioactivity -Bioactive glass & its applications. Non absorbable or bio inert ceramics-degradable or resorbable ceramics-bioactive ceramics-deterioration of ceramics-Bioactive ceramic manufacturing techniques- applications in medicine. Strengthening ceramics for biomedical applications.	9
2	Polymers - Polymerization and Basic Structure- Polymers Used as Biomaterials- Polyvinylchloride (PVC) -Polyethylene (PE) - Polypropylene-(PP) –Poly methyl methacrylate (PMMA). Biodegradable Polymers-surface Modifications for Improving Biocompatibility- Chemo gradient Surfaces for Cell and Protein Interaction- Synthetic degradable and non-bio degradable polymers.	9

	Natural polymers and their derivatives for medical use- Polyelectrolyte Complexes (PECs)- Overview - Biomedical Applications-Formation of PECs-Types of PECs- Applications of PECs-Wound healing, drug delivery.	
3	Composites- Natural composites-bone, cartilage - Structure – properties- anisotropy - particulate composites - fibrous composites - porous materials - biocompatibility of composites. Design and fabrication methods for biocomposites- Production techniques for biocomposite parts- Conventional composite processing techniques- Extrusion and injection for thermoplastic materials- Filament winding - compression - infusion - autoclaving - Hard tissue applications of biocomposites-soft tissue applications of biocomposites.	9
4	3-D printing systems for Tissue Engineering - Basics of 3D printing technology- Bioinks for 3D printing of artificial extracellular matrices- Printing technologies used in three dimensional printing-Application of 3D printing- Scaffold based bioink materials-hydrogels- Nanotechnology based bioink-Artificial extracellular matrix-3D-printed polymeric scaffolds for bone tissue engineering- 3D-printed natural and synthetic polymeric scaffolds for bone tissue engineering- Synthetic 3D-bioprinted scaffolds- polymer based composite scaffolds-Cellulosic materials as bioinks for 3D printing applications-Cellulose for biomedical applications- wound healing- Bioresponsive implants-drug delivery.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify materials which can be used as biomaterials based on their properties.	K2
CO2	Examine the mechanical properties of biomaterials and use strengthening mechanisms to enhance the properties.	K2
CO3	Explore possible applications of materials as implants/assistive devices.	K3
CO4	Analyze the host and implant response with respect to selected biomaterials.	K4
CO5	Explore the use of 3 -D printing technology in tissue engineering.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				1	1		2			2
CO2	3	3	2			1	1		2			3
CO3	3	3	3			2	2		2			3
CO4	3	3	3			2			2			3
CO5	3	3	3		1	1			2			3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomaterials: Principles and Applications	Park JB, Bronzino JD	CRC press	2002
2	Biomedical composites	Ambrosio L, editor	Woodhead Publishing	2017
3	Advanced 3D-printed systems and nanosystems for drug delivery and tissue engineering	Du Toit L, Kumar P, Choonara Y, Pillay V	Elsevier	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advances in biomaterials for biomedical applications.	Tripathi, Anuj, Jose Savio Melo	Springer	V6ol. 6. 2017
2	Biomaterials: an introduction	Park, Joon, and Roderic S. Lakes	Springer Science & BusinessMedia	2007
3	Biomaterials science and engineering.	Park JB	Springer Science & BusinessMedia	2012
4	Fundamentals of biocompatibility	Black J. Biological performance of materials	CRC Press	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_bt28/preview
2	https://onlinecourses.nptel.ac.in/noc24_bt28/preview
3	https://onlinecourses.nptel.ac.in/noc24_bt28/preview
4	https://onlinecourses.nptel.ac.in/noc24_bt28/preview
5	https://onlinecourses.nptel.ac.in/noc22_bt61/preview

SEMESTER S4

BIOSTATISTICS

Course Code	PEEBT416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop proficiency in using various statistical inference techniques, including hypothesis testing, confidence intervals, and p-value calculations, to make informed decisions based on sample data.
2. Acquire the skills to design experiments, collect data, and apply ANOVA techniques to assess the significance of differences between group means.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic concepts: Measures of central tendency – measure of dispersion – properties. Random variables – discrete and continuous – probability density function – binomial, Poisson and normal distributions – Joint probability density function – marginal and conditional distributions. Statistical inference: Interval estimation – mean and variance – testing of hypothesis – single population mean – difference between two population means.	9
2	Hypothesis testing for categorical data – Fisher exact test – chi square distribution – Goodness of fit. Non parametric tests – Wilcoxon sign rank test; Multisample inference –Introduction to analysis of variance – one way analysis of variance (fixed effects model) – hypothesis testing in one-way anova (fixed effects model) – comparison of specific groups in one-way anova	9
3	Regression and correlation: Fitting of regression line (least squares method) – linear regression – statistical inference on parameters from regression line – correlation coefficient– statistical inference on correlation coefficient – multiple regression – partial and multiple correlation – rank correlation coefficient.	9

4	Scope of statistics in biomedical data analysis - Statistical design of experiments for clinical and laboratory data – Study design - Measure of effect for categorical data – confounding and standardization – methods of inference for stratified categorical data – power and sample size estimation for categorical data	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts in statistics, types of data and probability distributions.	K2
CO2	Apply the different methods of statistical inference	K3
CO3	Perform analysis of variance on data obtained from a designed experiment	K4
CO4	Apply the principles of regression and correlation	K3
CO5	Understand the scope of statistics in biomedical data analysis	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2						2		3
CO2		3	3							2		3
CO3		3	3	2						2		3
CO4		3	3							2		3
CO5	3	3	3							2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biostatistics- A Foundation for Analysis in the Health Sciences	Wayne W. Daniel and Chad L. Cross	Wiley	11th edition , 2018
2	Fundamentals of Biostatistics,	B. Rosner	Cengage Learning.	8th edition 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Medical Statistics	Martin Bland,	Oxford University Press	2015
2	Principles and Practice of Biostatistics	Antonisamy B, Premkumar P S & Christopher S	Elsevier (Relex India Pvt. Ltd.), New Delhi,	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/102101056
2	https://archive.nptel.ac.in/courses/102/106/102106051/

SEMESTER S4
EMBEDDED SYSTEM DESIGN

Course Code	PEEBT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course introduces students to the embedded system concepts, including hardware and software aspects of Embedded System Design.
2. Students will learn to model and solve real-world problems using Embedded System Design

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Embedded system: Embedded Hardware Architecture, Communication Interface Standards, Embedded system development process, Embedded Operating systems, Types of Embedded Operating systems. Embedded Processors - CPU architecture of ARM processor(ARM9) - CPU Bus Organization and Protocol –Five stage pipelining – Instruction sets- ARM instruction set-Thumb instruction set.	9
2	Integration and Testing of Embedded Hardware and Firmware- Embedded firmware design approaches, Embedded System Development Environment – IDEs, Cross Compilers, Disassemblers, Decompilers, Simulators, Emulators and Debuggers.	9
3	RTOS based Design – Basic operating system services-structure-kernel, process, tasks and threads. Process management-memory management-device management, Task scheduling algorithms - Non pre-emptive scheduling and Pre-emptive scheduling.	9
4	Inter Process Communication and Synchronization –Shared data– Inter process communication -Signals – Semaphore – Message Queues – Mailboxes – Pipes –Sockets – Remote Procedure Calls (RPCs).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply knowledge of embedded system hardware architecture, communication standards, development processes, and types of embedded operating systems to design and develop embedded solutions effectively.	K3
CO2	Evaluate efficient solutions using ARM processor architecture and optimize embedded software performance.	K5
CO3	Develop solutions utilizing RTOS fundamentals and implementing effective task scheduling algorithms.	K4
CO4	Design and implement robust inter-process communication mechanisms and synchronization techniques to ensure reliable operation of embedded systems.	K3
CO5	Apply embedded system concepts and tools in the development, integration, and testing of complex embedded projects. Demonstrate proficiency in project management, teamwork, and documentation, adhering to industry standards and best practices.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2	-	-	-	-	2	-	3
CO2	3	2	3	3	3	-	-	-	-	2	-	3
CO3	3	3	3	2	3	-	-	-	-	2	-	3
CO4	3	2	3	2	2	-	-	-	-	2	-	3
CO5	3	2	3	2	2	-	-	-	-	2	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computers as Components: Principles of Embedded Computing System Design.	Wayne Wolf	Morgan Kaufman Publishers	Third Edition, 2008
2	Arm system developers guide", Designing and Optimizing system Software.	Andrew N Sloss, D. Symes, C. Wright	Morgan Kaufman Publishers /Elsevier	2006
3	Introduction to Embedded Systems.	Shibu K.V.	McGraw Hill Education (India),	First Edition, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Embedded Software Primer.	David E. Simon	Pearson Education Asia	First Edition 2000
2	Embedded / Real time systems: Concepts, Design and Programming	Dr. K. V. K. K. Prasad	Dream Tech press	Second Edition, 2003
3	Embedded systems: An Integrated Approach.	Lyla B. Das	Pearson Education India	First Edition 2012
4	Embedded Systems Architecture, Programming and Design.	Rajkamal	McGrawHill Education	Third Edition 2017
5	Programming for Embedded Systems.	Dreamtech Software Team	Wiley Deamtech	First Edition 2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106105193
2	https://onlinecourses.nptel.ac.in/noc20_cs16/preview
3	https://nptel.ac.in/courses/106105159
4	https://onlinecourses.nptel.ac.in/noc24_ee68/preview

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	6

2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics	6

	and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.

- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create
CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019

5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g.,

sustainable building design, water management project, infrastructure project).

- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4
LINEAR INTEGRATED CIRCUITS LAB

Course Code	PCEBL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize the students with the linear integrated circuits.
2. To design, implement and analyze circuits using op-amps and other ICs.

Expt. No.	Experiments (Minimum 12 experiments mandatory)
1	Voltage follower, inverting and non-inverting amplifiers using op-amps - Frequency response.
2	Summing and difference amplifiers using op-amps.
3	Active integrator and differentiator circuits.
4	Active first and second order high pass & low pass filters.
5	Instrumentation amplifier.
6	Notch filter.
7	Comparator circuits using op-amp.
8	Window comparator using op-amp.
9	Digital to Analog Converter.
10	Analog to Digital Converter.
11	Precision half wave rectifiers.
12	Precision full wave rectifiers.
13	Astable and monostable multivibrator using op amp.
14	Astable multivibrator using IC555.
15	Monostable multivibrator using IC555.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design linear analog circuits op amp.	K4
CO2	Implement circuits for analog biosignal processing.	K4
CO3	Implement Digital to analog converters, analog to digital converters and filters.	K4
CO4	Design integrators, differentiators and oscillators using op-amps.	K4
CO5	Develop circuits for biomedical applications using specialized ICs.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2		2	1	1			1		1
CO2	2	2	2		2	1	1			1		1
CO3	2	2	2		2	1	1			1		1
CO4	2	2	2		2	1	1			1		1
CO5	2	2	2	1	2	1	1			1		1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op-Amps and Linear Integrated Circuits	Ramakant A. Gayakwad	Pearson Education Asia	Fourth edition, 2015
2	Operational Amplifiers - Theory and Design	Johan Huijsing	Springer International Publishing	Third illustrated edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op Amps for Everyone	Bruce Carter, Ron Mancini	Elsevier Science	Fifth illustrated edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/J_YH_Reb_GM?si=tl5DYQwLRpHuhoFZ
2	https://www.youtube.com/watch?v=5EcfaNt9MY&list=PLg0tpLGKUoxs_5xL5qm_APwcMO2yYsGhk&index=19
3	https://www.youtube.com/watch?v=NneanGrfilg&list=PLg0tpLGKUoxs_5xL5qm_APwcMO2yYsGhk&index=11
4	https://www.youtube.com/watch?v=BeOpnJeJ0w8&list=PLg0tpLGKUoxs_5xL5qm_APwcMO2yYsGhk&index=14

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

MICROCONTROLLER PROGRAMMING AND INTERFACING LAB

Course Code	PCEBL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. This course aims to provide an experience to program microcontrollers using assembly language and C.
2. Students will learn to interface external hardware devices with microcontroller.
3. Additionally, the course fosters teamwork, effective documentation, and communication skills for professional technical reporting and collaborations

Expt. No.	Experiments
1	Familiarization of 8051 based kit and simulation platforms using assembly language programs for data transfer.
2	Assembly language program using arithmetic and logical instructions
3	Assembly language program using compare and rotate instructions
4	Assembly language program using program flow control instructions
5	Assembly language program to generate waveforms using Timers
6	8051 C programming examples for data transfer, arithmetic and logic operations
7	8051 C programming examples for compare and rotate operations
8	C program - Time delay and I/O operations
9	Interfacing Stepper motor with 8051
10	Interfacing LCD with 8051
11	Interfacing keyboard with 8051
12	Interfacing ADC with 8051
13	Assembly language program using bit processing instructions
14	Assembly language program for programming ports of 8051
15	Assembly language program examples for interrupt programming
16	Assembly language program examples for serial communication with different baud rates

- A minimum of 15 different types of programs shall be done for the first 4 experiments. The programs shall be:
 - a) moving single data between RAM & ROM memory locations
 - b) block transfer of data
 - c) addition of two 16-bit numbers, first n natural numbers and BCD data
 - d) sorting of data in ascending and descending order and finding the largest and the smallest in an array
 - e) checking data whether odd or even
 - f) finding the square root, factorial, LCM, HCF etc
 - g) generation of special number sequences like Fibonacci sequence
- For experiment 5, timers can be used to generate square wave with equal and unequal duty cycles and to generate sawtooth and triangular waveforms. Any three shall be done in the lab.
- A minimum of 10 different types of programs shall be done using 8051 C programming (6, 7 & 8)
- For interfacing, any two shall be done.
- Bit processing, Port, Interrupt and serial communication related programs (13, 14, 15 & 16) shall be supplementary.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*

- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop 8051 assembly language programs.	K3
CO2	Develop 8051 C programs.	K3
CO3	Interface 8051 with peripherals	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		3	2		2	2		3	
CO2	3	3	2		3	2		2	2		3	
CO3	3	3	2		2	2		2	2		3	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	8051 Microcontroller and Embedded Systems using Assembly and C	Muhammed Ali Mazidi and Janice GillispieMazidi	Pearson Education	2 nd Edition
2	The 8051 Microcontroller	Kenneth J Ayala,	Thomson Delmar Learning	3 rd Edition
3	Microcontrollers and Applications	RamaniKalpathi and Ganesh Raja	Sanguine Technical Publishers	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microcontrollers and Applications	Ramani Kalpathi and Ganesh Raja	Sanguine Technical Publishers	2009
2	Programming Arduino: Getting Started with Sketches (Tab) 2nd Edition	Simon Monk	McGraw Hill TAB	2nd Edition, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=scdMQFzhsyc
2	https://www.youtube.com/watch?v=odEQH9ML4RQ
3	https://www.youtube.com/watch?v=NhurqshD0HA
4	https://www.youtube.com/watch?v=RWZBgWv_01U

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.

- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

**ELECTRONICS & BIOMEDICAL
ENGINEERING**

SEMESTER S5
HOSPITAL ENGINEERING

Course Code	PCEBT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Students will be able to apply engineering solutions to optimize hospital operations, enhance patient care, and ensure safety and compliance with relevant standards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Role of Engineering in healthcare sector: Definition: Bio-Engineering, Biomedical Engineering, Clinical engineering & Hospital engineering. Role & responsibilities. Concept of smart hospital- Increasing staff productivity- optimizing cost efficiencies- Improving patient outcomes- Integrated building management platform- Architecture- Regulations Planning and design of various departments. (Radiology Dept, Nuclear Medicine, ICU, Central Sterilization and Operation theatres) – Smart hospitals -case study	11
2	Power systems in Hospitals: Electrical power systems in hospitals –Design of sub stations. Stabilized and uninterrupted power supply systems. Protective systems: Over voltage and over current protectors, circuit breakers, Surge protectors, EMI filters. Introduction to hospital electrical safety and its ISO standards- Safety measures & standards for radiation rooms-planning & installation. Patient safety – electrical shocks and hazards – micro and macro shocks – effects of electrical current on human body – leakage currents – types & measurements	11
3	Centralized medical gas supply and sterilization systems in hospital: Hospital gas supply systems - Centralized supply of air, nitrous oxide, vacuum & oxygen - principle of production of liquid oxygen. Air quality in hospital environment – Heating, Ventilation and Air Conditioning (HVAC)	11

	- Basic concepts and standards in different hospital areas - Refrigeration systems, Air changes, filtering & sterility. Sterilization systems in hospitals: Principles and techniques of sterilization - Steam & EO sterilization. Autoclaves, Incinerators. Hospital seamless flooring and antibacterial painting, Biomedical waste management and solutions	
4	Healthcare technology management: Medical Equipment Planning in healthcare - purchase & contract procedures (CMC and AMC) - selection, testing and calibration, installation of equipment. Training of medical staff. Repair & maintenance of medical equipment, Preventive maintenance. Healthcare technology management with RFID system-based asset tracking- medical equipment IT networking and communication with Hospital Information Systems (HIS)/Radiology Information System (RIS)/ Picture Archival and Communication System (PACS). Medical preparedness for disaster management: Principles of hospital disaster planning- Emergency medical services-Role of clinical and biomedical engineers in disaster preparedness and management-organizational, technological and professional recommendations. Introduction to ITIL framework	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

[‡]*Students are required to identify an equipped hospital with working medical equipments spread over different departments of the hospital. During their academic break from the second to third year, students shall make a hospital visit (not less than 30 hours) to the various departments in the hospital. The objective is to familiarize themselves with the facilities and the equipments in various departments.*

Subsequently, the students must prepare a report and present it before the evaluators. This shall be marked as the assignment for the course Hospital Engineering in the fifth semester.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the role of a biomedical engineer in hospitals and healthcare	K3
CO2	Describe hospital electrical systems and safety standards.	K2
CO3	Analyse the centralized medical gas supply and sterilization systems in hospitals	K3
CO4	Analyse the current technology management systems implemented in hospitals and evaluate the role of clinical engineering in disaster preparedness and response.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3			2	3	2	1	2	2		2
CO2	3	2				3	3	2		2		2
CO3	3	2				3		1		2		2
CO4	3	2				3	1	2	1	3		2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Handbook for Clinical and Biomedical Engineers	Azzam Taktak, Paul Ganney, David Long and Paul White (Eds.)	Academic press	2010.
2	Clinical Engineering from Devices to Systems	Roberto Miniati, Ernesto Iadanza, Fabrizio Dori	Elsevier	2005
3	Emerging Horizons of Clinical Engineering in Disaster preparedness and Management: Proposal for an expanded professional identity, Global Clinical Engineering	F. Hose	Journal	2020.
4	Medical Preparedness, Response and Homeland Security	Arora Rajesh, Arora Preeti Disaster Management	Oxfordshire	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Radio frequency identification applications in smart hospitals	H. Al Nahas and J. S. Deogun	The Proceedings of IEEE International Symposium on Computer-Based Medical Systems	pp. 337-342, 2007
2	Basic Refrigeration and Air Conditioning	Anantha Narayanan	2nd edition, TMH 1996. Switzerland 2015A	LNCS 8700, pp. 1–20, 2015.DOI: 10.1007/978-3-319- 16226-3_
3	Disaster medicine.	Hogan, David E., and Jonathan L. Burstein, eds	Lippincott Williams & Wilkins	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/arp20_ap01/preview
2	https://www.youtube.com/watch?v=NU4k8KUd2ho
3	https://www.youtube.com/watch?v=87BkSyTW0Vg&t=1765s
4	https://onlinecourses.nptel.ac.in/noc19_ar12/preview

SEMESTER S5
MEDICAL IMAGING TECHNIQUES

Course Code	PCEBT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course enables the students to understand the principle of image formation in various imaging modalities currently used in medicine. Students will be able to analyse the systems and techniques used in various diagnostic imaging methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Ultrasound imaging: Properties of ultrasound- Doppler effect. Principle of image formation- constructional details of US probes- Transducer Design – Considerations-Transducer arrays- Modes of scanning-A mode-B mode-M mode scanners-instrumentation-axial and lateral resolutions- Duplex scanning-applications of each. Doppler effect- Principles of 3D/4D ultrasound. (Assignment only -Functional Ultrasound Imaging, Clinical applications- therapeutic uses of ultrasound)	11
2	Computed Tomography: Principle of CT imaging, Generations of scanners-system instrumentation-X-ray tubes for CT applications-pencil beam and cone beam projections-Image reconstruction algorithms for pencil beam, Image characteristics-CT numbers-Nuclear imaging-PET-SPECT- Rectilinear Scanner, Gamma camera	11
3	Magnetic Resonance Imaging: Principle of Nuclear Magnetic Resonance- Relaxation-MR instrumentation-Contrast mechanisms in MRI- pulse sequences- FLASH-GRASS-principle of image reconstruction, Spatial	11

	encoding systems- k-space - Functional MRI- BOLD - Spatial and temporal Resolution of fMRI. (Assignment only: diffusion weighted imaging, MR spectroscopy)	
4	Hybrid Imaging: Instrumentation MR-PET Instrumentation- MR Compatible PET Detector Technology, MR-PET System Architecture. PET-CT, SPECT-CT. Intraoperative imaging Intraoperative imaging and image guided therapy-X-Ray Hybrid Modalities for Image Guidance-Technology of Ultrasound-Guided Therapy- CT Guided Interventions: Current Practice and Future Directions	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the principles of diagnostic Ultrasound imaging	K4
CO2	Analyze the CT and nuclear imaging techniques	K4
CO3	Analyze the Nuclear Magnetic Resonance imaging system and nuclear imaging systems	K4
CO4	Analyze hybrid imaging systems and intra operative imaging systems	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1			1				1		3
CO2	3	3	1			1				1		3
CO3	3	3				1				1		3
CO4	3	3				1				1		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Diagnostic ultrasound: physics and equipment. CRC Press, 2019	Hoskins, Peter R., Kevin Martin, and Abigail Thrush	CRC Press	2019.
2	Diagnostic ultrasound imaging: inside out	Szabo, Thomas L	Academic Press	2004
3	Webb's Physics of Medical Imaging	M Flower	Taylor & Francis	2016
4	Computed Tomography-E-Book: Physical Principles, Clinical Applications	Seeram, Euclid	Elsevier Health Sciences	2015
5	Intraoperative Imaging and Image-Guide Therapy	Ferenc A. Jolesz	Springer	2008
6	How does MRI work? An Introduction to the Physics and Function of Magnetic Resonance Imaging	Weishaupt, Dominik, Köchli, Victor D, Marincek, Borut	Springer	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Computerized Tomographic Imaging	Avinash C Kak, Malcolm Slaney	IEEE	2001
2	MRI made Easy	Hans H Schild		2003
3	Ultrasound imaging and therapy,	Fenster, Aaron, and James C. Lacefield	Taylor and Francis	2015
4	Nuclear Medicine Textbook, Methodology and Clinical Applications	DuccioVolterraniPaola Anna ErbalgnasiCarrióH. William Strauss Giuliano Mariani	Springer	2019
5	Nuclear medicine and PET/CT: technology and techniques	Christian, Paul E., and Kristen M. Waterstram-Rich	Elsevier	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_bt50/preview
2	https://archive.nptel.ac.in/courses/108/105/108105091/
3	https://archive.nptel.ac.in/courses/102/106/102106094/
4	https://onlinecourses.nptel.ac.in/noc21_bt50/preview

SEMESTER S5

THERAPEUTIC EQUIPMENTS

Course Code	PCEBT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Learn the basic principles of different therapeutic equipment used in clinical environments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cardiac Assist Devices: Cardiac pacemakers: different modes of operation, external and implantable pacemakers. Defibrillators: AC and DC defibrillator – basic principle. Cardioverter defibrillator – block diagram. Implantable defibrillator and automated external defibrillator (AED)– functional block diagram. Catheterization- IABP, Stents.	6
2	Respiratory Aids: Ventilators - Block diagram, ventilator settings, types of ventilators-Pressure, Volume, Time controlled, Flow, Patient Cycle Ventilators, Modes of ventilation: Non-Invasive- CPAP, BiPAP; Invasive modes - Controlled, Assisted, SIMV, APRV, Pressure Support; Flow sensors and FiO2 sensor. Anesthetic machines - Block diagram and working Gas blending and vaporizers, Anesthesia delivery system, breathing circuit. Computer aided anesthesia control.	8
3	ExtraCorporeal Devices: Dialysis machine - haemodialysis unit, block diagram and working. Heart lung machine (HLM) - Heart lung machine (HLM) – principle of operation-functional block diagram -types of oxygenators. Extracorporeal membrane oxygenation (ECMO) machine. Lithotripsy - Concept of lithotripter systems, Extracorporeal shock wave lithotripsy (ESWL). Drug delivery systems – Different approaches for	12

	drug delivery, Drug infusion devices -Gravity Drip Systems, Infusion pump, Syringe pump, Implantable pumps, Patient Controlled Analgesia (PCA) Pumps, Elastomeric Infusers. Recent developments in drug infusion systems.	
4	Diathermy, Medical Stimulators & Special Diagnostic Aids: Surgical diathermy - Short-wave diathermy, Ultrasonic therapy, Interferential current therapy, Neuro drills, Neuro navigation systems, Intra operative nerve monitors, Deep brain stimulators. Transcutaneous electrical nerve stimulation (TENS) - applications. Transcranial magnetic stimulation (TMS) – applications. Endoscopy – Principles, different types & applications. Block diagram of a fibre optic endoscope with integral TV cameras. Cryo-surgery techniques and application.	10

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognise the design concepts of cardiac pacemakers and defibrillators.	K2
CO2	Analyse the working principles of ventilators and anaesthetic machines.	K4
CO3	Explain the working principle of heart lung machines, dialysis machine, lithotripsy and drug delivery devices.	K4
CO4	Examine the therapeutic applications of electric current.	K2
CO5	Identify the components of endoscopes used for different applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	-	-	-	-	-	-	-	3
CO2	3	3	1	2	-	-	-	-	-	-	-	3
CO3	3	2	2	2	-	-	-	-	-	-	-	3
CO4	3	2	1	2	-	-	-	-	-	-	-	3
CO5	3	2	1	2	-	-	-	-	-	-	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical engineering fundamentals.	Bronzino, Joseph D and Donald R. Peterson.	CRC press	2014
2	Handbook of medical technology.	Kramme, Rüdiger, Klaus-Peter Hoffmann, and Robert Steven Pozos, eds. Springer	Springer Science & Business Media	2011

3	Medical Ventilator System Basics: A Clinical Guide.	Lei, Yuan.	Oxford University Press	2017
4	Handbook of biomedical instrumentation.	Khandpur, Raghubir Singh.	McGraw-Hill Education	2016
5	Introduction to biomedical instrumentation: The technology of patient care.	Christe, Barbara L.	Cambridge University Press	2018
6	Encyclopedia of medical devices and instrumentation. Vol. 4.	Webster, John G.	Wiley-Interscience	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical instrumentation: application and design.	Webster, John G., ed.	John Wiley & Sons	2009
2	"Automatic ventilation of the lungs."	Mushin, William Woolf	Elsevier	1980
3	Introduction to Biomedical Equipment Technology	Joseph J. Carr, John M. Brown	MGH, 2nd Edition	2006
4	Principles of Applied Biomedical Instrumentation	Geddes & Baker	Wiley	2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=Y0LtPEyTHgE&list=PLyUFBC_UMpje19E8ekFLfog2qkE6ldiZT&index=1&t=93s
2	https://www.youtube.com/watch?v=ilpqHq1APn8&list=PLyUFBC_UMpje19E8ekFLfog2qkE6ldiZT&index=2&t=10s
3	https://www.youtube.com/watch?v=YviI4AjKHsg&list=PLyUFBC_UMpje19E8ekFLfog2qkE6ldiZT&index=3
4	https://www.youtube.com/watch?v=kHONXxQK3iE&list=PLyUFBC_UMpje19E8ekFLfog2qkE6ldiZT&index=4

SEMESTER S5

INTRODUCTION TO BIOMEDICAL SIGNAL PROCESSING

Course Code	PBEBT504	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course makes students familiarized with basics of digital signal processing & concepts of digital filters & its design and applications of biomedical signal processing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of biomedical signal processing: Introduction & relevance of biomedical signal processing. Biomedical signals –ECG, EMG, EEG, PCG, VAG, carotid pulse, speech signal, concurrent signals. Signal representation, Aliasing-Sampling theorem. Elementary signals. Classification of discrete signals - Energy and Power signals – periodic and aperiodic, even and odd signals. Discrete time systems - Properties of discrete systems. LTI system- Impulse Response of a Linear Time-Invariant System -convolution-correlation - difference equation representation of discrete systems	9
2	Fourier Analysis: Fourier Analysis for Continuous Time Processes-Synthesis of an ECG signal using pure sinusoid. Frequency Domain Analysis of discrete time signals: Discrete Fourier series - Discrete Time Fourier Transform- Discrete Fourier Transform-properties - Fast Fourier Transform- FFT algorithms (Radix 2)- decimation in - time - decimation in - frequency.	9
3	Spectrum analysis & Z Transform: Spectrum analysis of bio signals – Introduction to parametric & non parametric methods. Z transform – ROC, properties of ROC, Properties of Z transform: Linearity, Time shifting, Scaling, Time reversal, Differentiation in Z domain, Convolution,	8

	Correlation, Multiplication, Parseval's relation, initial value theorem-rational Z transform-poles & zeros-LTI system function - inverse Z transform-long division-partial fraction expansion method-residue method-convolution method.	
4	Digital filters: Concept of digital filtering, types of digital filters. Comparison with analog filters. FIR filter design using Fourier series - use of window functions, IIR Digital Filters - Analog filter approximations - Butterworth approximation - Frequency transformation techniques - impulse invariant transformation - Matched Z transform technique- bilinear transformation	10

Suggestion on Project Topics

Students are asked to form groups of maximum five members. Each group should identify a project on the areas specified below and choose suitable data base. Develop a methodology to implement the project. Develop the algorithm and code for the project

Areas of Projects Topics:

- Filtering for removal of artefacts: Implementation of time domain Filters & Frequency domain filters for removal of high frequency and low frequency and periodic artefacts
- Event detection in biomedical signals – ECG: QRS wave detection, P wave and T wave detection, PPG: diastolic notch detection - Heart sound detection- EEG rhythm detection
- Frequency Domain characterisation of Biomedical signal: Power spectrum density estimation of biomedical signals
- Wave shape and Waveform Complexity analysis: Normal and Ectopic beat, Arrhythmia analysis, Analysis of exercise ECG, Analysis of respiration

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the basics of discrete time signals & systems and their time domain processing techniques with specific reference to biomedical signals.	K4
CO2	Apply the different methods of analysing digital signals using frequency domain analysis.	K3
CO3	Analyse the discrete time system using Z Transform.	K4
CO4	Design of different types of digital filters such as FIR & IIR filters.	K3
CO5	Analyse the signal processing techniques for different biomedical applications	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	1	2	-	-	-	-	-	-	-
CO2	3	2	-	1	2	-	-	-	-	-	-	-
CO3	3	2	2	1	2	-	-	-	-	-	-	-
CO4	3	2	2	1	2	-	-	-	-	-	-	-
CO5	3	2	3	2	2	2	2	2			1	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal Processing- Principles, Algorithms and Applications	John G Proakis & Dimitris G Manolakis	PHI	4 th Edition, 2016
2	Digital Signal Processing	P. Ramesh Babu	Scitech Publications, India	6th Edition, 2014
3	Biomedical Signal Analysis	Rangaraj M Rangayyan	John Wiley	2 nd Edition, 2015
4	Signals and Systems	Alan V Oppenheim, Alan S Willsky	Prentice Hall India	2 nd Edition, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Signals & Systems in Biomedical Engineering	Suresh R Devasahayam	Springer	2 nd Edition, 2013
2	Digital Signal Processing	Andreas Antoniou	Prentice Hall of India	4 th Edition, 2006
3	Digital Filters Analysis & Design	Andreas Antoniou	Prentice Hall of India	2018
4	Theory & Application of Digital Signal processing	R Rabiner & B. Gold	PHI	3 rd Edition, 2000
5	Digital Signal Processing – A Practical Guide for Engineers and Scientists	Steven W. Smith	Elsevier India Pvt Ltd	2006
6	Digital Signal Processing	Sanjit K. Mithra	Tata McGraw Hill	4 th Edition, 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	LINK ID
1	https://archive.nptel.ac.in/courses/117/105/117105134/ https://archive.nptel.ac.in/courses/108/101/108101174/ https://nptel.ac.in/courses/117102060
2	https://archive.nptel.ac.in/courses/117/105/117105134/ https://archive.nptel.ac.in/courses/108/101/108101174/ https://nptel.ac.in/courses/117102060
3	https://archive.nptel.ac.in/courses/108/101/108101174/ https://archive.nptel.ac.in/courses/117/105/117105134/ https://nptel.ac.in/courses/117102060
4	https://archive.nptel.ac.in/courses/117/105/117105134/ https://archive.nptel.ac.in/courses/108/101/108101174/ https://nptel.ac.in/courses/117102060

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

MEASUREMENTS AND INSTRUMENTATION

Course Code	PEEBT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The basic objective of this course is to introduce the concepts of electronics measurements and instrumentation and to provide the comprehensive understanding of principles of measurement of basic circuit parameters and electrical quantities using instrumentation, to ensure accurate reliable data acquisition in engineering applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Block Schematics of Measuring Systems: Performance characteristics, Static characteristics, Accuracy, Precision, Resolution, Types of Errors, Dynamic Characteristics, Repeatability, Reproducibility, Fidelity. Measurement of Voltage and current: Moving Coil and Moving Iron types, Extension of Range-shunts and multipliers, DC Voltmeters, D'Arsonval Movement, DC Current Meters, AC Voltmeters and Current Meters, Ohmmeters, Multimeters.	9
2	Bridges: Measurement of Resistance: Wheat Stone Bridge, Kelvin Bridge; Measurement of Inductance: Maxwell's Bridge; Measurement of Capacitance: Schering's Bridge. Measurement Of Physical Parameters: Flow measurement, displacement meters, liquid level measurement, measurement of humidity and moisture, velocity, force, pressure, temperature- measurements, data acquisition systems.	9
3	Signal Generators: Sine wave generator, Frequency – Synthesized Signal Generator, Sweep frequency Generator. Pulse and square wave generators.	9

	Function Generators. Signal Analyzers: AF, HF Wave Analyzers, Harmonic Distortion, Heterodyne Wave Analyzers, Spectrum Analyzers, Power Analyzers, Capacitance –Voltage Meters, Oscillators.	
4	Transducers: Classification, Strain Gauges; Force and Displacement Transducers, Resistance Thermometers- RTD, LVDT, Thermocouples, Hotwire Anemometers, Piezoelectric Transducers. Oscilloscopes: Cathode Ray Tube, Vertical and Horizontal Deflection Systems, Delay lines, Probes and Transducers, Specification of an Oscilloscope. Oscilloscope measurement Techniques, Special Oscilloscopes – Storage Oscilloscope, Sampling Oscilloscope.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the fundamental concepts and principles of Measurements and instrumentation.	K2
CO2	Apply suitable methods for the measurement of basic circuit parameters voltage and current.	K3
CO3	Apply suitable methods for the measurement of various quantities, resistance, inductance and capacitance.	K3
CO4	Understand functioning, specification and application of signal generation and signal analysing instruments.	K2
CO5	Describe the working principle, selection criteria and applications of various sensors, transducers in relation to measurement systems.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	2
CO2	3	3	-	-	-	2	-	-	-	-	-	2
CO3	3	3	-	-	2	-	-	-	-	-	-	2
CO4	3	2	-	-	2	-	-	-	-	-	-	2
CO5	3	2	3	-	3	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Course in Electrical and Electronics instrumentation and Measurements	K. Sawhney	Dhanpat Rai & Co.	2023
2	A Course in Electrical and Electronics instrumentation and Measurements	J B Gupta	S K Kataria & Sons	14 th Edition, 2014
3	Electronic Instrumentation	H.S.Kalsi	Tata McGraw Hill	4 th Edition 2019
4	Electronics Instrumentation and Measurements	David A. Bel	Oxford	3 rd Edition 2013
5	Electrical Measurements and Measuring Instruments	Golding E. W and Widdis	Wheeler Publishers	3 rd Edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern electronic instrumentation and measurement techniques	Albert D. Helfrick and William D. Cooper.	Prentice Hall of India	2016
2	Basic Electrical Measurements	Stout M. B.	Prentice Hall of India	2012
3	Electronic Measurements and Instrumentation	Oliver, Cage	McGraw Hill	2017
4	Electronics Measurements and Instrumentation	P. Purkrait, B. Biswas, S. Das, C.Koley	McGraw Hill Education (India) Pvt. Ltd	7 th Edition 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_ee44/preview https://nptel.ac.in/courses/108105153
2	https://onlinecourses.nptel.ac.in/noc19_ee44/preview https://nptel.ac.in/courses/114106049
3	https://nptel.ac.in/courses/114106049
4	https://onlinecourses.nptel.ac.in/noc23_ee105/preview

SEMESTER S5

ADVANCED MICROPROCESSORS & MICROCONTROLLERS

Course Code	PEEBT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2.30
Prerequisites (if any)	PBEBT 404	Course Type	Theory

Course Objectives:

1. Students will acquire an understanding of basic architecture of advanced microprocessors and microcontrollers for system development.
2. Students will be able to design and develop microprocessor & controller-based solution for solving real world problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	8086 Microprocessor: Register organization – Architecture- Physical memory organization & segmentation-Signal descriptions of 8086 chip – General bus operation – Addressing capability- Assembler directives - Addressing Modes - Introduction to Stack, Interrupts & Interrupt service routines, Minimum and Maximum mode 8086 system operation.	9
2	80386 Microprocessor: Salient features of 80386 DX -Architecture and Signal Description- Register Organization-Memory addressing in Real Address mode, Protected mode, Segmentation, Paging & Virtual modes. Pentium - General features, pipelining and super scalar architecture. Introduction to multicore processors: Concept of multi core processing - Homogeneous and heterogeneous multicore processors -- Major issues in multicore processing - Internal architecture of Intel Core2 Duo (Simple block diagram representation only) - Technological features of IA processors.	9
3	High performance RISC ARM architecture: ARM Processor Fundamentals- ARM architecture - ARM7TDMI programmer's model (basic level only)- Memory format – Embedded Applications. ARM based SoC- General features- Basic system-on-chip (SoC) architecture- System level interconnection – Memory organisation- SoC design approach. <i>(Class Assignment/Seminar topics: SoC Interconnect Customization and</i>	9

	<i>Configuration, Trade-off analysis on reconfigurable Parallelism, Case studies on design approach)</i> Introduction to high performance computing- GPU hardware architecture- Comparison with CPU – GPU programming mode- GPU in Radiation imaging.	
4	Introduction to ESP32 Microcontroller: Features of ESP32 development board – Architecture - Pin configuration – ESP32 v/s Arduino – Programming of ESP32 with Arduino IDE (only basic level). Introduction to STM32: Fundamentals of 32 -bit STM controllers – Architecture - STM boot modes - Interfacing of STM32 with stepper motor and 7 segment display.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Microprocessors and Peripherals	K.M Bhurchandi and A. K Ray	Tata Mc Graw -Hill Publishing Company Limited	2009
2	The Pentium Microprocessor	James L Antonakos	Pearson Education	2009
3	General Purpose Graphic Processor Architectures	Tor M. Aamodt, Wilson Wai Lun Fung, Timothy G. Rogers	Morgan & Claypool Publishers	2018
4	Exploring Raspberry Pi- Interfacing to the Real world with Embedded Linux	Derek Molloy	Wiley	2016
5	GPU based High Performance Computing in Radiation Therapy	Xun Jia, Steve B. Jiang	CRC Press	2018
6	Advanced Programming with STM32 microcontrollers	Majid Pakdel	Elektor Publication	2020
7	ESP32 Formats & Communication: Application of Communication Protocols with ESP32 Microcontroller	Neil Cameron	Apress	2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Intel Microprocessors- Architecture, Programming and Interfacing	Barry B Brey	Pearson Prentice Hall, 8 th Edition	2008
2	ARM System Developers Guide	Andrew N SLOSS, Dominic SYMES, Chris WRIGHT	Elsevier	2012
3	Microprocessors and Interfacing	D. V. Hall	MGH, 2nd Edition	2006
4	Microprocessors & Microcontrollers	Latha C, Murugeshwari B	SCITECH Publications	2015
5	ARM System on Chip Architecture	Steve Furber	Addison Wesley Professional, 2 nd Edition	2000
6	Getting Started with STM32 Nucleo Development	Agus Kurniawan	Agus Kurni	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108/105/108105102/
2	https://onlinecourses.nptel.ac.in/noc23_cs113/
3	https://onlinecourses.nptel.ac.in/noc21_cs09/
4	https://onlinecourses.swayam2.ac.in/aic20_sp04

SEMESTER S5

INTRODUCTION TO BIONANOTECHNOLOGY

Course Code	PEEBT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Students will be able to understand the principles and techniques in nanobiotechnology:
2. Students will be able to explore the applications and implications of nanobiotechnology in various fields:

Module No.	Syllabus Description	Contact Hours
1	Introduction: Introduction to nanotechnology and bio nanotechnology, time frame of nanobiotechnology, Biocompatibility and its assessment, Cell-nanomaterial interactions, toxicity and environmental impact	9
2	Nano particles preparation methods: Synthesis of metallic,ceramic,polymeric nanoparticles &its application in biofield. Biomimetic based nano structure. Biomolecules synthesis-DNA,viruses.Dendrimers Green synthesis methods, Recombinant DNA technology, Monoclonal antibodies	9
3	Analytical tools in nanobiotechnology: Tools in Nanobiotechnology: Microscopy and imaging techniques such as SEM, TEM, AFM and STM Elemental composition by EDS, WDX, ICPMS, SIMS etc. Morphology and crystal structure by XRD and XAS. Fluorescent microscopy and confocal microscopy	9
4	Applications of nanobiotechnology: Applications of nano-biotechnology in Biosensing, bioimaging for medical diagnosis, drug delivery, regenerative medicine . Nano-biotechnology in cancer, nanotherapeutics: gene therapy and immunotherapy. New materials and devices in medicine-implants and coatings, chemotherapy administration, nanobots. Future goals& Potential effects on global economics- Sustainable agriculture, sensors, nano carriers	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Differentiate nanobiotechnology and bionanotechnology	K2
CO2	Analyse nanoparticles and their preparation methods	K4
CO3	Describe the working principle of analytical tools in nanobiotechnology	K2
CO4	Evaluate future goals and concerns of nanobiotechnology	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2		-	1	-	-	-	2	-	3
CO2	2	3	2		-	1	-	-	-	2	-	3
CO3	2	3	2		-	1	-	-	-	2	-	3
CO4	2	3	2		-	1	-	-	-	2	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanobiotechnology: Concepts, Applications and Perspectives	Christof M.Niemeyer (Editor), Chad A. Mirkin	Wiley,	1 st edition, 2004.
2	Nanobiotechnology-II more concepts and applications	Chad A Mirkin and Christof M. Niemeyer,	Wiley VCH,	1 st edition, 2007
3	Nanotechnology in Biology and Medicine: Methods, Devices, and Applications	Tuan Vo-Dinh,	CRC Press,	2 nd edition, 2019.
4	Bio-Nanotechnology: A Revolution in Food, Biomedical and Health Sciences,	Debasis Bagchi (Editor), Manashi Bagchi (Author), Hiroyoshi Moriyama (Author), Fereidoon Shahidi (Author),	Wiley-Blackwell,	1 st edition 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanobiomaterials in Antimicrobial Therapy, Applications of Nano-biomaterials	Alexandru Grumezescu (Editor),	William Andrew,	1st edition, 2016.
2	Prospects and applications of nanobiotechnology: A medical perspective	Md Fakruddin, Zakir Hossain & Hafsa Afroz,	Journal of Nanobiotechnology volume 10, Article number: 31 (2012).	DOI: https://doi.org/10.1186/1477-3155-10-31
3	Perspective on Analytical Sciences and Nanotechnology, in Advanced Environmental Analysis	Deepali Sharma, Suvardhan Kanchi, Krishna Bisettya and Venkatasubba Naidu Nuthalapathib,	Applications of Nanomaterials,	Volume 1, 2016, pp. 1-34 DOI: 10.1039/9781782623625-00001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/118/107/118107015/
2	https://archive.nptel.ac.in/courses/118/104/118104008/
3	https://onlinecourses.nptel.ac.in/noc22_mm18/preview
4	https://archive.nptel.ac.in/courses/118/107/118107015/

SEMESTER S5
BIOMEDICAL OPTICS & BIOPHOTONICS

Course Code	PEEBT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. This course will give an understanding of the light -tissue interactions and the different applications of lasers in imaging, diagnosis, and therapy.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Bio photonics and Tissue Optics: Overview, Light transport in Tissue- Radiation transport theory, radiation transport model-basic parameters, scattering phase function, radiation transport equation, Tissue properties-refractive indices, scattering properties, absorption properties. Optothermal interactions-temperature rise and tissue damage, optothermal and opto-acoustic effects. Lasers for bio photonics: Basic laser action, laser diodes, CO2 lasers, Nd-YAG laser, optical fiber lasers.	9
2	Bio imaging: Near field optical microscopy-basic principles, instrumentation-optical probes, biological applications-investigation of cell material, Optical Coherence tomography imaging-principles of operation, image generation in OCT, OCT system, Ultrahigh-Resolution Optical Coherence Tomography. Hologram and Medical Applications: Types of Hologram, Advantages & Disadvantages.	9
3	Lasers in Diagnosis: Bio photonics for blood analysis, optical inference and control of blood physiology – glucose sensing, haemoglobin oxygenation, tissue oxygen analysis- fundamentals, clinical applications of NIR oxygenation monitoring, early optical diagnosis of pressure ulcers-clinical diagnostics – evaluation of tissue blood perfusion, local tissue metabolism and oxygen supply, local tissue structure atrophy.	9

4	Lasers in Treatment: Laser tissue welding-principles, benefits and limitations, mechanism of laser tissue welding-photochemical and photo thermal welding, therapeutic application of lasers in gastroenterology-lasers for destruction of tumors, lasers for lithotripsy, Lasers in restorative dentistry, laser ablation of hard tissues.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the basics of bio photonics and lasers used to explore the light-tissue interactions in diagnosis and therapy.	K3
CO2	Analyse the applications of photonics in bioimaging.	K4
CO3	Describe the diagnostic applications of lasers.	K4
CO4	Describe the diagnostic and therapeutic applications of lasers.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	-	-	1	1	-	2
CO2	2	2	2	-	-	-	-	-	1	1	-	2
CO3	2	2	2	-	-	-	-	-	1	1	-	2
CO4	2	2	2	-	-	-	-	-	1	1	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biophotonics- Concepts to Applications	Gerd Keiser	Springer	2016
2	Laser fundamentals	William T Silfvast, 2 nd edition	Cambridge University Press	2008
3	Lasers and Optoelectronics Fundamentals, Devices and Applications	Anil K Maini	Wiley	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Photonics Handbook: Therapeutics and Advanced Biophotonics	Tuan Vo-Dinh	CRC Press	2014
2	Handbook of Biophotonics	Jürgen Popp, Valery V. Tuchin, Arthur Chiou, Stefan H. Heinemann	Wiley-VCH	2012
3	Biophotonics for medical applications	Meglinski, Igor	Elsevier	2015
4	Introduction to Biophotonics	Paras N Prasad	Wiley- Interscience	2003
5	Lasers- Fundamentals and applications	K Thyagarajan, Ajoy Ghatak 2nd edition	Springer	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/
2	https://onlinecourses.nptel.ac.in/
3	https://onlinecourses.nptel.ac.in/
4	https://onlinecourses.nptel.ac.in/

SEMESTER S5

VLSI DESIGN

Course Code	PEEBT526	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCEBT303	Course Type	Theory

Course Objectives:

This course aims to:

1. Introduce the fundamentals of VLSI fabrication and the implementation of digital circuits using MOS transistors, covering MOSFET operation, MOS circuit design principles, including layout design rules and combinational logic circuits.
2. Explore basic circuit concepts and review power dissipation in CMOS circuits, focusing on strategies for power reduction.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic IC fabrication technology: Basic fabrication steps- Wafer preparation, Oxidation, Diffusion, Etching process, Ion implantation techniques, Isolation techniques. Comparison and performance evaluation of bipolar and MOS technology. Fabrication process of MOS devices: NMOS and PMOS IC technology, CMOS n-well and p-well process.	9
2	MOS transistor and geometrical effects: Structure of MOSFET, Operation of MOSFET, current – voltage characteristics, channel length modulation, short channel effects, narrow channel effects, scaling factors for device parameters. Analogy of MOSFET capacitance. MOS inverters: Basic circuit of NMOS inverter-voltage transfer characteristics, inverters with n- type MOSFET load. CMOS inverter - voltage transfer characteristics.	9

3	Combinational MOS logic circuits: Circuit operation of NMOS NOR & NAND gates, CMOS NOR & NAND gates, compound circuits using CMOS logic. Basic circuit concepts: sheet resistance, area capacitance, delay unit, inverter delays – driving large capacitive loads, super buffers.	9
4	MOS circuit design processes: Stick Diagram, Layout design rules for MOS circuits – Lamda based rules, Layout examples of Inverter, NAND and NOR. Review of power dissipation in CMOS Circuits: Sources of power dissipation- static, dynamic and leakage power dissipation. Dynamic power reduction techniques - Supply voltage scaling approaches-parallelism, pipelining (Basic Concepts).	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the design hierarchy and fabrication process of IC	K2
CO2	Understand the MOS structure, MOSFET operation, current-voltage characteristics, and the effects of scaling on device parameters	K2
CO3	Analyze the basic circuit structures and operation using MOS transistors	K4
CO4	Understand the layout design rules to design MOS circuits and power dissipation and reduction strategies in CMOS circuits	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	1	-	-	-	-	-	-	-	-	3
CO2	3	-	1	-	-	-	-	-	-	-	-	3
CO3	3	-	1	-	-	-	-	-	-	-	-	3
CO4	3	-	1	-	-	-	-	-	-	-	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CMOS Digital Integrated Circuits Analysis & Design	Sung Mo Kang, Yusuf Leblebici	McGraw- Hill Higher Education	41/e, 2002
2	Basic VLSI Design	Douglas A Pucknell, Kamran Eshraghian	PHI Learning	3/e, 2015
3	Digital Integrated Circuits - A Design perspective	Jan M. Rabaey, A. Chandrakasan, B. Nikolic	Pearson	2/e, 2003
4	Principles of CMOS VLSI Design: A Systems Perspective	Neil H E Weste and Kamran Eshraghian	Pearson Education	2/e, 1993

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	VLSI Engineering	Thomas E. Dillinger	Pearson education	1/e, 1988
2	VLSI Technology	S M Sze	McGraw Hill Education	2/e, 2017
3	Integrated Circuits	K R Botkar	Khanna Publishers	10/e, 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/101/108101089/ https://www.youtube.com/watch?v=vbWNy2ZpXLM
2	https://archive.nptel.ac.in/courses/108/107/108107129/
3	https://archive.nptel.ac.in/courses/108/107/108107129/
4	https://archive.nptel.ac.in/courses/117/106/117106149/

SEMESTER S5

ARTIFICIAL NEURAL NETWORKS

Course Code	PEEBT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course type	Theory

Course Objectives:

1. Enable the student to understand the basic concepts of different types of neural networks
2. Apply the knowledge gained to implement machine learning concepts to applications in Biomedical field.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental Concepts of Artificial Neural Networks: Structure of biological neurons relevant to ANNs– Comparison Basic building blocks of ANN. Activation functions, McCulloch-Pitts Neuron Model Learning Rules- Hebbian Learning Rules, Perceptron, Delta, Competitive learning	9
2	Single layer and Multilayer networks: Perceptron networks, Gradient descent method, Back Propagation algorithm Architecture, training algorithm. Counter Propagation Network, Adaptive resonance Theory, ART 1 and ART 2 Learning, stability plasticity concept., Self-Organising feature maps, Kohonen network, SOM learning	9
3	Associative memory networks: Algorithms for pattern association Hetero associative networks, Auto associative memory networks Bidirectional associative memory networks Energy Function. Hopfield networks. Support vector machines: hard-margin and soft-margin SVMs, concepts of kernels and feature spaces, Applications in Biomedical Engineering	9
4	Convolutional Neural Networks: - Optimization Algorithms, Regularization Methods: L1 and L2 Regularization. Recurrent Neural Networks: RNN, LSTM, GRU. Transformers: BERT. Generative Models: GAN. Applications in Biomedical Engineering	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. **Problem Definition (K4 - 5 points)**
 - a. Clearly defines the problem.
 - b. Examine and identifies relevant factors.
2. **Problem Analysis (K4 - 5 points)**
 - a. Break-down and presents a well-reasoned solution approach.
3. **Evaluate (K5 - 5 points)**
 - a. Thoroughly evaluate the proposed solutions.
4. **Conclusion (K3- 3 points, K5 – 2 points)**
 - a. Summarizes findings and insights. State which solution is most appropriate for the problem

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the basic concepts of machine learning algorithms, neural network architectures and its applications.	K4
CO2	Develop different single layer/multiple layer networks	K4
CO3	Develop networks to understand the concepts of ART and SOM	K4
CO4	Evaluate the concept of associative networks and support vector machine	K5
CO5	Implement algorithms for convolutional neural networks	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1							1		2
CO2	3	2		2						1		2
CO3	3	2		2						1		2
CO4	3	2	2	1						1		3
CO5	3	2		2						1		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of neural networks: architectures, algorithms and applications.,	Fausett, Laurene V.	Pearson Education India	2006
2	Neural networks and deep learning.	Aggarwal, Charu C	Springer	2018.
3	Pattern Recognition and Machine Learning	Bishop, C. M	Springer, New York	2006
4	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2015
5	Neural Networks and Learning Machines,	S. O. Haykin	Pearson Education (India),	3rd Edition 2016
6	Neural Networks and Deep Learning	Michael A. Nielsen,	Determination Press	, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Machine Learning,	E. Alpaydin	Prentice Hall (India)	3 rd Edition, 2015.
2	Machine Learning for Dummies	John Paul Mueller, Luca Massaron	Wiley	2021
3	Deep learning techniques for biomedical and health informatics	Dash, Sujata, Biswa Ranjan Acharya, Mamta Mittal, Ajith Abraham, and Arpad G. Kelemen	Springer	2020.
4	Deep Learning for Biomedical Applications	D. Jude Hemanth, Omer Deperlioglu, Utku Kose	CRC Press	2021
5	Machine Learning for Biomedical Applications With Scikit-Learn and PyTorch	Maria Deprez, Emma C. Robinson	Elsevier	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117105084
1	https://ocw.mit.edu/courses/9-641j-introduction-to-neural-networks-spring-2005/
2	https://nptel.ac.in/courses/106106184
3	https://nptel.ac.in/courses/106105215
4	https://nptel.ac.in/courses/108103192
4	https://nptel.ac.in/courses/106106139

SEMESTER S5
MEDICAL ELECTRONICS LAB

Course Code	PCEBL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCEBT302 & PCEBT402	Course Type	Lab

Course Objectives:

1. The student will be capable of designing and implementing of instrumentation for medical equipment using Op-amp, specific sensors and specialized ICs.

Expt. No.	Experiments
1	Study of temperature transducer ICs - AD590 and LM35.
2	Sample and hold circuit using op-amps and LF398.
3	Real time ECG Acquisition Using Arduino Board and ECG Module (8232) .
4	UJT relaxation oscillator.
5	DC power control using SCR.
6	Design of pacemaker circuits (fixed mode) using op-amp.
7	Design of ECG simulator circuit
8	Design & set up a circuit for following thermistor characteristics. a) V-I characteristics b) V-T Characteristics c) R-T characteristics
9	Study of LDR & Plot the following characteristics. a) Resistance (R) Vs I(lux) b) Voltage(V) Vs I(lux)
10	Design and set up a of biotelemetry circuit using IC 4046. (Transmitting PCG/ ECG signals).

11	Study of Electro Surgical Unit
12	Study of Sphygmomanometer- manual and automated
13	Study of Patient monitoring / recording device (any one). a. ECG Machine b. EEG Machine
14	Study of IVD instruments (Any one) 1. Colorimeter 2. Biochemistry analyser 3. Haematology analyser
15	Study of life assist device (Any one) ● Defibrillator ● Ventilator

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design and implement circuits for biomedical applications using Op amp	K4
CO2	Design the circuits using transducers for biosignal acquisition	K4
CO3	Develop circuits using specialized ICs for biomedical applications	K4
CO4	Describe the basic operations of medical equipment.	K4
CO5	Design the circuits for the Determination of physiological signals.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	3	3	3		1	1	3	3		2
CO2	3	1	3	3	3		1	1	3	3		2
CO3	3	2	3	3	3		1	1	3	3		2
CO4	3	2	3	3	3		1	1	3	3		2
CO5	3	2	3	3	3		1	1	3	3		2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Devices and Circuit Theory	R E Boylestad and L Nashelsky	11 th ed, Pearson Education.	2012
2	Electronic Devices & Circuits,	Allan Mottershead	Prentice Hall of India, New Delhi	2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pulse, digital and Switching Waveforms	Millman and Taub	Tata McGraw Hill	2007
2	Handbook of Biomedical Instrumentation,	R S Khandpur	3 rd Edition, McGraw-Hill Education / India	2014
3	Encyclopedia of Medical Devices and Instrumentation	Dr. John G. Webster	2 nd Edition, Wiley-Interscience	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ee55/preview

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.

- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

BIOMEDICAL SIGNAL PROCESSING LAB

Course Code	PCEBL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. This course enables the students to develop and implement algorithms for analysis & processing of biomedical signals

Experiments (10 Mandatory)

Expt. No.	Experiments
1	Familiarization of Python & QT Octave programming- Generation of elementary signals and its basic operations
Programs in Python/QT octave	
2	Basic signal processing operations (Convolution, Correlation, Digital filters etc.)
3	Analysis of ECG (calculation of heart rate & parameters related to QRS complex, P wave & T wave)
4	Analysis of respiratory signal & calculation of respiratory rate
5	Analysis of PPG signal and calculation of pulse rate
6	Time & frequency domain analysis of HRV from ECG and using HRV software
7	Time domain Filtering of bio signal for removal of baseline drift & high frequency noise
8	Power line interference rejection from bio signals
9	Detection of Premature Ventricular Contraction in ECG
10	Estimation of dicrotic notch from PPG signal
11	Event detection in EEG signal
12	Spectral analysis of EEG Signal
13	Time and frequency domain analysis of EMG Signal
14	Concurrent signal analysis with ECG, PPG and PCG
15	Familiarisation of python toolbox (biosppy, neurokit2 etc.) for bio signal processing
17	Arrhythmia classification -using machine learning techniques
18	Analysis of speech signal (identification of voiced, unvoiced and silent portion etc.)

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement the basic signal processing operations in Python & QT octave	K3
CO2	Implement time domain and frequency domain filtering techniques	K3
CO3	Develop algorithms to detect events in biomedical signal	K5
CO4	Develop algorithms to extract features of normal biomedical signals	K5
CO5	Develop algorithm to analyse biomedical signal and detect abnormalities	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	3	-	-		3	2	-	1
CO2	3	2	-	-	3	-	-		3	2	-	1
CO3	3	2	1	2	3	-	-		3	2	-	1
CO4	3	2	3	3	3	-	-		3	2	-	1
CO5	3	2	1	2	3	-	-		3	2	-	1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Signal Analysis	Rangaraj M Rangayyan	John Wiley	2 nd Edition, 2015
2	Signals & Systems in Biomedical Engineering	Suresh R Devasahayam	Springer	2 nd Edition, 2013
3	Practical Guide for Biomedical Signals Analysis Using Machine Learning Techniques	AbdulhamitSubasi	Academic Press	1 st Edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical Biomedical Signal Analysis Using MATLAB	<u>Katarzyna J. Blinowska,</u> <u>Jarosław Żygierewicz</u>	<u>CRC Press</u>	2021
2	Biosignal and Medical Image Processing	John L.Semmlow, Benjamin Griffel	<u>CRC Press</u>	3 rd Edition, 2021
3	Biomedical Signal Processing and Artificial Intelligence in Healthcare	<u>Walid A. Zgallai</u>	Academic Press	2020
4	Python for Signal Processing	José Unpingco	Springer	2013
5	Signal Processing for Neuroscientists: An Introduction to the Analysis of Physiological Signals	Wim van Drongelen	Elsevier Science	2 nd Edition ,2018

Video Links (NPTEL, SWAYAM...)	
SL No	Link ID
1	https://archive.nptel.ac.in/courses/108/105/108105101/
2	https://bmshp-coep.vlabs.ac.in/exp/electrocardiogram-waveform/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

**ELECTRONICS & BIOMEDICAL
ENGINEERING**

SEMESTER S6

BIOMECHANICS & DESIGN OF PROSTHETIC DEVICES

Course Code	PCEBT601	CIE Marks	40
Teaching Hours/Week (L:3 T:1 P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NONE	Course Type	PC

Course Objectives:

1. This course enables students to understand the biomechanics of human tissues. This will help them to design prosthetic devices and implants.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Joint prosthesis-components-materials-Biomechanical analysis of joints – Spine, Hip, Knee-Human gait-phases-gait analysis methods-sensors- gait analysis-health and disease	11
2	Mechanics of Hard Tissues - structure and mechanical properties of bone - cortical and cancellous bones, Elasticity-stress, strain, elastic modulus, Poisson's ratio, yield strength, ultimate tensile strength, failure strength, strain to failure, Wolff's law of bone remodeling, models of viscoelasticity Soft Tissues - Structure, functions, material properties – tendon function, elasticity in a tendon, models of non-linear elasticity in a tendon – physiological and non-physiological regimes, Davis' law of soft tissue remodeling.	11
3	Neural prosthesis- Retinal Prosthetic Devices - Spatial, Temporal, and Contrast-Resolutions Obtainable With Retina Implants- Subretinal Artificial Silicon Retina Microchip Implantation in Retinitis Pigmentosa-Development of an Intraocular Retinal Prosthesis-Boston retinal implant-cochlear implants- Myoelectric prosthesis	11

4	Models of Peripheral Circulation-Vascular Resistance-Vascular Capacitance-Lumped Parameter Model of Peripheral Circulation- Venous Return Curve-Windkessel Simplifications-Heart as a Pump- Length –Tension Relationship in a Skeletal Muscle, P-V relationship in Ventricle-Model of Heart- Variable Capacitor model-Cardiac Cycle in P-V Plane-Iso Volumetric Contraction-Contractility- Heart Lung Pumping Unit-Open Chest Model-Effect of Intra Thoracic Pressure-Modelling of intact Cardiovascular System-Cardiac Output under abnormal conditions-Graphical Solution.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To analyse the biomechanical principles of human locomotion	K4
CO2	To analyse the structure property relationships of hard and soft tissues	K4
CO3	To apply the principles of mechanics for the design of neural prosthetics	K3
CO4	To understand the mechanics of heart and vascular system	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	-	1	-	-	-	1	-	3
CO2	3	3	1	-	-	1	-	-	-	1	-	3
CO3	3	3	2	-	-	1	-	-	-	1	-	3
CO4	3	3	2	-	3	1	-	-	-	1	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Orthotics and prosthetics in rehabilitation-e-book.	Lusardi, Michelle M., Millee Jorge, and Caroline C. Nielsen.	Elsevier Health Sciences	2013.
2	Visual prosthesis and ophthalmic devices: new hope in sight.	Tombran-Tink, Joyce, Colin J. Barnstable, and Joseph F. Rizzo, eds.	Springer Science & Business Media,	2007.
3	Mechanics of biomaterials: fundamental principles for implant design.	Pruitt, Lisa A., and Ayyana M. Chakravartula.	Cambridge University Press,	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to neural engineering for motor rehabilitation.	Farina, Dario, Winnie Jensen, and Metin Akay,	John Wiley & Sons,	2013.
2	Design of artificial human joints & organs.	Pal, Subrata.	Springer	2014.
3	Orthotics and Prosthetics in Rehabilitation E-Book	Chui, Kevin C., Milagros Jorge, Sheng-Che Yen, and Michelle M. Lusardi.	Elsevier Health Sciences	2019.
4	The Hip Joint.	Iyer, K. Mohan	CRC Press	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/fCqKhggM120?si=Cy55Lj3t_kh8pb0p
2	https://onlinecourses.nptel.ac.in/noc24_bt06/preview
3	https://youtu.be/wWjuZvFlh5Q
2	https://ocw.mit.edu/courses/16-423j-aerospace-biomedical-and-life-support-engineering-spring-2006/9cef3c5c3b668f8762a27e2ef6044be3_cardio_mech.pdf

SEMESTER S6

PRINCIPLES OF RADIO DIAGNOSIS & RADIOTHERAPY

Course Code	PCEBT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Students will acquire knowledge of the physics and technology behind radiological imaging and radiotherapy.
2. Students will be able to understand principles of radiation physics, including the interaction of radiation with matter.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Radiographic system: X-Ray production, interactions in the anode-Bremsstrahlung and characteristic radiations. Stationary and rotating anode X-ray tubes. Electrical circuit for an X-ray unit- Autotransformer, Timer, HT generator, rectification, filament circuit, meters, automatic exposure controls, generator ratings, factors influencing the X ray- spectra, beam quantity and quality, scattered radiation control in radiography- collimators and grids.	9
2	Image receptors: CR image receptors - Photostimulable phosphor (PSP), PSP screen-film cassette imaging plate (IP), CR reader. Digital radiography (DR) image receptors- indirect conversion- phosphorescent screen- Cesium Iodide or Gadolinium Oxysulfide, active-matrix array (AMA)- amorphous silicon TFTs. Direct-capture DR-active-matrix array- amorphous selenium. Special Radiographic Techniques: Fluoroscopy- components of a digital fluoroscopy imaging system-dynamic flat-panel detector (DFPD) units and charge-coupled device (CCD) camera. Interventional Radiology-Digital	9

	subtraction angiography- image formation- temporal and energy subtraction, Mammography- Features of a dedicated Mammography System, dental radiology- Intra-Oral Radiography, Panoramic Radiography and Cone beam CT.	
3	Types of radio therapy --external beam radiation therapy (teletherapy), brachytherapy. Teletherapy sources- Kilovoltage and Megavoltage radiations- the unit of absorbed dose, percentage depth dose curves, isodose charts. Teletherapy machines: Kilovoltage and megavoltage X-Ray Units- Linear Accelerators, Betatron, Cyclotron, collimators and beam direction devices, Gamma ray Units-Cobalt Machines. Radiation sources (radionuclides) for brachytherapy. Intensity-modulated radiotherapy- multileaf collimator, stereotactic techniques. Gamma knife, proton beam therapy. Therapy planning: Isodose curves and dose mapping, wedges, tissue compensators and intensity modulators, practical features of a treatment planning system.	9
4	Instrumentation for Dosimetry: Ionization chambers, film dosimetry, thermo luminescent and optically stimulated luminescence dosimetry, semiconductor dosimeters. Radiation protection: Permissible doses, design features of radiotherapy facilities, materials used for shielding, primary and secondary barriers. Advisory and regulatory agencies. Introduction to Radiology information system (RIS): PACS, digital imaging and communication in medicine (DICOM).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Evaluate the performance of different components of radio diagnosis systems.	K2
CO2	Interpret the unique features of fluoroscopy, mammography, angiography and dental X-ray units.	K2
CO3	Distinguish the important radiotherapy machines based on their mode of operation and identify the use of various tele therapy and brachytherapy.	K3
CO4	Identify the use of modern receptors for digital radiography and radiation detectors for dosimetry	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2			2			2	2		2
CO2	3	2	2			2			2	2		2
CO3	3	2	2			2			2	2		2
CO4	3	2	2			2			2	2		2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Radiography in the Digital Age	<u>Carroll Quinn B.</u>	Charles C. Thomas	2014
2	Radiologic Science for Technologists	Stewart Carlyle Bushong	Elsevier	2017
3	Digital Radiography in practice	<u>Quinn B. Carroll</u>	Charles C Thomas	2019
4	Physics for diagnostic radiology	P P Dendy, B Heaton	<u>Taylor & Francis</u>	2011
5	Practical radiotherapy: physics and equipment	P Cherry, AM Duxbury	John Wiley & Sons	2019
6	Handbook of radiotherapy physics: theory and practice	P Mayles, A Nahum, JC Rosenwald	<u>CRC Press</u>	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of diagnostic radiology	William E. Brant, Clyde Helms	Lippincott, Williams & Wilkins	2012
2	Fundamental physics of radiology	WJ Meredith, JB Massey	Butterworth-Heinemann	2013
3	Diagnostic Radiology Physics A handbook for teachers and students	David R. Dance	International Atomic Energy Agency	2014
4	Principles of radiographic imaging: an art and a science	RR Carlton, AM Adler, V Balac	Cengage Learning	2019
5	Fundamental Physics of Radiology	W.J. Meredith, J.B. Massey	CRC Press	2013
6	Radiation Detectors for Medical Imaging	Jan S. Iwanczyk	CRC Press, Taylor & Francis Group	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/tQEjUCkiYhl?si=MRb-ONQnPNalMBYi
2	https://youtu.be/U8mInQlwwN8
3	https://youtu.be/BR2w0v6yuPw
4	https://youtu.be/vQ5HrubRYVo

SEMESTER S6

ELECTRICAL NETWORKS & ANALYSIS

Course Code	PEEBT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GXEST104	Course Type	Theory

Course Objectives:

1. **Apply** advanced circuit analysis techniques including mesh and node analysis, and utilize key network theorems to both DC and AC circuits, demonstrating problem-solving and critical thinking skills
2. **Analyze** transient and steady-state responses of RL, RC, and RLC circuits using Laplace Transforms and initial condition methods, enhancing students' ability to interpret and evaluate circuit behavior.
3. **Design** and characterize two-port networks and passive filters, integrating knowledge to create effective signal processing solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mesh and Node Analysis and Network Theorems: Mesh and node analysis of network containing independent and dependent sources. Supermesh and Supernode analysis. Steady-state AC analysis using Mesh and Node analysis. Thevenin's theorem, Norton's theorem, Superposition theorem, Reciprocity theorem, Maximum power transfer theorem. (applied to both dc and ac circuits having dependent source)	9
2	Network Analysis: Transients - DC and sinusoidal response of RL, RC and RLC circuits, Initial and final conditions, Review of Laplace Transforms and Inverse Laplace Transforms, Transformation of basic signals and circuits into s-domain. Transient analysis of RL, RC, and RLC networks with	9

	impulse, step and sinusoidal inputs (with and without initial conditions).	
3	Transformed circuits in s-domain: Transform impedance/admittance of R, L and C - Transfer Function representation – Poles and zeros. Analysis of Coupled Circuits: – Dot polarity convention – Sinusoidal steady state analysis of coupled circuits - Linear Transformer as a coupled circuit - Analysis of coupled circuits in s-domain.	9
4	Two port networks and Passive filters: Characterization of two port networks using Z, Y, Hybrid and Transmission parameters, Tand π representations. Interrelationship among parameter sets. Series and parallel connections of two port networks. Passive filters – Filter fundamentals, Classification of Filters, Characteristic impedance, Design of Constant K - Low Pass, High Pass & Band Pass filters.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply advanced circuit analysis techniques to solve complex electrical circuits involving both independent and dependent sources, utilizing mesh and node analysis and network theorems for precise circuit simplification and resolution.	K3
CO2	Analyze the DC and sinusoidal responses of RL, RC, and RLC circuits, applying Laplace Transforms to evaluate transient behavior under impulse, step, and sinusoidal inputs with and without initial conditions.	K4
CO3	Analyze and design circuits in the s-domain by calculating impedance and admittance, representing transfer functions with poles and zeros, and evaluating coupled circuits using dot polarity conventions and sinusoidal steady-state analysis.	K4
CO4	Develop and analyze two-port networks and passive filters by examining Z, Y, Hybrid, and Transmission parameters, understanding their interrelationships, and applying filter fundamentals and characteristic impedance to create effective solutions for practical network configurations and signal processing.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2								2
CO2	3	3	2	2								2
CO3	3	3	2	2								2
CO4	3	3	2	2								2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Network Analysis	Valkenburg, V.	Pearson	3rd Edition, 2019
2	Circuits and Networks: Analysis and Synthesis	Sudhakar, A., Shyammohan, S. P.	McGraw Hill	5th Edition, 2015
3	Theory and Problems in Electric Circuits	Edminister, J. A., Nahvi, M	McGraw Hill	5th Edition, 2010
4	Network Analysis and Synthesis	Singh, R. R.	McGraw-Hill Education	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Circuits – Schaum’s Outline Series	Edminister, J.	McGraw-Hill	2009
2	Electric Circuits and Networks	Suresh Kumar, K. S.	Pearson	2008
3	Network Analysis with Applications	Stanley, W. D.	Pearson	4th Edition, 2006
4	Network Analysis and Synthesis	Singh, R. R.	McGraw-Hill	2nd Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/106/108106172/
2	https://archive.nptel.ac.in/courses/108/105/108105159/
3	https://www.youtube.com/watch?v=ZHGeR2CyN5s
4	https://archive.nptel.ac.in/courses/108/105/108105159/

SEMESTER S6

IOT & BIOMEDICAL APPLICATIONS

Course Code	PEEBT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of Internet of Things (IoT) concepts, including key protocols, architectural principles, and sensor technologies, with a focus on their theoretical applications and implications.
2. Examine the various IoT protocols and architectures, understanding their roles and interactions within IoT ecosystems, and evaluating their effectiveness in different contexts.
3. Explore the theoretical aspects of wearable health monitoring systems, including their design, functionality, and integration with data analytics and cloud technologies to address healthcare challenges.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT and Enabling Technologies: Introduction to Internet of Things – Definition and Characteristics of IoT, Physical Design of IoT – IoT Protocols, Logical Design of IoT - IoT communication models, IoT Communication APIs. IoT enabled Technologies - Wireless Sensor Networks, Cloud Computing, Embedded Systems. Sensor Interfaces – ECG, EEG, PPG, Pulse Oximeter, Temperature Sensors, and Pressure Sensors. Challenges of using IoT in Healthcare.	9
2	IoT Protocols, Architecture, and Standards: IoT Protocols & Architecture: Protocols - Protocol Standardization for IoT – Efforts – M2M and WSN Protocols – SCADA and RFID Protocols – Issues with IoT	9

	Standardization – Unified Data Standards – IEEE802.15.4 – BACNet Protocol – Modbus – KNX – Zigbee. Network and Security Architecture. IoT Open Source Architecture (OIC) - OIC Architecture & Design Principles. IoT Devices and Deployment Models - IoTivity: An Open-source IoT Stack - Overview - IoTivity Stack Architecture - Resource Model and Abstraction.	
3	Wearable Health Monitoring Systems and IoT Implementation: Wearable Health Monitoring System (WHMS): Basics of WHMS, Need for Wearable Systems, Applications of WHMS - Home Healthcare, Sports/Fitness. Design of Wearables: Introduction, Architecture, and Attributes. Design of Smart Textiles and Wearable Electronics. Issues of WHMS & Factors Inhibiting Growth. Building IoT with Arduino & Raspberry Pi: Programming – Python Program with Raspberry Pi, Controlling Hardware - Controlling AC Power Devices with Relays, Controlling Servo Motor, DC Motor. Sensors - Light Sensor, Motion Detection Sensors, Wireless Bluetooth Sensors.	9
4	Data Handling, Analytics, and Cloud for IoT: Data Handling & Analytics: Introduction, Big Data, Types of Data, Characteristics of Big Data, Data Handling Technologies, Data Acquisition, Data Storage. Introduction to Data Analytics, Types of Data Analytics, Local Analytics, Cloud Analytics, and Applications. IoT Physical Servers and Cloud Offerings – Introduction to Cloud Storage Models and Communication APIs. Webserver – Web Server for IoT, Cloud for IoT, Python Web Application Framework, Designing a RESTful Web API.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a clear understanding of various IoT protocols and architectural frameworks, including their roles and functionalities within IoT ecosystems.	K2
CO2	Apply theoretical designs for IoT systems using Raspberry Pi, focusing on system integration and hardware control concepts.	K3
CO3	Apply knowledge of wearable health monitoring systems to design and evaluate their components and functionalities in healthcare settings.	K3
CO4	Analyze data management practices in IoT, including data acquisition, storage, and analytics, to understand their impact on system performance and effectiveness.	K4
CO5	Apply concepts to deploy IoT applications and integrate them with cloud platforms for effective data management and analysis.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	3					2		2
CO2	3	2	2	1	3					2		2
CO3	3	2	2	1	3					2		2
CO4	3	2	2	2	3					2		2
CO5	2	2	3	1	3					2		2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things – A Hands-on Approach	Arshdeep Bahga, Vijay Madisetti	Universities Press	2015
2	From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence	Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle	Academic Press	1st Edition, 2014
3	Getting Started with Raspberry Pi	Matt Richardson, Shawn Wallace	O'Reilly Media, Inc.	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rethinking the Internet of Things: A Scalable Approach to Connecting Everything	Francis daCosta	Apress Publications	1st Edition, 2013
2	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, Jerome Henry	Cisco Press	2017
3	Internet of Things and Personalized Healthcare Systems	Venkata Krishna, Sasikumar Gurumoorthy, Mohammad S. Obaidat	Springer Briefs in Applied Sciences and Technology, Forensic and Medical Bioinformatics	2019
4	Architecting the Internet of Things	Dieter Uckelmann, Mark Harrison, Michahelles Florian (Eds)	Springer	2011
5	Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects	Michael Margolis	O'Reilly Media	2nd Edition, 2011
6	Principles of Internet of Things (IoT) Ecosystem: Insight Paradigm	Sheng-Lung Peng, Souvik Pal, Lianfen Huang (Eds)	Springer International Publishing	2020
7	Internet of Things (IoT) for Automated and Smart Applications	Yasser Ismail (Ed)	BoD–Books on Demand	2019
8	Internet of Things: For Things, and by Things	Abhik Chaudhuri	CRC Press	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105166/
2	https://onlinecourses.swayam2.ac.in/arp20_ap03/preview
3	https://onlinecourses.swayam2.ac.in/arp20_ap03/preview
4	https://onlinecourses.nptel.ac.in/noc22_cs53/preview

SEMESTER S6

ARTIFICIAL ORGANS & IMPLANTS

Course Code	PEEBT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Students will acquire knowledge in basic principles and designing of various types of artificial and tissue derived implants.
2. Students will be able to understand the use of 3D printing technology in implantation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cochlear implants: Basics of speech signal – Normal hearing mechanism – Deafness- Functional block diagram of cochlear implants-Implant characteristics & it's classification -Auditory prostheses using deep brain & spinal cord stimulation-Physiological challenges-I/V Stimulation-Single-multiple source systems-Real time programming.	9
2	Design of artificial organs: Heart mechanism- Design of heart valves- Heart assist devices-Pulsatile pumping systems-Multi stage multi actuated pump system -Kidney and it's artificial replacement-hemodialyzers& it's types- Anatomy of skin - Design of artificial skin- Liver and it's artificial replacement-The medical device market and ethical issues of implants.	9
3	Micro electrode technologies for Deep Brain Stimulators (DBS): Implant site -Stimulation parameters - Closed loop control of DBS- Bionic micro stimulation- First and second generation RF powered Bionic micro	9

	stimulators-Clinical applications-Brain control and sensing of artificial limbs-Stimulation of peripheral nerve interfaces-Construction of implant-wireless communication-prosthetic master control unit.	
4	Fundamentals of Tissue Engineering: Engineering biomaterials for tissue engineering- Nano biomaterials for tissue engineering – Drug delivery & Gene therapy mechanism- Vascularization of Engineered Tissues-Tissue micro environments (Basic level of tissue engineering of nervous system, liver, kidney & cartilage). Fundamentals of 3 D printing and its applications in Biomedical engineering: Concept of 3D printing -3D printing process- Materials for 3D printing - Biomedical applications of 3 D printing technology- maxillofacial surgery & upper limb prosthetics.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand design aspects of neural prostheses.	K2
CO2	Explore different kind of artificial organs.	K2
CO3	Analyse design aspects of deep brain stimulators.	K4
CO4	Review applications of tissue engineering on various organs.	K4
CO5	Evaluate the use of 3 D printing technology in Biomedical Engineering.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-	1	-	-	-	1	-	3
CO2	3	3	2	1	-	1	-	-	-	1	-	3
CO3	3	3	2	1	-	1	-	-	-	1	-	3
CO4	3	3	2	1	-	1	-	-	-	1	-	3
CO5	3	3	3	1	-	1	-	-	-	1	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of artificial human joints & organs	Pal S.	Boston, MA: Springer US	2014
2	3D Printing in Biomedical Engineering	Singh S, Prakash C, Singh R	Springer	2020
3	Tissue engineering: principles and practices	Fisher JP, Mikos AG, Bronzino JD, Peterson DR	CRC Press	2012
4	Implantable neural prostheses	Zhou, David D., and Elias S. Greenbaum	Springer Science+ Business Media	2009
5	Principles of Tissue engineering	Robert Lanza, Anthony Atala	Textile eBook	2018
6	3D Printing for Tissue Engineering & Regenerative Medicine	Murat Guvendiren, Vahid Serpooshan	MDPI Books	2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced 3D-printed systems and nanosystems for drug delivery and tissue engineering	Du Toit L, Kumar P, Choonara Y, Pillay V	Elsevier	2020
2	Principles and Applications in Engineering Series: Tissue Engineering	Bernhard Palsson, Jeffrey A. Hubbell, Robert Plonsey, Joseph D. Bronzino	CRC Press	2003
3	Biomaterials, artificial organs and tissue engineering	Larry L. Hench, Julian R. Jones	Woodhead Publishing Limited	2005
4	Principles of Heart Valve Engineering	Kheradvar, Arash	Academic Press	2019
5	Engineering Neural Tissue from Stem cells	Dr. Stephanie Willerth	CCRM	2017
6	Textbook of Radiology for Cochlear Implants	Mohnish Grover, Gaurav Gupta, C. Preetam	Thieme	2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me130
2	https://onlinecourses.nptel.ac.in/noc21_mg60
3	https://onlinecourses.nptel.ac.in/noc22_ee87
4	https://onlinecourses.nptel.ac.in/noc22_bt61

SEMESTER S6

BIOMEDICAL SIGNAL PROCESSING & APPLICATIONS

Course Code	PEEBT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBEBT504	Course Type	Theory

Course Objectives:

1. To familiarize with concepts of advanced digital signal processing techniques which includes adaptive filtering, wavelet analysis, power spectrum, estimation and multivariate analysis

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Power spectrum analysis: Estimation of power density spectrum, non-parametric methods- Periodogram - Bartlett, Welch, Blackman and Tukey Methods-comparison. Parametric models, estimation Auto regressive (AR), Moving average (MA) Autoregressive moving average (ARMA) models. Parametric model based linear prediction theory-Forward and backward linear prediction Estimation of parameters - spectrum estimation of ECG, Power spectrum estimation of Heart rate variability data. Application, Spectrum estimation of EEG	9
2	Wavelet Analysis Introduction to wavelet transforms – CWT & DWT Orthogonal wavelet decomposition, Orthonormal wavelets - filter banks. Applications of DWT: wavelet denoising, ECG feature detection & Heart sound Analysis.	9
3	Adaptive filters: Optimal and adaptive filters, Weiner filters, Adaptive signal processing - Steepest descent algorithm LMS adaptive algorithm, Adaptive noise canceller – cancellation of 50 Hz signal in ECG- cancellation of maternal ECG in foetal electrocardiography	9

4	Multivariate Analysis: Principal Component Analysis – denoising of ECG & EEG signals using PCA, Independent Component Analysis-EEG denoising. Computation of diagnostic signal parameters of ECG: Heart rate and QRS detection using Multivariate analysis (PCA and ICA).	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse parametric and nonparametric methods of power spectrum estimation	K3
CO2	Apply the methods of wavelet analysis in biomedical signal processing	K3
CO3	Investigate the techniques in adaptive filters	K4
CO4	Implement various signal processing techniques in biomedical applications	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2								
CO2	3	3	2	2								
CO3	3	3	2	2								
CO4	3	3	3	2					1	1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal Processing- Principles Algorithms and Applications	John G Proakis& Dimitris G Manolakis	PHI	4th Edition, 2016
2	Biomedical Signal Analysis,	Rangaraj M Rangayyan	John Wiley	2 nd Edition, 2015
3	Signals & Systems in Biomedical Engineering	Suresh R Devasahayam	Springer	2 nd Edition, 2013
4	Insight into wavelets: from theory to practice,	K P Soman, K I Ramachandran, N G Resmi	PHI	3rd Edition, 2015
5	Discrete Wavelet Transforms- Biomedical Applications	Hannu Olkkonen,	In Tech,	1 st Edition, 2011
6	Adaptive Signal Processing	Bernard Widrow and Samuel D. Stearns	Person Education	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical signal Processing,	D C Reddy	Tata McGraw-Hill, New Delhi,	1 st Edition,2005
2	Biosignal and Biomedical Image Processing – Matlab Based Applications,	John L.Semmlow	Marcel Dekker Inc. New York	2004.
3	Wavelet Transforms- Introduction to Theory and Applications	Raghuveer M Rao et al	Pearson Education Asia	1999
4	Statistical and Adaptive Signal Processing	Monalokis, Ingle and Kogon	Artech House INC	2005
5	Adaptive Filter theory,	Simon Haykin	Pearson Education India	5thEdition, 2013
6	Principal Component Analysis	I.T Jolliffe	Springer	2ndEdition, 2002
7	Independent Component Analysis for Audio and Bio signal Applications,	Ganesh R. Naik	In Tech	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ee136/preview
3	https://onlinecourses.nptel.ac.in/noc24_ee149/preview

SEMESTER S6

COMPUTATIONAL METHODS IN BIOMEDICAL ENGINEERING

Course Code	PEEBT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Introduce practical computational methods for data analysis, modelling and simulation of biomedical engineering.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Computational Methods: Overview of computational methods in biomedical engineering, Basic programming skills (Python/MATLAB) Variables and operations, conditional statements (if, elif, else) and loops (for, while), Defining and calling functions, Data Structures, File handling, Libraries for Numerical Computing and Data analysis, Data Visualization	9
2	Numerical Methods for Solving Equations: Root finding methods (bisection, Newton-Raphson) Solving linear and nonlinear equations Applications in biomedical engineering.	9
3	Introduction to modeling and simulation: Ordinary differential equations (ODEs) and their applications, Modeling of biological processes (neuronal dynamics, cell electrophysiology) Partial Differential Equations (PDEs): Describing spatial and temporal variations in phenomena such as heat transfer in tissues and fluid flow in blood vessels.	9
4	Bioinformatics: Sequence Analysis: Algorithms for DNA, RNA, and protein sequence alignment. Structural Bioinformatics: Molecular modeling and simulation techniques for studying protein structures and interactions. Genomics and Proteomics: Computational tools for analyzing genomic and proteomic data to understand gene expression and protein functions.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the basics of programming concepts of MATLAB and Python for computing in biomedical field	K2
CO2	To apply numerical methods for solving biomedical engineering problems.	K3
CO3	To develop computational models for biological systems.	K6
CO4	To analyze and interpret biomedical data using computational tools.	K4
CO5	To gain hands-on experience with software tools commonly used in biomedical engineering.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	1	3					2		2
CO2	3	2	2	1	3					2		2
CO3	3	2	2	1	3					2		2
CO4	3	2	2	1	3					2		2
CO5	1	1	3	1	3					2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical Methods in Biomedical Engineering	Stanley Dunn, Alkis Constantinides, Prabhas V. Moghe	Elsevier	2005
2	Python for data analysis.	McKinney, W.	O'Reilly Media, Inc	2013
3	Numerical and Statistical Methods for Bioengineering: Applications in MATLAB	Michael R. King, Nipa A. Mody	Cambridge University Press	2010
4	Bioinformatics For Dummies	Jean-Michel Claverie, Cedric Notredame	John Wiley & Sons	2011
5	Bioinformatics: Sequence and Genome Analysis	David W. Mount	Cold Spring Harbor Laboratory Press	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Python Data Science Handbook : Essential Tools for Working with Data	Jake VanderPlas	O'Reilly Media, Inc	2016
2	Bioinformatics: Sequence and Genome Analysis	David W. Mount	Cold Spring Harbor Laboratory Press	2004
3	An Introduction to Computational Systems Biology Systems-Level Modelling of Cellular Networks	Karthik Raman	CRC Press	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/3uY5N3rgoLE
2	https://www.youtube.com/watch?v=0H7i2sJn47Y&t=3s
3	https://onlinecourses.swayam2.ac.in/nou24_bt08/preview
4	https://nptel.ac.in/courses/102103056
4	https://nptel.ac.in/courses/102105100

SEMESTER S6
MEDICAL ROBOTICS

Course Code	PEEBT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBEBT504	Course Type	Theory

Course Objectives:

1. Gain comprehensive knowledge of the basic principles of robotics, including core components such as sensors, actuators, and drive mechanisms.
2. Examine the kinematics and dynamics of robotic systems, focusing on their practical applications in healthcare also develop and implement programs for controlling robots to improve medical procedures and patient care.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Components of robotic system: History of medical robots, Classification of medical robots, Present status and future trends. Basic components of robotic system. Basic terminology- Accuracy, Repeatability, Resolution, Degree of freedom. Mechanisms and transmission, End effectors, Grippers different methods of gripping, Mechanical grippers-Slider crank mechanism, Screw type, Rotary actuators, Cam type gripper, Magnetic grippers, Vacuum grippers, Air operated grippers; Specifications of robot.	9
2	Drive systems and Sensors Drive system- Pneumatic Drives-Hydraulic Drives-Mechanical Drives, Electrical Drives-D.C. Servo Motors, Stepper Motors, A.C. Servo Motors-Salient Features, Applications and Comparison. Sensors in robots – Touch sensors, Tactile sensor, Proximity and range sensors, Robotic vision sensor, Force sensor, Light sensors, Pressure sensors	9
3	Robot kinematics and robot programming Forward Kinematics, Inverse Kinematics and Difference; Forward Kinematics and Reverse Kinematics of	9

	manipulators with Two, Three Degrees of Freedom (in 2 Dimension), Four Degrees of freedom (in 3 Dimension) Jacobians, Velocity and Forces Manipulator Dynamics, Trajectory Generator, Manipulator Mechanism Design-Derivations and problems. Lead through Programming, Robot programming Languages-VAL Programming	
4	Robots in healthcare: Tele operated Robot-Assisted Minimally Invasive Surgery, Robot design concepts Robot assisted Image-Guided Interventions: CT, MR, Ultrasound, Other applications: Robotic catheters for heart electrophysiology, Humanoids, Micro/Nano robots for biomedicine - Delivery, surgery, sensing, and detoxification. Case Studies: Characterization of gestures to the design of robots- Design methodologies- Technological choices - Security. Cardiac, abdominal, and urologic procedures with tele-operated robots (3 procedures each), Neurosurgery and Orthopaedic surgery with robots.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of robotics and its components	K2
CO2	Review sensors and drive mechanisms in robotics	K2
CO3	Illustrate the Kinematics and Dynamics of robotics	K3
CO4	Analyse applications of robots in healthcare and develop programs for controlling robots.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2			2				2		2
CO2	2	2	2			2				2		2
CO3	2	2	2			2				2		2
CO4	3	3	3			2				2		2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Modeling and Control	Mark W. Spong, Seth Hutchinson, and M. Vidyasagar	Wiley Publishers	2006
2	Medical Robotics: Minimally Invasive Surgery	Paula Gomes	Woodhead Publishing	2017
3	Industrial Robotics: Technology, Programming, and Applications	Mikell P. Groover and Mitchell Weiss	McGraw-Hill	2012
4	Introduction to Robotics: Analysis, Control, Applications	Saeed B. Niku	Wiley	2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical Robotics	Achim Schweikard and Floris Ernst	Springer	2015
2	Medical Robots	Daniel Faust	Rosen Publishers	2016
3	Introduction to Robotics: Mechanics and Control	John J. Craig	Pearson Education	2008
4	Robotics Technology and Flexible Automation	Deb, S.R.	Tata McGraw Hill Book Co.	1994

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/a6_fgnuuYfE
2	https://youtu.be/k5Er0-HD-qw

SEMESTER S6

MEDICAL IMAGE PROCESSING

Course Code	PBEBT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To provide an overview of the nature of human visual perception, the elements of an image-processing system which includes sampling, quantisation and two dimensional transforms.
2. To make the students familiar with image enhancement, restoration, segmentation, classification, machine learning and deep learning techniques for image analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Image perception -MTF of the visual system –monochrome vision models - color vision model. Image sampling and quantization: Two-dimensional sampling theory, Image quantization - uniform quantizer. Image transforms - Two dimensional orthogonal and unitary transforms - properties of unitary transforms, 2D DFT, DCT, Hadamard, Haar, KL transforms.	9
2	Image enhancement: contrast enhancement, clipping and thresholding - digital negative, intensity level slicing - bit extraction. Histogram processing, Magnification and interpolation. Image Enhancement in Spatial domain: Smoothing and sharpening operations in time domain, Filtering in frequency domain: Smoothing and sharpening frequency domain filters, Homomorphic filtering.	9
3	Image restoration: Image degradation, Noise, Blurring, Modelling image degradation, Inverse filtering, Wiener filtering. Image segmentation- Thresholding- optimal, adaptive, Region-based methods- region growing and	9

	region merging, Boundary-based methods- edge detection and linking, boundary tracking.	
4	Feature recognition and classification: Features - describing the contents and the shape of objects. Classification- Supervised and unsupervised learning. Discriminant analysis, k-nearest-Neighbor (kNN) classifier. Unsupervised- k-means clustering. Deep Learning for Medical Image Analysis, Convolutional Neural Networks. Image fusion for hybrid imaging, Super resolution imaging.	9

Suggestion on Project Implementation

- In the first class, students shall be given an awareness and/ or guidelines on how to implement and learn this course through a project and are asked to form groups of maximum five members in the first week.
- Each group should identify a project where different image processing techniques can be used for biomedical image processing applications and perform a literature survey.
- Choose a suitable dataset, proper processing techniques and suitable machine learning, deep learning or computer vision algorithms for your application.
- Develop the block diagram, the algorithms, the flowchart and necessary codes and present them before the jury within the first month after the commencement of classes.
- Implement the necessary algorithms.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method **(CIE: 60 marks, ESE: 40 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 subdivisions. • Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Infer the nature of processes involved in human visual image perception and also in image-processing systems which includes sampling, quantisation and two dimensional transforms.	K2
CO2	Apply methods of image enhancement both in time and transform domains.	K3
CO3	Analyse the techniques for feature extraction, image segmentation and image restoration.	K4
CO4	Analyse machine and deep learning algorithms for medical image analysis.	K4
CO5	Evaluate the performance of image processing techniques for specific problem domains.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										1
CO2	3	2			1							1
CO3	3	2	2	2	1							2
CO4	2	2	2	2	1							2
CO5	2	2	2	2	2				2			2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Digital Image Processing	Jain Anil K	Prentice Hall of India	1989
2	Digital Image Processing for Medical Applications	Geoff Dougherty	Cambridge University Press	2007
3	Digital Image Processing	Gonzalez Rafael C	Fourth Edition	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning for Medical Image Analysis	Edited by S. Kevin Zhou Hayit Greenspan Ding gang Shen	Academic Press, Elsevier	Ist Edition, 2017
2	Computer Vision Algorithms and Applications	Richard Szeliski	Springer	2011
3	Hands-On One-shot Learning with Python	Shruti Jadon Ankush Garg	Packt Publishing	2019
4	Python Deep Learning	Ivan Vasilev Daniel Slater Gianmario Spacagna Peter Roelants Valentino Zocca	Packt Publishing	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/7Pzq9IwscbE?si=4yMyNurWD3dfbQF5 , https://youtu.be/nfEaPWZ_dRQ?si=r-Ith_KIUfLDboO52 , https://youtu.be/7LXi7_ISJn0?si=dxNRVXnfr3t9yd7m ,
2	https://youtu.be/ArKe6zMkXnk?si=v21e56QYZWgr5h5W ,
3	https://youtu.be/ljCZM8mpj8I?si=yXQIEOrtXMwOP1lx ,
4	https://youtu.be/Y_zQ9B4rWMY?si=DSVaLNRVDsVPaFRu , https://youtu.be/11WBbnlMkwg?si=etrvyD6j0WGYZRD , https://youtu.be/ljCZM8mpj8I?si=yXQIEOrtXMwOP1lx ,

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

BIOSENSORS & TRANSDUCERS

Course Code	OEEBT611	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	40
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Learn the basic principles of biosensors, transducers & their applications.
2. Provide students an insight into biopotentials and electrodes.
3. Introduce Nano biosensors & their applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	TRANSDUCERS: Classification of transducers, characteristic of transducers. Temperature transducers: Resistance temperature detector (RTD), Thermistor, Thermocouple, chemical thermometry. Displacement transducers: potentiometer, resistive strain gauges, inductive displacement, capacitive displacement transducer. Pressure transducer: variable capacitance pressure transducers, LVDT transducers, strain gauge transducers, catheter tip transducers, Piezoelectric transducer. Photoelectric transducers: photo-emissive tubes, photovoltaic cell, photoconductive cell, photodiodes.	9
2	BIO-POTENTIAL ELECTRODES: Electrode theory, Skin Contact Impedance. Types of Electrodes: Microelectrodes- Metal microelectrodes, micropipette (metal filled glass and glass micropipette electrodes). Depth and needle electrodes- Depth electrode, Needle electrode. Surface electrodes- Metal plate surface electrodes, Suction Cup electrode, Adhesive tape electrode, Multipoint electrode. BIO-POTENTIALS: ECG -	9

	Generation of cardiac action potentials- Characteristics of ECG Signal, Einthoven's triangle -Lead Systems. Electrodes for ECG acquisition. EEG – Electroencephalogram - Brain action potentials - Characteristics of signal- Electrodes and 10-20 system for EEG signal acquisition. Evoked Potentials. EMG - Electrical activity of muscles – Characteristics of EMG signal. Electrodes for EMG acquisition. (Instrumentation not needed for ECG, EEG & EMG)	
3	FUNDAMENTALS OF BIOSENSORS -Basic Principle, Characteristics of a Biosensor. Types - Based on Transducer: Electrochemical- Amperometric, Potentiometric; optical-Surface plasmon resonance, Evanescent Wave Fluorescence Biosensors; piezo electric; calorimetric biosensors. Based on Bio recognition elements in biosensors- natural-semisynthetic-synthetic.Immobilization methods ISFET, Enzyme electrodes. Principles of Fiber-Optic Sensing - Definitions, Classification, Working Principles. Applications of Biosensors -Biosensors for clinical diagnostics-Glucose detection- cancer detection.	9
4	NANO BIOSENSORS: Significance, Types of Nanomaterials - Gold Nano-particles, Quantum Dots, Carbon Nanotubes & Graphene (Key features). Synthesis of Nanomaterials-Top-down Approach & Bottom down Approach. Types of nano biosensors: nanotubes-based sensors, nanoparticles-based biosensors, nanowire-based sensors, quantum dots-based sensors (Basics of sensing only). Application of Nanomaterial based biosensors : Metal nanoparticle-based biosensors, carbon nanotube-based biosensors.	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Ex-1	Internal Ex-2	Total
5	15	10	10	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the principle of transduction, classifications and the characteristics of different transducers and study its biomedical applications.	K3
CO2	Understand the concepts, types, working and practical applications of electrodes for acquiring and recording various physiological parameters	K2
CO3	Understand the fundamentals of biosensors and its applications	K3
CO4	Identify and Evaluate Various Types of Nanomaterials and its applications.	K3
CO5	Design and implement simple projects for data acquisition and display the output	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	1	-	-	-	-	-	-
CO2	3	2	2	-	-	1	-	-	-	-	-	-
CO3	3	2	2	-	-	1	-	-	-	-	-	-
CO4	3	2	2	-	-	1	-	-	-	-	-	-
CO5	3	3	3	-	3	3	-	-	3	3	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of Medical Physiology	Guyton and Hall	Saunders, an imprint of Elsevier Inc	12e/ 2011
2	Biosensors Fundamentals and Applications	Bansi Dhar Malhotra, Chandra Mouleypandy	Smithers Rapra	1e/ 2017
3	Introduction to Biosensors_ From Electric Circuits to Immunosensors	Jeong-Yeol Yoon (auth.)	Springer International Publishing	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Applied Biomedical Instrumentation	Geddes and Baker	Wiley Inter science	1989
2	Biomedical Transducers and Instruments	Tatsuo Togawa, Toshiyo Tamura and P. Ake Öberg	CRC Press	2000
3	Fiber-Optic Sensors for Biomedical Applications	Daniele Tosi Guido Perrone	Artech House Applied Photonics Series	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLp6ek2hDcoNBrYuh8TYc3YNQUvKanqiRa&si=s0dTnwc4f63NzELM
2	https://www.youtube.com/watch?v=0EDwazhDRQk https://drmcet.digimat.in/nptel/courses/video/108108167/L10.html
3	https://youtu.be/kQ6CY1qpGjY?si=Hh-S5INJwkw2jUbT
4	https://archive.nptel.ac.in/courses/127/105/127105235/

SEMESTER S6
BIOMECHANICS

Course Code	OEEBT612	CIE Marks	40
Teaching Hours/Week (L:3 T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NONE	Course Type	OE

Course Objectives:

1. This course enables students to understand the biomechanics of human tissues and apply the engineering knowledge for design of prosthetic devices and implants

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Biomechanics of Hard Tissues - structure and mechanical properties of bone - cortical and cancellous bones, Elasticity-stress, strain, elastic modulus, Poisson's ratio, yield strength, ultimate tensile strength, failure strength, strain to failure, Wolff's law of bone remodeling, models of viscoelasticity - Soft Tissues - Structure, functions, material properties – tendon function, elasticity in a tendon-mechanical properties	9
2	Joint prosthesis-components-materials- shoulder, Hip, Knee-Human gait-phases-gait analysis methods-sensors- postural stability-gait analysis-human gait in health and disease.	9
3	Fracture fixation devices-pins, screws, intramedullary devices, plates-materials for fracture fixators-Dental implants-types-materials-heart valves-types – mechanical and tissue heart valves-materials-testing of heart valves	9
4	Basics of 3D printing technology- Bioinks for 3D printing of artificial extracellular matrices-Printing technologies used in three dimensional printing-Application of 3D printing- Scaffold based bioink materials-hydrogels- Nanotechnology based bioink-Artificial extracellular matrix-3D-printed polymeric scaffolds for bone tissue engineering- 3D-printed natural	9

	and synthetic polymeric scaffolds for bone tissue engineering-Synthetic 3D-bioprinted scaffolds-polymer based composite scaffolds	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To analyse the structure property relationships of hard and soft tissues	K4
CO2	To analyse the biomechanical principles of human locomotion	K4
CO3	To explore the use of assistive devices and implants	K2
CO4	To understand the use of 3D printing technology in the design of implants	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	-	1	-	-	-	1	-	3
CO2	3	3	1	-	-	1	-	-	-	1	-	3
CO3	3	3	2	-	-	1	-	-	-	1	-	3
CO4	3	3	2	-	3	1	-	-	-	1	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced 3D-printed systems and nano systems for drug delivery and tissue engineering.	Du Toit L, Kumar P, Choonara Y, Pillay V, editors	Elsevier	2020 Mar 8.
2	Biomechanics- Mechanical properties of living tissues	Y.C Fung,	Springer	1993
3	Mechanics of biomaterials: fundamental principles for implant design.	Pruitt, Lisa A., and Ayyana M. Chakravartula.	Cambridge University Press,	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of artificial human joints & organs.	Pal, Subrata.	Springer	2014.
2	Orthotics and Prosthetics in Rehabilitation E-Book	Chui, Kevin C., Milagros Jorge, Sheng-Che Yen, and Michelle M. Lusardi.	Elsevier Health Sciences	2019.
3	The Hip Joint.	Iyer, K. Mohan	CRC Press	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/fCqKhggM120?si=Cy55Lj3t_kh8pb0p
2	https://youtu.be/BhVTcwiv1_Y?si=_U-KpPe0mGV8GqAR

SEMESTER S6

BIOSIGNALS & SIGNAL PROCESSING

Course Code	OEEBT613	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	40
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Understand the structure and function of excitable cells, explore the ionic basis of cardiac action potentials and the specialized conductive system of the heart.
2. Learn to generate, interpret, and analyze ECG waveforms, as well as understand EEG brain rhythms, signal characteristics, and electrode systems. Study EMG signals, along with the characteristics of heart sounds.
3. Identify and mitigate various types of noise in biopotential recordings using time domain filters and frequency domain filters. Perform power spectrum analysis and conduct spectrum analysis of bio signals.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cell Potentials: Excitable cells - cell membrane - structure. Nernst potential- Resting membrane potential - Goldman Hodgkin Katz equation. Specialized excitatory and conductive system of heart - rhythmicity- ionic basis of cardiac action potentials – plateau region – pacemaker. Nerve action potentials – ionic basis of generation, properties and refractory period. Synapse -Neurotransmitters – types	9
2	ECG: Generation – ECG – waves – duration – intervals – segments - Lead systems-unipolar & bipolar. EEG - Brain rhythms & waves - characteristics of signal - Electrode system. EMG: Membrane potentials & Action potentials in smooth muscles –Characteristics of EMG signal. Heart sounds –	9

	Phonocardiography. PPG – characteristics - dicrotic notch. Introduction to other biopotentials – ENG& EGG (Instrumentation not needed for ECG, EEG & EMG)	
3	Event detection in ECG - QRS detection -Derivative based method-Pan Tompkins algorithm. Detection of P Wave. Dicrotic notch detection. EEG rhythm detection -Correlation based method. Detection of S1 &S2 heart sound. EMG envelop extraction Noise: Random Noise -structured noise-physiologic interference, Time domain filters: synchronised averaging- moving averaging-derivative based operators. Noise removal in ECG: random noise removal -baseline drift removal. Frequency domain filters: Butterworth low pass &low filter. Adaptive filters – cancellation of 50 Hz signal in ECG	9
4	Power spectrum analysis: Estimation of power density spectrum, non-parametric method-Periodogram &modified periodogram. Parametric methods -ARMA model-based PSD estimation. Spectrum analysis of bio signals- spectrum estimation of ECG-Power spectrum estimation of Heart rate variability data. Spectrum estimation of EEG	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Ex-1	Internal Ex-2	Total
5	15	10	10	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand bioelectric potentials and analyse the characteristics of bio signals in time and frequency domain.	K4
CO2	Apply signal processing techniques for biomedical event detection	K3
CO3	Apply filtering for artifact removal	K3
CO4	Analyse parametric and nonparametric methods of power spectrum estimation	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	2	1	-	-	-	-	-	1
CO2	3	3	2	1	2	2	-	-	-	-	-	1
CO3	3	3	2	2	2	1	-	-	-	-	-	1
CO4	3	3	2	2	2	2	-	-	-	-	-	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of Medical Physiology	Guyton and Hall	Saunders, an imprint of Elsevier Inc	12e/ 2011
2	Biomedical Signal Analysis	Rangaraj M. Rangayyan	The Institute of Electrical and Electronics Engineers	Second Edition, 2015
3	Digital Signal Processing Principles, Algorithms and Applications	John G Proakis & Dimitris G Manolakis	PHI	4 th Edition, 2016
4	Adaptive Signal Processing	Bernard Widrow and Samuel D. Stearns	Person Education	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Physiology: From Cells to Systems	Lauralee Sherwood	Brooks/Cole, Cengage Learning	7 th Edition, 2008
2	Signals & Systems in Biomedical Engineering	Suresh R Devasahayam	Springer	2nd Edition, 2013
3	Biomedical signal Processing	D C Reddy	Tata McGraw-Hill	1st Edition, 2005
4	Bio signal and Biomedical Image Processing	John L. Semmlow	Marcel Dekker Inc	New York, 2004.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=uttY5TGkCCI&t=4s
2	https://www.youtube.com/watch?v=sni74I8SAIM
3	https://www.youtube.com/watch?v=JPIGYAtfU30
4	https://www.youtube.com/watch?v=gR9MAxJaMWE&t=5s
5	https://www.youtube.com/watch?v=ahQF4Gs3xdA&t=1s
6	https://www.youtube.com/watch?v=vnqQlh8fJWg
7	https://www.youtube.com/watch?v=aC1Ztc99X1w
8	https://www.youtube.com/watch?v=mC-3puK5v2I

SEMESTER S6

BIOMATERIALS

Course Code	OEEBT614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Study the classification, bonding, crystal structure, mechanical behaviour, and applications of metals, polymers, and ceramics in medical implants.
2. Explore the structure, properties, and applications of collagen-based biomaterials in tissue engineering, focusing on mechanical behaviour and tissue regeneration.
3. Evaluate the factors influencing biocompatibility and blood compatibility of implant materials, and apply appropriate testing methods for their design and selection.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Metals for implants- Classification – Metals-Stainless steel-Co-Cr alloys-Ti based alloys-Bonding and crystal structure-interstitial sites-crystallographic planes and directions-theoretical shear strength Imperfections in metals and alloys-defects-metal processing for improved material properties-corrosion processes-Applications of metals in medical implants. Composites- Structure-properties-anisotropy particulate composites-fibrous composites-porous materials-biocompatibility of composites sterilization, and materials selection for implant design-Successful device performance and implant design.	9
2	Polymers-Bonding and crystal structure-Molecular weight distribution in polymers- Structure-Polymers used as biomaterials-sterilization-surface	9

	modification to improve biocompatibility-chemo gradient surfaces for cell and protein integration- Mechanical behaviour of polymers Polymer processing Polymers in medical implants- Ceramic materials- Bonding and crystal structure-Mechanical behaviour of ceramics-Fracture behaviour and toughening mechanisms-thermal stresses-processing of ceramics ceramics in medical implants	
3	Tissue derived biomaterials-Structure and Properties of Collagen and Collagen-Rich Tissues Biotechnology of Collagen-Design of a Restorable Collagen-Based Medical Implant-Tissue Engineering for Tissue and Organ Regeneration. Mechanical behaviour of structural tissues microstructure and mechanical properties of the basic building blocks of structural tissues-hierarchical organization of dental tissues, bone, tendons, ligaments, cartilage, skin, and blood vessels-macroscopic mechanical performance of tissues based on their microstructure -compare and contrast the mechanical performance of various structural tissues	9
4	Biocompatibility – Definition. Factors affecting biocompatibility – Carcinogenicity, mutagenicity, cytogenicity, toxicity. Blood Compatibility – Factors affecting blood compatibility. Material response: Swelling and leaching, Corrosion and Dissolution, Reaction of Biological molecules with biomaterial surfaces. Testing of implants: Methods of test for biological performance- In vitro and In vivo implant test methods. Clinical testing of implant Materials, Design and selection of implant materials – Design Process.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the structure-property relationship of biomaterials.	K2
CO2	Examine the applications of biomaterials	K3
CO3	Discuss the usage of tissue derived biomaterials.	K2
CO4	Apply factors affecting the biocompatibility and blood compatibility of implant materials.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1			1						2
CO2	3	2	1			1						2
CO3	3	2	2			2						2
CO4	3	2	2			2						2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomaterials: Principals & Practice	Wong, Joyce Y., Joseph D. Bronzino, and Donald R. Peterson	CRC Press	2013
2	Biomaterials: an Introduction	J B. Park, Roderic S. Lakes	Plenum Press	1992
3	Biomaterials Science: An introduction to Materials in Medicine	Ratner, Hoffman, Schoen and Lemons	Elsevier Academic Press	2004.
4	Biological Performance of materials	Jonathan Black, Marcel Decker	CRC Press	1981

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Encyclopedic handbook of biomaterials and bioengineering	Donald L. Wise	CRC Press	1995
2	Biomechanics and biomaterials in orthopaedics	Poitout, Dominique G	Springer London	2004
3	Biomaterials The intersection of biology and materials science	J. S Temenoff, A G Mikos	Pearson Education	2008
4	Polymeric Biomaterials	Piskin and A S Hoffmann	Martinus Nijhoff Publishers	1986

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://rb.gy/b0iagj
2	https://rb.gy/74thln

SEMESTER S6

BIO INSTRUMENTATION LAB

Course Code	PCEBL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any):	PCEBL507	Course Type	Lab

Course Objectives:

1. Familiarization of bio signal acquisition systems.
2. Analysis of bio signals
3. Circuit Analysis and troubleshooting of biomedical equipment.
4. Familiarise testing and calibration of biomedical equipment

Expt. No.	Experiments
1	QRS Detector
2	Patient Isolation Circuit
3	Flash ADC
4	X – Ray Timer
5	ESU Waveform Generation
6	Study of 7107 IC
7	Acquisition of Muscle And Nerve action potential, Measurement of Nerve Conduction Velocity using EMG acquisition system.
8	Acquisition of EEG and analysis of Visual and auditory Evoked potentials
9	Acquisition of ECG and PPG signals, Concurrent analysis of signals (Eg.Pulse transit time)

10	System level analysis (Any two) i. Colorimeter ii. Defibrillator iii. Ultra Sound Scanner iv. Power supply Unit v. X Ray Machine
11	Study of Spirometer and Pulmonary function Analysis
12	Trouble shooting of medical equipment and Simulation of circuits i. Colorimeter ii. pH meter iii. Electrosurgical Unit
13	Electrical safety analysis (Any one) i. ECG/EMG/EEG Machine ii. Electro Surgical Unit
14	Testing and Maintenance of Medical Equipment – Overview of testing instruments and standards Performance testing (Any one) i. Patient monitor ii. Defibrillator iii. Ventilator iv. Infusion pump
15	Acquisition and analysis of PCG signal: Segmentation of systolic and diastolic phases, detection of abnormalities (e.g. murmur, stenosis)
16	Acquisition and analysis of IR images: Segmentation and detection of anomalies,

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design and implement circuits for biomedical applications using Op amp	K4
CO2	Design the circuits using transducers for bio signal acquisition	K4
CO3	Develop circuits using specialized ICs for biomedical applications	K4
CO4	Describe the basic operations of medical equipment.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	3	3	3		1	1	3	3		2
CO2	3	1	3	3	3		1	1	3	3		2
CO3	3	2	3	3	3		1	1	3	3		2
CO4	3	2	3	3	3		1	1	3	3		2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Signal Analysis case study approach	Rangaraj M Rangayyan	PHI	2004
2	Electronic Devices & Circuits,	Allan Mottershead	Prentice Hall of India, NewDelhi	2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pulse, digital and Switching Waveforms	Millman and Taub	Tata McGrawHill	2007
2	Handbook of Biomedical Instrumentation,	R S Khandpur	3 rd Edition, McGraw- Hill Education / India	2014
3	Encyclopedia of Medical Devices and Instrumentation	Dr. John G. Webster	2 nd Edition, Wiley- Interscience	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ee55/preview
2	https://nptel.ac.in/courses/108105101

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

**ELECTRONICS & BIOMEDICAL
ENGINEERING**

SEMESTER S7
POWER ELECTRONICS & APPLICATIONS

Course Code	PEEBT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Introduce various power electronic devices
2. Design & analyze power conversion circuits having applications in Biomedical field

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Power electronic devices: Power diodes, Power transistors, MOSFETS and IGBTs. Structure, fabrication, Regions of operations, Switching characteristics- turn ON and turn OFF switching waveforms, Safe operating area, Paralleling of transistors. Thyristor family: SCRs, DIAC, TRIAC and GTOs. V I characteristics of SCR. Introduction to classification of power electronic circuits: AC-DC converters, DC-DC converters, DC-AC converters and AC-AC converters. Application of power devices in renewable energy systems and Electric vehicles.	9
2	AC-DC converters: Single phase-controlled rectifiers – half wave, full-wave, half-controlled and fully controlled. typical waveforms with R, RL, RL with diode and RL with voltage source - voltage and current equation for half-wave-controlled rectifier. Three phase half-wave and full-wave controlled rectifier with R load, waveforms.	9

3	DC-DC converters: Basic chopper operation using semiconductor switches, Buck, Boost and Buck-Boost converters, Selection of devices, Electrical and thermal stress, Isolated converters, Half bridge and full bridge converters, Understanding governing equations and switching waveforms. DC-AC Inverters, Square wave operations, half bridge and full bridge inverters.	9
4	Carrier based unipolar and bipolar PWM inverters, Effect of harmonics and its elimination, Sinusoidal PWM inverters. Power electronic systems: Solar power plants, ON grid and OFF grid types, Traction control in EVs, Forward motoring and generating modes, PWM control techniques: voltage mode, current mode and gated oscillator controls, Closed loop responses and stabilization. Gate driver circuits, Isolated and non-isolated, types.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify suitable power semiconductor devices suitable for a given biomedical application.	K2
CO2	Analyze the performance of single phase and three phase controlled rectifiers.	K4
CO3	Design and implement chopper circuits and inverters suitable for various applications.	K3
CO4	Implement various control methods and drive circuits for power electronics applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table: (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3									1	2
CO2	3	3									1	2
CO3	3	3									1	2
CO4	3	3									1	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power Electronics - Essentials and applications	L Umanand	Wiley India Pvt Ltd	1e/ 2009
2	Power Electronics Circuits, Devices and Applications	Muhammad H. Rashid	Pearson Education	4e/ 2017
3	Power Electronics	M.D. Singh, K.B. Khanchandani	TataMcgraw Hill	2e/ 2017
4	Power Electronics	Dr. P. S. Bimbhra	Khanna Publishers	7e/ 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power electronics: converters, applications, and design	Ned Mohan, Tore M. Undeland	John Wiley & Sons	3e/ 2007
2	Power Electronics	Daniel. W. Hart	Pearson Education	1e/ 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/102/108102145/
2	https://archive.nptel.ac.in/courses/108/101/108101126/

SEMESTER S7
HUMAN FACTORS IN ENGINEERING AND DESIGN

Course Code	PEEBT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the basic concepts and techniques employed in broad area of human factors engineering focusing on anthropometric principles in design of work places and to optimize the interaction between human and systems.
2. Integrate human factor considerations in the design of medical devices and health care tools for improved user experience and safety.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Human factors Engineering – Definition, Human Factors & Systems - Aspects of Human-machine systems. A model of Human information processing. Process of seeing- Visual capabilities-Accommodation, Visual acuity and Contrast sensitivity. Factors affecting visual capabilities. Human factor aspects of hard copy text and computer screen text. Graphic representations of Text & Data. Human factors in designing Symbols & Codes	9
2	Visual Displays - Qualitative visual displays, Quantitative visual displays, Signals & Warning lights, Representational Displays-Aircraft Bank Angle displays-Principles of aircraft position displays. Auditory Displays - Process of hearing. Principles of auditory displays, Auditory Displays for Specific Purposes. Cutaneous Senses- Tactual Displays- , Substitute for Hearing, seeing, Tactual graph. Compatibility – Types of Compatibility-Conceptual, movement, spatial and Modality Compatibility. Movement compatibility- Rotary controls and rotary displays in same plane and	9

	different plane.	
3	Muscle Physiology- Work Physiology. Measure of physiological strain, Physical Workload, Strength & Endurance. Manual Material Handling (MMH), Bio-mechanical recommended limits of MMH. Reducing the risk of MMH overexertion. General principles of Biomechanical Design. Anthropometry -Anthropometric design principles. Static and Dynamic Anthropometry. Case study of Anthropometric Design- Viewing Angle determination of Cardiac Output Monitor. Work place and Seating Design- Work space envelope, Factors in design of workspace surfaces, Principles of seat design.	9
4	Classification of human errors- Dealing with human errors. Accidents - Theories of accident causation-CFAC model. Occupational Safety and Health - Industrial Safety and Health, Hazards and Control Measures, Warnings and Safety Programs. Human factors for Mobile medical device design - General and special considerations, Design principles for home healthcare; Consideration related to Hand tool design in health care. Case studies exemplifying design for device mobility and use in motion: Infusion pumps, Portable X-Ray machine, Portable Ultrasound Device, Transport ventilator.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the human factor aspects of visual capabilities and their implications for the design.	K2
CO2	Design and evaluate auditory, tactual displays based on human sensory processes and compatibility principles to enhance user interaction and performance in various applications.	K3
CO3	Identify the role of anthropometry and Ergonomics in product design.	K3
CO4	Analyse and Apply principles of muscle physiology, biomechanics to design ergonomic work spaces and optimize the occupational settings.	K4
CO5	Classify and manage human errors, apply theories of accident causation, implement occupational safety and incorporate human factors principles in the design of mobile medical devices and health care tools.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3							2		2
CO2	3	2	3							2		2
CO3	3	2	3							2		2
CO4	3	2	3							2		2
CO5	3	2	3							2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Factors in Engineering and Design	Mark S. Sanders & Ernest J. McCormic	McGraw Hill	International Edition, 1993
2	Handbook of Human Factors in Medical Device Design	Matthew Bret Weinger, Michael E. Wiklund, Daryle Jean Gardner-Bonneau	CRC Press	2011
3	Introduction to Human Factors and Ergonomics for Engineers,	Mark Lehto, Steven J. Landry	CRC Press	2 nd Edition 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Factors Engineering in System Design	Terence S. Andre, Aaron W. Schopper	British Columbia Teacher	1997
2	Human Factors Design Handbook	Wesley E. Woodson,	McGraw-Hill Professional	2 nd Edition ,1992
3	A Guide to Human factors and Ergonomics	Martin Helander	Taylor and Francis Group	2 nd Edition ,2005
4	Human Factors Engineering and Ergonomics - A Systems Approach,	Guastello, Stephen J.-	CRC Press	2 nd Edition, 2013
5	Human Factors Methods – A Practical Guide for Engineering and Design,	Neville Stanton, Paul Salmon, Guy Walker, Christopher Baber, Dan Jenkins	Ashgate Publishing Limited	2 nd Edition, 2015
6	Handbook of Cognitive Task Design,	Erik Hollnagel	Lawrence Erlbaum Publishers	1 st Edition 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/110105162
2	https://nptel.ac.in/courses/110105162
3	https://nptel.ac.in/courses/110105162
4	https://nptel.ac.in/courses/110105162

SEMESTER S7
FUNDAMENTALS OF BIOMEMS AND MICROFLUIDICS

Course Code	PEEBT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Student will get the fundamental understanding of BioMEMS, micro-systems and a variety of diagnostic and therapeutic applications including intelligent biochips and sensors.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	MEMS Material and Fabrication: Typical MEMs and Microsystems, Compare MEMS and BioMEMS, materials for MEMS - Silicon and its compounds, Silicon piezo resistors, Gallium Arsenide, quartz, polymers. Micromachining techniques -photolithography, thin film deposition, doping-diffusion, ion implantation, etching, Bulk machining, LIGA process, wafer bonding.	9
2	Sensors And Actuators: Mechanical sensors and actuators – beam and cantilever –microplates, strain, pressure and flow measurements. Thermal sensors and actuators -actuator based on thermal expansion, thermo couples, thermal resistor. Electrostatic sensors and actuators - Parallel plate capacitor, pull in effect, Inertia sensor. Piezoelectric sensor and actuator: Properties of piezoelectric materials, – inchworm motor, Pressure sensor.	9
3	Microfluidic Systems: Fluid dynamics, continuity equation, momentum equation, equation of motion, laminar flow in circular conduits, fluid flow in micro conduits, fluid actuation methods. Dielectrophoresis, microfluid dispenser, microneedle, micro pumps, micro valves.	9

4	Application of BioMEMS: CAD for MEMS, micro total analysis systems (MicroTAS)-Lab-on-a-Chip- detection and measurement, microsystem approaches to polymerase chain reaction (PCR). MEMS based drug delivery -Drug delivery to the brain, Ocular drug delivery, Gastrointestinal tract drug delivery – capsule endoscopy, Transdermal drug delivery, Implantable subcutaneous drug delivery.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand concept of MEMS and BioMEMS	K2
CO2	Describe various materials used for MEMS fabrication	K2
CO3	Compare different types of sensors and actuators and their principles of operation at the micro scale level.	K3
CO4	Analyze the role of microfluidics in MEMS drug delivery	K4
CO5	Explore the application of MEMS in healthcare.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2							2		2
CO2	3	3	2							2		2
CO3	3	3	2							2		2
CO4	3	3	2							2		2
CO5	3	3	2							2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	MEMS and Microsystems Design and Manufacture	Tai Ran Hsu	1 st edition, Tata McGraw Hill Publishing Company, New Delhi	2017
2	BioMEMS: Technologies and Applications	WanJun Wang, Stephen A. Soper,	1 st edition, CRC Press, New York	2007
3	Electromechanics and MEMS	Thomas B. Jones	1 st edition, Cambridge University Press	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Microfabrication: the Science of Miniaturization	Marc J. Madou	2 nd edition, CRC Press	2002
2	An introduction to Microelectromechanical Systems Engineering,	Nadim Maluf, Kirt Williams	2 nd Edition, Artech House Inc, MA,	2004
3	Fundamentals of BioMEMS and Medical Microdevices	Steven S. Saliterman	1 st edition, Wiley Blackwell	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117105082
2	https://archive.nptel.ac.in/courses/108/108/108108113/
3	https://nptel.ac.in/courses/112105187
4	https://onlinecourses.nptel.ac.in/noc20_ce52/preview

SEMESTER S7
ASSISTIVE MEDICAL DEVICES

Course Code	PEEBT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the basic concepts of technology behind assistive devices for persons with different kinds of disabilities
2. Introduce the different types of assistive technologies currently available to provide a better quality of life

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Definitions and Principles of Assistive Technology, The HAAT Model - Applications, Universal and Specialized designs, Modified or Adapted Commercial Technology. A Functional Framework for Assistive Technologies: Allocation of functions, Hard and Soft assistive technologies, Application of ethical concepts in assistive technology. Human/ Technology Interfaces: Anatomic Sites for Control of AT, Elements of the Human/Technology Interface.	9
2	Control Interfaces for AT: Characteristics, Control interfaces for Direct and– Keyboards- types, Enhancing keyboard control, Touch screens, Electronic Pointing interfaces, Joysticks, Eye-Gaze systems, BCI, Control interfaces for indirect selection - Switches, Switch arrays, Discrete Joysticks, and Chord Keyboards. Multiple versus Integrated CI. Assistive Technologies for Seating and Positioning: Biomechanical principles, Sitting posture and center of pressure, Seating technologies. Wheeled Mobility: Supporting structure, Frame types, Propelling structure,	9

	Specialized bases, Smart Wheelchairs, Factors considered for selecting a wheelchair, Safe transportation of individuals in wheelchairs.	
3	Assistive Technologies for manipulation: Low-technology Aids for manipulation, Special-purpose electromechanical aids for manipulation, Electronic aids to daily living, Robotic aids - Principles of Assistive Robots, Robots as personal assistants, Mobile assistive robots, Robot use by children. Assistive Technologies for Vision: Reading aids, Optical and non-optical aids, Electronic aids, Braille displays, Recorded audio materials, Automatic reading of text, Mobility and orientation Aids – Canes, Alternative mobility devices, Electronic Travel Aids for Obstacle Detection, Orientation, and Mobility – Electronically augmented canes, Head-mounted glasses, Navigation aids for the blind – GPS-based, Special purpose visual aids.	9
4	Assistive Technologies for hearing impaired and deaf: Hearing aids – Types – Body worn, Behind the ear, Spectacles aid, In the ear, Bone conduction, Middle ear implant, Cochlear implant, Auditory brain stem implant aids, Audio frequency induction loop systems, IR systems, Alarm and alerting devices – audio, visual and tactile alarms, Devices for deafblind – Braille devices, fingerprinting. Assistive Technologies for Cognitive Augmentation: Cognitive disorders – Congenital – Intellectual, learning, ADHD, Autism, Acquired disorders – Dementia, Brain injury, Stroke, Cognitive Assistive Technologies – Devices to aid memory, Time management, Prompting/Cueing/Coaching Devices.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic principles of assistive technology and HAAT model.	K2
CO2	Identify the control interfaces and assistive technologies for seating, positioning and wheeled mobility.	K2
CO3	Discuss the use of robotic technologies in assistive technology and assistive technologies for visually impaired.	K2
CO4	Identify the assistive devices for hearing impaired and deaf people and role of assistive technology in aiding cognitive functions	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1				2						
CO2	3	1				2						
CO3	3	1				2						
CO4	3	1				2						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Assistive Technologies: Principles And Practice	Albert M. Cook and Janice M. Polgar	Mosby Publications	4e/ 2015.
2	Assistive Technology for the Hearing-impaired, Deaf and Deafblind	Marion A. Hersh, Michael A. Johnson	Springer	2e/ 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	E-Health, Assistive Technologies and Applications for Assisted Living Challenges and Solutions	Martina Ziefle and Carsten Rcke	GI Global	1e/ 2011
2	Assistive Technology for People with Disabilities	Diane Pedrotty Bryant, Brian R Bryant,	Pearson	2e/ 2012
3	Assistive Technology Intervention in Healthcare	Shruti Jain, Sudip Paul	CRC Press	2022
4	Choosing Assistive Devices A guide for users and professionals,	Helen Pain Lindsay, McLellan Sally Gore	Jessica Kingsley Publishers	1e/ 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://ocw.mit.edu/courses/6-811-principles-and-practice-of-assistive-technology
2	https://nptel.ac.in/courses/109106168
3	https://onlinecourses.swayam2.ac.in/aic19_ge01

SEMESTER S7

DEEP LEARNING TECHNIQUES

Course Code	PEEBT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic concepts of neural networks.	Course Type	PE

Course Objectives:

1. Understanding Deep Learning Architectures: Learn about popular deep learning architectures such as AlexNet, VGG16, GoogLeNet, ResNet, and DenseNet. Understand their design principles, layer types, and applications.
2. Hands-on Experience with Deep Learning Platforms: Gain practical experience with deep learning platforms like Keras, TensorFlow, and PyTorch. Explore data types, tensor operations, model configuration, training, and testing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Convolutional Neural Network (CNN) , Convolution layer, Activation function, Pooling Layer, Dense Layer, Transfer learning, Deep CNN architectures. Feedforward Neural Networks , Backpropagation, Gradient Descent, Loss functions, Optimizers, Train/Val/Test dataset distributions, Performance matrix. Hyper parameter tuning, Bias Variance trade-off, Problem of overfitting, Regularization, Batch normalization, Dataset augmentation. Applications	9
2	Deep learning Networks: History-AlexNet, GoogLeNet, VGGNet, ResNet, Dense Net Deep learning platforms: Keras, Theano, TensorFlow and PyTorch. Tensor Flow: data Types, Tensors, create Tensors, operations on Tensors, Tensor Board Keras: Common Functional Modules, Model Configuration, Training, and Testing.	9

3	Recurrent Neural Networks Forwards and Backward pass. Gradient propagation. Backpropagation through time (BPTT) algorithm. Many-to-one RNN model (sentiment analysis), Many-to-many RNN model (language model), Vanishing gradients in Recurrent networks LSTMs, Bidirectional RNNs, Gated RNNs, Encoder-Decoder Models, Attention Networks Issues in RNN/LSTM.	9
4	Deep Autoencoders: Unsupervised Deep Learning, Autoencoders Undercomplete autoencoders, Regularized autoencoders - Applications Generative Modelling: Variational Autoencoder (VAE). Variational Learning, VAE: network architecture. Generative Adversarial networks (GANs), Boltzmann Machines, Restricted Boltzmann Machines. Deep reinforcement learning: Reinforcement learning RL Process Reward hypothesis, Observations/States Space Action Space Rewards. The Policy π : the agent's brain Policy-Based Methods, Value based methods. Deep reinforcement learning-Q-learning, actor-critic, policy gradient, experience replay. Applications.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the concepts of various deep learning networks.	K4
CO2	Compare modelling aspects of various deep neural network architectures.	K4
CO3	Examine the performance criteria of different deep learning models	K3
CO4	Implement deep learning algorithms and solve real-world problems.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		3							2
CO2	3	3	2		3							1
CO3	3	3	2		2							1
CO4	3	3	3	2	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep learning	Goodfellow, Ian, Yoshua Bengio, and Aaron Courville	MIT press	2016
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems.	Géron Aurélien	O'Reilly Media, Inc	2 nd , 2022
3	Neural networks and deep learning.	Nielsen and Michael A.	Determination press	2015
4	Computational methods for deep learning.	Yan W.	Springer	2021
5	Natural Language Processing with Transformers: Building Language Applications with Hugging Face	Lewis Tunstall, Leandro von Werra, and Thomas Wolf	O'Reilly Media	2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Machine Learning with Python: A Guide for Data Scientists	Andreas C. Müller and Sarah Guido	O'Reilly Media	2023
2	Learning TensorFlow: A guide to building deep learning systems.	Hope, Tom, Yehezkel S. Resheff, and Itay Lieder	O'Reilly Media, Inc.	2017
3	Handbook of medical image computing and computer assisted intervention.	Zhou, S. Kevin, Daniel Rueckert, and Gabor Fichtinger	Academic Press	2019
4	Deep Learning for Biomedical Applications.	Kose, Utku, Omer Deperlioglu, and D. Jude Hemanth	CRC Press	2021
5	Handbook of Deep Learning in Biomedical Engineering and Health Informatics.	Julie, E.G., Robinson, Y.H. and Jaisakthi, S.M.	CRC Press	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=4TC5s_xNKSs
2	https://www.youtube.com/watch?v=Pu35DppbXO8
3	https://onlinecourses.swayam2.ac.in/imb24_mg115/preview
4	https://www.youtube.com/@reinforcementlearning1077

SEMESTER S7

ADVANCED MEDICAL IMAGING & IMAGE PROCESSING TECHNIQUES

Course Code	PEEBT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The students will be introduced to the advanced image processing techniques for medical images as well as texture based analysis method for biomedical applications.
2. The students will be able to understand active shape models used for medical image data and the image archival systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Multi-modal images and Hybrid Imaging: Hybrid PET/CT Systems, PET/MRI Systems. Super resolution imaging. Image Fusion: Preprocessing measures-geometrical transformations and image interpolation, Distortion identification- disparity analysis. Image Registration- Global Similarity, Transform Identification, Registration Evaluation and Approval. Three-Dimensional Data from Two-Dimensional Slices, Panorama Fusion, Stereo Surface Reconstruction.	9
2	Texture in Biomedical Images: Image intensity versus image texture, Characterizing the Texture, From Markov Random Fields to Fractals, From Markov Random Fields to Gibbs Distributions, Local binary patterns (LBPs), Co-occurrence Matrices, Generalized Co-occurrence Matrices, Orientation Histograms, Textons, From Spatio-Frequency to Spatio-Structural Space, Statistical Spatio-Structural Space, Monogenic Signal, From Monogenic Signal Back to Gabor Functions. Deep Learning Techniques on Texture Analysis- Lung nodule detection in CT scans.	9
3	Active Shape and Active Appearance Models: Creating an ASM, Using ASMs for Segmentation, The Active Appearance Model, Deficiencies of	9

	Statistical Shape Models, Changing the Parametric Model, Using a Part-Based Model, Using a Non-parametric Model, Physically-Based Shape Models, Mass-Spring Models, Finite Element Models, Shape, Appearance and Pose Priors for Segmentation, Atlas-Based Segmentation, Combining Shape Information with Level Set Segmentation, Solutions Based on Graph Cuts, Graph Cuts on Pre-segmented Images	
4	Image Storage and Transfer: Image Management: Archiving- Image Storage and Transfer, Information Systems in a Hospital, The DICOM Standard, Establishing DICOM Connectivity, The DICOM File Format, Technical Properties of Medical Images, Displays and Workstations, HL7 standard, Principles of Image Compression for Archiving and Communication- Philosophy of Image Compression, Generic Still-Image Compression. System, Principles of Lossless Compression, Predictive Coding, Principles of Lossy Compression, Pixel-Oriented Methods, Block-Oriented Methods, Global Compression Methods.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. **Problem Definition (K4 - 5 points)**
 - a) Clearly defines the problem.
 - b) Examines and identifies the relevant factors.
2. **Problem Analysis (K4 - 5 points)**
 - a) Break-down and presents a well-reasoned solution approach.
3. **Evaluate (K5 - 5 points)**
 - a) Thoroughly evaluate the proposed solutions.
4. **Conclusion (K3- 3 points, K5 – 2 points)**
 - a) Summarizes findings and insights. State which solution is most appropriate for the problem

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
- Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the sequenced steps needed for medical image fusion.	K4
CO2	Examine the potential of the texture-based analysis methods for biomedical applications.	K4
CO3	Apply the shape and appearance models to augment data-driven segmentation.	K3
CO4	Interpret the way in which the medical images are archived and distributed in a hospital.	K3
CO5	Evaluate the performance of advanced image processing techniques for specific problem domain.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	1							1
CO2	2	3	2	2	1							1
CO3	2	2	2	1	2							1
CO4	2	2	2	1	2							1
CO5	2	2	2	1	2				2	2		1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied Medical Image Processing	Wolfgang Birkfellner	CRC Press	2014
2	Guide to Medical Image Analysis- Methods and Algorithms	Klaus D. Toennies	Springer	Second Edition, 2017
3	Biomedical texture analysis: fundamentals, tools and challenges	A Depeursinge, S Omar, K Al, JR Mitchell	Elsevier Ltd	2018
4	Visual computing for medicine: theory, algorithms, and applications	B Preim, CP Botha	Elsevier	2013
5	Medical Image Processing, Reconstruction and Analysis: Concepts and Methods	J Jan	CRC Press	Second Edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Image Processing for Medical Applications	Geoff Dougherty	Cambridge University Press	2007
2	Biomedical Imaging- Principles And Applications,	Edited By Reiner Salzer	John Wiley & Sons, Inc.,	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/ZK8BbrlyAww?si=L6kIDiPob4O-Sic3
2	https://youtu.be/SCNDpDMKsRg?si=Ebr4tGntABSBS410
3	https://youtu.be/Wp9sAT6-3XI?si=ZTXu6TtDQ8pgjJUe https://youtu.be/nm3EKVs0qT4?si=cvG3jpxVMxj57OR_
4	https://youtu.be/7Pzq9IwscbE?si=f6SsyI11ewuF3OXz

SEMESTER S7
COMMUNICATION TECHNIQUES

Course Code	PEEBT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of both analog and digital communication systems, including fundamental principles, modulation techniques, and key components of advanced communication systems like microwave, satellite, and optical communications.
2. Gain proficiency in digital modulation schemes, understand the evolution of mobile communication technologies from 1G to 5G, and learn about modern wireless communication systems and multiple access techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to communication systems: Definition, Block level representation. Analog and digital communication systems: Comparison, Need for modulation Amplitude Modulation: Representation of AM, Modulation index, Frequency spectrum, Power relations in AM wave, AM Generation, Envelope AM Detection, Superheterodyne AM receiver, Balanced modulator Frequency modulation: Mathematical representation, waveforms, frequency deviation, bandwidth requirement, FM Generation– Varactor Diode Modulator & Armstrong method, Pre-emphasis and De-emphasis, Balanced slope detector, FM transmitters and FM receivers (block diagram)	9
2	Pulse Modulation: Need for pulse modulation, Different types, Pulse Amplitude Modulation, Pulse Width Modulation, Pulse Position Modulation and Pulse Code Modulation – principles of operation. Digital Modulation Schemes: BPSK, QPSK, BFSK, Differential Phase Shift Keying (DPSK), QAM. Modems: Classification, Modes of operation, Modem interfacing	9
3	Microwave communication system: Introduction- transmitter- repeater-	9

	receiver, frequency bands. Satellite communication system: Introduction-transponder, Kepler's laws, Satellite sub systems. Optical communication system: Features: fiber optic technology- total internal reflection- Snell's law- light transmission through an optical fiber- numerical aperture, Optical fiber communication system. Cellular communications: Principles of cellular networks, frequency reuse, operation of cellular system, hand off strategies, roaming, improving capacity in cellular system.	
4	Wireless communication systems: Introduction to modern wireless communication systems, Wireless local area networks, Bluetooth and Personal Area networks. Introduction to Multiple Access: FDMA, TDMA, Spread Spectrum Multiple Access, Space division Multiple Access, CDMA, OFDM. GSM services, architecture of GSM network. Overview of Mobile Communication Generations (1G to 4G), Evolution and Need for 5G, 5G Network Architecture and Components	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a thorough understanding of the fundamental principles and concepts of both analog and digital communication systems, including the necessity and differences of various modulation techniques.	K4
CO2	Gain proficiency in various digital modulation schemes such as BPSK, QPSK, BFSK, DPSK, and QAM, and understand modem classification and interfacing.	K3
CO3	Explore the key components and functioning of advanced communication systems, including microwave, satellite, optical communication systems, and principles of cellular communications.	K2
CO4	understand the evolution of mobile communication technologies from 1G to 4G, understand the need for and architecture of 5G networks, and learn about modern wireless communication systems and multiple access techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-	1	-	-	-	1	-	3
CO2	3	3	2	1	-	1	-	-	-	1	-	3
CO3	3	3	2	1	-	1	-	-	-	1	-	3
CO4	3	3	2	1	-	1	-	-	-	1	-	3
CO5	3	3	3	1	-	1	-	-	-	1	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Analog Communication Systems	Dr. Sanjay Sharma	Katson Books	2009
2	Communication Systems (Analog and Digital)	Dr. Sanjay Sharma	Katson Books	2009
3	Advanced Electronic Communication Systems	Tomasi	Pearson	6th Edition, 2015
4	Wireless Communications and Networks	Mishra	McGraw Hill	2nd Edition, 2013
5	5G Mobile and Wireless Communications Technology	Afif Osseiran, Jose F. Monserrat, Patrick Marsch	Cambridge University Press	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Communication	Dennis Roody and John Coolen	Prentice Hall of India, New Delhi	4 th edition
2	Electronic Communication Systems	George Kennedy & Davis	Tata McGraw Hill	6th Edition
3	Digital and Analog Communication Systems	Sam Shanmugham	John Wiley & Sons	August 2006
4	Modern Wireless Communication	Simon Haykin, Michael Mohar	Pearson Education	2005
5	Satellite Communication	Dennis Roody	McGraw Hill	4th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ee11
2	https://onlinecourses.nptel.ac.in/noc21_ee74
3	https://onlinecourses.nptel.ac.in/noc22_ee85

SEMESTER S7
DESIGN OF BIOMEDICAL DEVICES

Course Code	PEEBT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Student will be able to get an overview of design concepts for a medical device starting from generating ideas, classifying medical devices and understanding the design procedures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Life cycle of Medical device; Determining and Documenting device requirements: Idea feasibility/Generating concept: Prototype development to prove the idea. Requirements: Product specification, Specification review, design specification Software/Hardware requirement specification, Software/Hardware design description, Device History Record, Device Master Record, Design history files.	9
2	Design Phase – Risk analysis- hazards review, risk trace matrix, Risk assessment of Medical device-FMEA: Design and system FMEA, Hardware design, Software design, design reviews, environmental protection, product misuse, Biocompatibility – sterilization requirements (sterilization and packaging), human factors engineering, Bill of materials preparation, Need of CAD in product design. Application of AI tools in device designing.	9
3	Verification and Validation: Validation and verification meaning and differences, Design verification plan, protocol, software-hardware test plan for medical devices, system testing- types of testing. Manufacturing supply chain: Product manufacturing, IQ/OQ/PQ, Design transfer process to	9

	manufacturing, Labelling-IFU, Quality assurance, audits.	
4	Fundamentals of Medical device regulation: Medical device definition, classification (FDA, EU and CDSCO); Medical device safety, Global framework for regulations, Regulatory tools, Medical device Standards: Development process, Conformity assessment, Quality system requirements; Basic awareness on ISO 13485, 14971, 10993, 11607, 62304, IEC 60601.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the needs of designing a product and generate design requirements	K2
CO2	Follow a deterministic engineering design process to create new products	K2
CO3	Identify steps in the verification and validation of the product design	K3
CO4	Utilize fundamental design principles, machine elements, manufacturing and assembly techniques	K3
CO5	Be aware of the principles of regulations for medical devices as per WHO guidelines	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1		2	2				1		3
CO2	3	1	3	2	2					1		3
CO3	3	3	3	2	2							3
CO4	3	3	3	1	2				2			3
CO5	3	3	3	3	2	2			2			3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of medical device design.	Richard C. Fries	CRC Press	2019
2	Reliable design of medical devices	Richard C. Fries	CRC Press	2012
3	Medical Device Design Innovation from Concept to Market	Peter Ogrodnik	Elsevier,	2013
4	Biomedical devices: materials, design, and manufacturing	Lam, Raymond HW, and Weiqiang Chen	Springer	2019
5	Design of Biomedical Device and Systems	Paul H. King, Richard C. Fries, Arthur T. Johnson	CRC Press, 3 rd Edition	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design controls for the medical device industry	Teixeira, Marie B	CRC press,	2019
2	Medical devices: regulations, standards and practices	Ramakrishna, Seeram, Lingling Tian, Charlene Wang, Susan Liao, and Wee Eong Teo	Woodhead Publishing	2015
3	Introduction to product design and development for engineers.	Jamnia, Ali.	CRC Press	2018
4	Handbook of Medical Device Regulatory Affairs in Asia	Jack Wong, Raymond Tong	2 nd Edition,	2018
5	Biodesign: the process of innovating medical technologies.	Yock, Paul G., Stefanos Zenios, Josh Makower, Todd J. Brinton, Uday N. Kumar, FT Jay Watkins, Lyn Denend, Thomas M. Krummel, and Christine Q. Kurihara.	Cambridge University Press,	2015
6	Medical device software verification, validation and compliance.	Vogel, David A.	Artech House	2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ge14/preview
2	https://nptel.ac.in/courses/112107217
3	https://archive.nptel.ac.in/noc/courses/noc19/SEM1/noc19-ge14/
4	https://archive.nptel.ac.in/courses/127/106/127106136/

SEMESTER S7
NEURAL PROSTHESIS AND IMPLANTS

Course Code	PEEBT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Comprehend the various challenges involved in the design and implementation of upper limb prostheses, including anatomical, functional, and technical constraints.
2. Investigate different types of upper limb prostheses and evaluate their effectiveness, considering the needs of diverse user groups and the limitations imposed by current technologies.
3. Utilize knowledge of the human motor system and control strategies to innovate and improve the development of upper limb prosthetics, ensuring better integration with the user's natural movements and enhancing overall usability.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Microelectronic Visual Prostheses: Microelectronic visual implant technologies: Retinal stimulation and retinal implant - Epiretinal, Subretinal and Extraocular implants, Visual stimulation in the brain, Optic Nerve Stimulation, Engineering challenges in the development of visual prostheses, Stimulation microelectrode arrays - types and electrode materials, optic nerve stimulation using penetrating microelectrode arrays, Neural electrical stimulator, DSP-based image processing system for visual prostheses	9
2	Modern Cochlear Implant: System Review - Architecture and functional block diagram of a modern cochlear implants, External unit, RF link, Internal unit, Electrodes, Safety Considerations, Auditory Prosthesis Using Deep Brain Stimulation: Development and Implementation-Design considerations, approaches for implementation, Electrode technologies, Stimulation Strategies.	9

3	Signals and Signal Processing for Myoelectric Control: Myoelectric Signal - Acquisition, Processing, Control of prosthetic function, Terminal device electronics	9
4	Upper Limb Prosthetics: Pre-prosthetic Assessment, Prosthetic Options, Components for the Upper Limb Prosthesis - Hands and Work Hands (Terminal Devices), Pinch Force, Weight, Work Hands, Cosmetic/Protective Covers, Electric wrist rotation, Wrist Flexion, Electronic Controllers, proportional Versus Digital control, Elbow components, All electric alternatives control Systems, Interface designs, Indications and contraindications, Training and outcome measurement	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the principles and engineering challenges of retinal and visual implant technologies	K4
CO2	Analyze the architecture and functional block diagrams of modern cochlear implants	K4
CO3	Analyze and Apply Myoelectric Signal Processing for Prosthetic Control	K4
CO4	Evaluate and Develop Upper Limb Prosthetic Solutions	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2			2	2					3
CO2	3	3	2			2	2					3
CO3	3	3	2			2	2					3
CO4	3	3	2			2	2					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Implantable Neural Prostheses Devices and Applications Powered Upper Limb	David D. Zhou Elias Greenbaum	Springere-ISBN 978-0-387-77261-5	2009
2	Prostheses: Control, Implementation and Clinical Application	L. McLean, R. N. Scott (auth.), Ashok Muzumdar	Springer-Verlag Berlin Heidelberg, ISBN: 978-3-642-62302-8, 978-3-642-18812-1	2004
3	Mechatronic Hands Prosthetic and Robotic Design	Paul H. Chappell	The Institution of Engineering and Technology, ISBN 978-1-78561-155-1 (PDF)	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Orthotics and Prosthetics in Rehabilitation	Kevin Chui, Milagros Jorge, Sheng-Che Yen, Michel M. Lusardi	Elsevier, ISBN: 978-0-323-60913-5	4 th Edition
2	Mechanics of Biomaterials Fundamental Principles for Implant Design,	Lisa A. Pruitt and Ayyana M. Chakravartul	Cambridge University Press	2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_bt37/preview
2	https://onlinecourses.nptel.ac.in/noc23_ee41/preview

SEMESTER S7

TISSUE ENGINEERING & BIOFABRICATION TECHNOLOGY

Course Code	PEEBT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. This course aims to introduce students to the foundational concepts and technologies of biofabrication, including the principles of tissue engineering, biomaterials, and various fabrication techniques. Students will develop a deep understanding of scaffold design, cell culture methods, and advanced biofabrication technologies, preparing them to innovate in the field of regenerative medicine.
2. This course focuses on the practical applications of tissue engineering in developing advanced tissue substitutes and regenerative solutions. Students will explore specific applications such as skin, bone, and cartilage engineering, and learn to utilize cutting-edge technologies including 3D bioprinting and nanocomposite scaffolds, equipping them with skills to tackle real-world challenges in tissue engineering.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental Of Tissue Engineering: Fundamentals Of Stem Cell Tissue Engineering; Growth Factors; Extracellular Matrix: Structure, Function And Tissue Engineering Application; Mechanical Forces On Cells; Cell Adhesion; Cell Migration. Tissue Engineering Enabling Technologies: Polymer Scaffold For Tissue Engineering Applications; Biomimetic Materials; Nanocomposite Scaffolds Tissue Engineering; Bioreactors; Regulatory Issues In Tissue Engineering.	9
2	Tissue Engineering Application: Bioengineering Of Human Skin Substitute; Nerve Tissue Engineering; Musculoskeletal Tissue Engineering; Bone Tissue Engineering; Cartilage Tissue Engineering; Temporomandibular Tissue Engineering; Smooth Muscle Tissue	9

	Engineering; Oesophagus Tissue Engineering. Vascular Graft Tissue Engineering Cardiac Tissue Engineering; Heart Valve Tissue Engineering; Urologic Organ Tissue Engineering; Hepatic Tissue Engineering; Renal Tissue Engineering; Dental Tissue Engineering; Tracheal Tissue Engineering.	
3	In vitro Biofabrications of tissues and organs: Scaffold free tissue engineering, classification of present scaffold free system, in vitro self-produced ECM rich scaffold free constructs, the rotating wall vessel Bioreactor system. Aggregation/Spheroid based approaches: Preparation of multicellular spheroids, Molding MCS, Bioprinting	9
4	Biofabrication of 3D tissue scaffolds: Conventional techniques for 3D scaffolds, Advanced techniques for 3D scaffolds. Applications of 3D tissue scaffolds: Nanofibrous scaffolds, Hydrogel scaffolds. Fabrication of Microscale Hydrogels: Photolithography, Soft photolithography, Bioprinting, Microfluidics. Laser assisted Bioprinting: Workstation requirement, Droplet ejection mechanism, The bio ink as a scaffold, Biopapers	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze various biofabrication techniques, including 3D bioprinting and hydrogel-based fabrication, to understand their applications in creating biological tissues and organs.	K4
CO2	Design and construct biomaterial scaffolds for specific tissue engineering applications, taking into account material properties, scaffold architecture, and biocompatibility.	K3
CO3	Apply tissue engineering principles to develop advanced solutions for applications such as skin, bone, and cartilage engineering, effectively addressing challenges in regenerative medicine.	K2
CO4	Understand advanced biofabrication technologies, including nanocomposite scaffolds and bioreactors, to create functional tissue constructs and overcome current limitations in the field.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-	1	-	-	-	1	-	3
CO2	3	3	2	1	-	1	-	-	-	1	-	3
CO3	3	3	2	1	-	1	-	-	-	1	-	3
CO4	3	3	2	1	-	1	-	-	-	1	-	3
CO5	3	3	3	1	-	1	-	-	-	1	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Tissue Engineering	Clemens van Blitterswijk	Academic Press	2008
2	Principles of tissue engineering	Lanza, R., Langer, R., Vacanti, J. P., & Atala, A. (Eds.)	Academic press.	2020

3	Principles and applications in engineering series. Tissue Engineering	Palsson, B., Hubbell, J. A., Plonsey, R., & Bronzino, J. D.	, CRC Press, Boca Raton, FL.	2003
4	Biofabrication, micro and nano biofabrication, printing patterning and assemblies	Gabor forgaces, We sun	Elsevier	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Biomaterials	C. Mauli Agrawal	CRC Press	2017
2	Biomaterials Science: An Introduction to Materials in Medicine	Buddy D. Ratner, Allan S. Hoffman, Frederick J. Schoen, Jack E. Lemons	Academic Press	2020
3	Regenerative Medicine and Tissue Engineering: Principles and Applications	Paul M. W. et al.	Wiley	2015
4	3D Bioprinting: Principles and Applications	Frank A. Doyle, Lisa C. H. Becker	Elsevier	2020
5	Bioprinting: Principles and Applications	William T. Hsu	Springer	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_bt33
2	https://onlinecourses.nptel.ac.in/noc22_bt61
3	https://onlinecourses.nptel.ac.in/noc22_bt50

SEMESTER S7

ADVANCED COMPUTER PROGRAMMING TECHNIQUES

Course Code	PEEBT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic programming concepts.	Course Type	Theory

Course Objectives:

1. Implement R programs to extract data from different sources, as well as to clean and manipulate it.
2. Develop advanced R programs for data analytics.
3. Develop advanced Python programs for machine and deep learning.
4. Analyse the ROS framework, architecture and interface.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Statistical programming language R: Downloading R and R Studio, Basics of Coding in R, Using Comments in R, Executing Commands in R, Importing Packages in R, Getting Data into R, Saving Output in R, Accessing Records and Variables in R, Slicing and Extracting Data, Variable Conversion, Data Selection and Manipulation, Math Function. Visualizations using the ggplot2 library, Create commonly used data visualizations for each data type including histograms, scatter plots, and box plots, and improve data visualizations using facets.	9
2	Advanced Data Processing-R programming-loading, manipulating, visualizing, and saving R Data Structures, sample-based statistics measuring central tendency and dispersion, different types of variables, scrapping data from public websites, examples of cohort-rebalancing. Statistical shape analysis using R, shape package, Shape distances in R, Tangent space coordinates in R, Calculation of mean shapes in R, Generalized Procrustes algorithms for shape analysis in R, tangent-based PCA. R packages for decision tree classification, random Forest R. R in Healthcare.	9

3	Coding in python: Data analysis with pandas - the data structure of pandas, Data Frame- Basics, Basics of deep learning with python, Gradient decent algorithm, ANN for regression, Over fitting and cross fitting validation, regularization, convolutional neural networks, RNN, Tensor Flow and keras for deep convolutional neural networks. Deep neural network and transfer learning using Python, Tensorflow, and Keras.	9
4	Robotics Programming ROS framework operating system , UNIX commands - Architecture of Operating System, catkin workspace, ROS navigation and listing commands Navigation through file system - Understanding and Debugging Nodes visualization using Gazebo Rviz URDF modelling Xacro launch files. Sensor Drivers and Motor Drivers for ROS Actuator Interfacing Case Studies: Navigation stack-creating transforms -odometer imu laser scan base controller robot configuration TurtleBot the low-cost mobile robot	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop R programs to extract data from different sources, as well as to clean and manipulate it.	K3
CO2	Develop advanced R programs for data analytics and elaborate the steps used in R to create and visualize decision trees	K4
CO3	Develop advanced Python programs for machine and deep learning. Analyse the matrix operations using pandas data frame and Apply transfer learning using Python, Tensorflow, and Keras	K4
CO4	Analyse the ROS framework, architecture and interface	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3				1	1		1
CO2	3	2	2	2	3				1	1		1
CO3	3	2	2	2	3				1	1		1
CO4	3	2	2	2	3				1	1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Analyse the ROS framework, architecture and interface	Hector, Andy	Oxford University Press	2021
2	Python crash course: A hands-on, project-based introduction to programming.	Matthes, Eric.	nostarch press,	2019
3	Statistical shape analysis: with applications in R. John Wiley & Sons	Dryden, Ian L., and Kanti V. Mardia.	John Wiley & Sons,	Vol. 995., 2016
4	Mastering python for data science.	Madhavan, Samir.	Packt Publishing Ltd	2015
5	Programming Robots with ROS: a practical introduction to the Robot Operating System	Quigley, Morgan, Brian Gerkey, and William D. Smart.	“O’Reilly Media, Inc.”	2015
6	Data science using Python and R.	Larose, Chantal D., and Daniel T. Larose.	John Wiley & Sons,	2019.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Operating System (ROS).	Koubâa, Anis, ed.	Cham: Springer,	Vol. 1, 2017.
2	Data Science and Predictive Analytics.	Dinov, Ivo D.	Springer, Ann Arbor	2018
3	Robot operating system (ros) for absolute beginners.	Joseph, Lentin.	Springer	2018
4	Learning ROS for robotics programming.	Fernandez, Enrique, Luis Sanchez Crespo, Anil Mahtani, and Aaron Martinez,	Packt Publishing Ltd,	2015.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1&2	https://onlinecourses.nptel.ac.in/noc24_mg113/preview
3	https://onlinecourses.nptel.ac.in/noc24_cs89/preview https://onlinecourses.nptel.ac.in/noc24_cs78/preview
4	https://onlinecourses.nptel.ac.in/noc24_me117/preview

SEMESTER S7

ADVANCED BIOMEDICAL SIGNAL PROCESSING

Course Code	PEEBT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Students will learn various ECG data compression techniques, including direct data compression and transformation compression, as well as other techniques. They will also be able to apply ECG signal processing methods
2. Students will gain knowledge about the fundamentals of speech signal processing and analyze speech signals through short-time speech analysis, frequency domain analysis, and understand key measures.
3. Students will understand the modeling of EEG signals, and will learn methods for reducing artefacts in EEG signals. They will be able to conduct both parametric and non-parametric spectral analysis, and perform EEG segmentation.
4. For EMG signal processing, students will explore applications in various areas, as well as techniques for amplitude estimation, spectral analysis, and conduction velocity estimation of surface EMG signals.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ECG Data Reduction: Direct data compression Techniques: Turning Point-AZTEC. Transformation Compression Techniques: Karhunen - Loeve Transform. Other data compression Techniques: Huffman coding ECG signal processing Techniques: Filtering: baseline removal by linear time variant filtering. Powerline interference removal by linear filter. Event detection: QRS detection basic steps – P Wave detection using time domain analysis. Morphological analysis of ECG- correlation coefficient -minimum	9

	phase component. Heart rate variability analysis: time domain measures - spectrum analysis of heart rate variability	
2	Speech signal Processing Speech Fundamentals: Articulatory Phonetics – Production and Classification of Speech Sounds- Articulation- Voicing- Articulatory Model Speech Analysis: -short Time Speech Analysis - short time energy- short time average magnitude- short time zero crossing rateshort time ACF. Frequency domain analysis: Filter Banks- STFT- Spectrogram- Formant Estimation & analysis- Cepstral Analysis-MFCC	9
3	Signal Processing Techniques in EEG Modelling the EEG signals: linear and stochastic EEG models -Time varying AR modelling. Artefacts in EEG: types – artefact reduction methods -linear filtering – Artifact cancellation using linearly combined reference signals -adaptive artefact cancellation. Non parametric spectral analysis-spectral parameters: Power in bands, spectral slope, peak frequency. Parametric spectrum estimation: Burg method. EEG segmentation-basics	9
4	EMG Signal Processing Techniques EMG applications: Diagnostic EMG, Kinesiology, Ergonomics. Prosthesis control. Amplitude Estimation in the Surface EMG-Signal Model and ML Estimation, spectral analysis of surface EMG- Conduction velocity estimation -intramuscular EMG signal decomposition	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. **Problem Definition (K4 - 5 points)**
 - a. *Clearly defines the problem.*
 - b. *Examines and identifies the relevant factors.*
2. **Problem Analysis (K4 - 5 points)**
 - a. *Break-down and presents a well-reasoned solution approach.*
3. **Evaluate (K5 - 5 points)**
 - a. *Thoroughly evaluate the proposed solutions.*
4. **Conclusion (K3- 3 points, K5 – 2 points)**
 - a. *Summarizes findings and insights. State which solution is most appropriate for the problem*

Scoring:

1. *Accomplished (4 points): Exceptional analysis, and depth of understanding.*
2. *Competent (3 points): Solid performance with minor areas for improvement.*
3. *Developing (2 points): Adequate effort but lacks depth or clarity.*
- Minimal (1 point): Incomplete or significantly flawed.*

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the techniques of ECG data compression and implement signal processing algorithms for ECG signal processing	K4
CO2	Apply different time & transform domain techniques for speech signal processing	K3
CO3	Develop various signal processing techniques in EEG analysis	K4
CO4	Apply various signal processing techniques in EMG analysis	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	2					1		1
CO2	3	3	3	2	3					2		2
CO3	3	3	3	2	3					2		2
CO4	3	3	3	2	3					2		2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bioelectrical Signal Processing in Cardiac and Neurological applications	Leif Sornmo and Pablo Laguna	Elsevier Academic Press	2005
2	Biomedical Signal Analysis	Rangaraj M Rangayyan,	John Wiley	1 st edition, 2002
3	Digital Processing of Speech Signals	Rabiner and Schafer	Prentice Hall	2 nd Edition
4	Discrete Time Processing of Speech Signals	Deller J. R. Proakis J. G. and Hanson J. H	Wiley Inter science	2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomedical Digital Signal Processing	Willis J. Tompkins	EEE, PH	2004
2	Discrete-Time Speech Signal Processing: Principles and Practice	. Thomas F. Quatieri	Prentice Hall	1 st Edition 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=5VdwVadGa-U
2	https://www.youtube.com/watch?v=PEnaQdcV30s

SEMESTER S7

BIOMEDICAL INSTRUMENTATION

Course Code	OEEBT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	OE/ ILE

Course Objectives:

1. Understand the principles of bioelectric potentials, including resting and action potentials, and their applications in clinical diagnostics such as ECG, EEG, and EMG.
2. Learn the fundamentals of various biomedical imaging and measurement technologies, such as X-ray radiography, MRI, and ultrasound, and their applications in diagnosing and treating medical conditions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Origin of bioelectric potentials – resting and action potentials - propagation of action potentials. Examples of bioelectric potentials - ECG, EEG, EMG, – Electrodes for measurement of biopotentials. Body surface electrodes, Microelectrodes, Transducers for measurement of temperature, pressure, displacement. Electrical activity of brain - Electro encephalogram – 10-20 lead system, Instrumentation of EEG machine. Evoked potential -types & applications.	9
2	Electrical activity of heart, electrocardiogram - lead systems -ECG machine – block diagram, Arrhythmia Monitor, Cardiac pacemakers – internal and external pacemakers, Defibrillators –principles of operation, Measurement of heart sounds –phonocardiography Measurement of blood pressure – sphygmomanometer & Oscillo metric methods. Photo plethysmography - for pulse rate measurement - Pulse oximeters, Patient monitoring systems. Heart-lung machine, Principle of operation, applications, Artificial heart valves. Electrical activity of Muscle – Electromyogram (EMG) –.	9

	Electromyograph	
3	Spirometer - measurement of respiratory parameters. Blood flow meters. Electromagnetic blood flow meter, Ultrasonic Doppler blood flow meter. Lithotripsy–principles, types- Ultrasonic lithotripsy, Extracorporeal shock wave lithotripsy (ESWL) & applications. Surgical diathermy – Principle of operation, applications, Ventilators- controls, modes, assisted mechanisms.	9
4	X-ray radiography - Principles of x-ray generation – Block diagram of x-ray machine – X-ray computed tomography - Principle of operation, sectional imaging, scanner configurations. Ultrasonic imaging – Basic principles - Ultrasonic transducers- modes of image display-Principles & Applications. Magnetic resonance imaging -MRI instrumentation, Imaging modes. Electric shock hazards – Leakage currents – Test instruments for checking safety parameters of biomedical equipment.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the origin of bioelectric potentials and principle of different transducers used in biomedical instrumentation.	K2
CO2	Infer the electrical activity of the brain, heart and muscles, and understand the principles and applications of biomedical devices.	K4
CO3	Examine the respiratory and blood flow measurement techniques, and understand the principles and applications of lithotripsy, surgical diathermy, and ventilators.	K3
CO4	Discuss the principles and instrumentation of medical imaging techniques as well as the safety considerations for biomedical equipment.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1			2						3
CO2	3	3	1			2						3
CO3	3	3	1			2						3
CO4	3	3	1			2						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Compendium of Biomedical Instrumentation	Khandpur, Raghbir Singh	McGraw-Hill Education	2020
2	Biomedical engineering fundamentals	Bronzino, Joseph D., and Donald R. Peterson	Biomedical engineering fundamentals	2014
3	Medical Instrumentation: Application and Design	John G. Webster	Wiley	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The biomedical engineering handbook	Bronzino, Joseph D, and Donald R. Peterson	CRC press	2018
2	Principles of biomedical instrumentation	Webb, Andrew G	Cambridge University Press	2018
3	Handbook of biomedical engineering	Kline Jacob	Elsevier	2012
4	Encyclopaedia of Medical Devices and Instrumentation	John G. Webster	Wiley	2006
5	Fundamentals of Biomedical Instrumentation and Measurement	Richard Aston and Steven A. Hunsaker	CRC Press	2017
6	Medical Imaging: Principles and Applications	Thomas L. Szabo	Springer	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_bt07/preview
2	https://onlinecourses.swayam2.ac.in/nou24_bt07/preview
3	https://onlinecourses.swayam2.ac.in/nou24_bt07/preview
4	https://www.youtube.com/watch?v=2O1nW_KzVEo

SEMESTER S7
ASSISTIVE MEDICAL DEVICES

Course Code	OEEBT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the basic concepts of technology behind assistive devices for persons with different kinds of disabilities
2. Introduce the different types of assistive technologies currently available to provide a better quality of life

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Definitions and Principles of Assistive Technology, The HAAT Model - Applications, Universal and Specialized designs, Modified or Adapted Commercial Technology. A Functional Framework for Assistive Technologies: Allocation of functions, Hard and Soft assistive technologies, Application of ethical concepts in assistive technology. Human/ Technology Interfaces: Anatomic Sites for Control of AT, Elements of the Human/Technology Interface.	9
2	Control Interfaces for AT: Characteristics, Control interfaces for Direct and– Keyboards- types, Enhancing keyboard control, Touch screens, Electronic Pointing interfaces, Joysticks, Eye-Gaze systems, BCI, Control interfaces for indirect selection - Switches, Switch arrays, Discrete Joysticks, and Chord Keyboards. Multiple versus Integrated CI. Assistive Technologies for Seating and Positioning: Biomechanical principles, Sitting posture and center of pressure, Seating technologies. Wheeled Mobility: Supporting structure, Frame types, Propelling structure, Specialized bases, Smart Wheelchairs, Factors considered for selecting a	9

	wheelchair, Safe transportation of individuals in wheelchairs.	
3	Assistive Technologies for manipulation: Low-technology Aids for manipulation, Special-purpose electromechanical aids for manipulation, Electronic aids to daily living, Robotic aids - Principles of Assistive Robots, Robots as personal assistants, Mobile assistive robots, Robot use by children. Assistive Technologies for Vision: Reading aids, Optical and non-optical aids, Electronic aids, Braille displays, Recorded audio materials, Automatic reading of text, Mobility and orientation Aids – Canes, Alternative mobility devices, Electronic Travel Aids for Obstacle Detection, Orientation, and Mobility – Electronically augmented canes, Head-mounted glasses, Navigation aids for the blind – GPS-based, Special purpose visual aids.	9
4	Assistive Technologies for hearing impaired and deaf: Hearing aids – Types – Body worn, Behind the ear, Spectacles aid, In the ear, Bone conduction, Middle ear implant, Cochlear implant, Auditory brain stem implant aids, Audio frequency induction loop systems, IR systems, Alarm and alerting devices – audio, visual and tactile alarms, Devices for deafblind – Braille devices, fingerprinting. Assistive Technologies for Cognitive Augmentation: Cognitive disorders – Congenital – Intellectual, learning, ADHD, Autism, Acquired disorders – Dementia, Brain injury, Stroke, Cognitive Assistive Technologies – Devices to aid memory, Time management, Prompting/Cueing/Coaching Devices.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic principles of assistive technology and HAAT model.	K2
CO2	Identify the control interfaces and assistive technologies for seating, positioning and wheeled mobility.	K2
CO3	Discuss the use of robotic technologies in assistive technology and assistive technologies for visually impaired.	K2
CO4	Identify the assistive devices for hearing impaired and deaf people and role of assistive technology in aiding cognitive functions	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1				2						
CO2	3	1				2						
CO3	3	1				2						
CO4	3	1				2						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Assistive Technologies: Principles And Practice	Albert M. Cook and Janice M. Polgar	Mosby Publications	4e/ 2015.
2	Assistive Technology for the Hearing-impaired, Deaf and Deafblind	Marion A. Hersh, Michael A. Johnson	Springer	2e/ 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	E-Health, Assistive Technologies and Applications for Assisted Living Challenges and Solutions	Martina Ziefle and Carsten Rcke	GI Global	1e/ 2011
2	Assistive Technology for People with Disabilities,	Diane Pedrotty Bryant, Brian R Bryant,	Pearson	2e/ 2012
3	Assistive Technology Intervention in Healthcare	Shruti Jain, Sudip Paul	CRC Press	2022
4	Choosing Assistive Devices A guide for users and professionals,	Helen Pain Lindsay, McLellan Sally Gore	Jessica Kingsley Publishers	1e/ 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://ocw.mit.edu/courses/6-811-principles-and-practice-of-assistive-technology
2	https://nptel.ac.in/courses/109106168
3	https://onlinecourses.swayam2.ac.in/aic19_ge01

SEMESTER S7
MEDICAL IMAGING TECHNIQUES

Course Code	OEEBT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course enables the students to understand the principle of image formation in various imaging modalities currently used in medicine. Students will be able to analyse the systems and techniques used in various diagnostic imaging methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Ultrasound imaging: Properties of ultrasound- Doppler effect. Principle of image formation- constructional details of US probes- Transducer Design – Considerations-Transducer arrays- Modes of scanning-A mode-B mode-M mode scanners-Duplex scanning -instrumentation-axial and lateral resolutions-applications of each.	9
2	Computed Tomography: Principle of CT imaging, Generations of scanners-system instrumentation-X-ray tubes for CT applications-pencil beam and cone beam projections-Image reconstruction algorithms for pencil beam, Image characteristics-CT numbers-	9
3	Magnetic Resonance Imaging: Principle of Nuclear Magnetic Resonance-Relaxation-MR instrumentation-Contrast mechanisms in MRI- pulse sequences- FLASH-GRASS-principle of image reconstruction,Spatial encoding systems- k-space - Functional MRI- BOLD - Spatial and temporal Resolution of fMRI.	9
4	Nuclear imaging -PET-SPECT-system level descriptions- Rectilinear Scanner, Gamma camera. Hybrid Imaging: Instrumentation MR-PET Instrumentation-MR Compatible PET Detector Technology, MR-PET System Architecture. PET-CT, SPECT-CT.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles of diagnostic Ultrasound imaging	K2
CO2	Analyze the system requirements for Computed tomographic imaging.	K4
CO3	Understand the principles of Magnetic Resonance imaging	K2
CO4	Analyze the nuclear imaging and hybrid imaging systems	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1			1				1		3
CO2	3	3	1			1				1		3
CO3	3	3				1				1		3
CO4	3	3				1				1		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Diagnostic ultrasound: physics and equipment. CRC Press, 2019	Hoskins, Peter R., Kevin Martin, and Abigail Thrush	CRC Press	2019
2	Diagnostic ultrasound imaging: inside out.	Szabo, Thomas L	Academic Press	2004
3	Webb's Physics of Medical Imaging	M Flower	Taylor & Francis	2016
4	Computed Tomography-E-Book: Physical Principles, Clinical Applications	Seeram, Euclid	Elsevier Health Sciences	2015
5	Intraoperative Imaging and Image-Guide Therapy	Ferenc A. Jolesz	Springer	2008
6	How does MRI work? An Introduction to the Physics and Function of Magnetic Resonance Imaging	Weishaupt, Dominik, Köchli, Victor D, Marincek, Borut	Springer	2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Computerized Tomographic Imaging	Avinash C Kak, Malcolm Slaney	IEEE	2001
2	MRI made Easy	Hans H Schild		2003
3	Ultrasound imaging and therapy,	Fenster, Aaron, and James C. Lacefield	Taylor and Francis	2015
4	Nuclear Medicine Textbook, Methodology and Clinical Applications	DuccioVolterraniPaola Anna ErbaIgnasiCarrióH. William Strauss Giuliano Mariani	Springer	2019
5	Nuclear medicine and PET/CT: technology and techniques	Christian, Paul E., and Kristen M. Waterstram- Rich	Elsevier	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_bt50/preview
2	https://archive.nptel.ac.in/courses/108/105/108105091/
3	https://archive.nptel.ac.in/courses/102/106/102106094/
4	https://onlinecourses.nptel.ac.in/noc21_bt50/preview

SEMESTER S7
ARTIFICIAL ORGANS & IMPLANTS

Course Code	OEEBT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Students will acquire knowledge in basic principles and designing of various types of artificial and tissue derived implants.
2. Students will be able to understand the use of 3D printing technology in implantation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cochlear implants: Basics of speech signal – Normal hearing mechanism – Deafness- Functional block diagram of cochlear implants-Implant characteristics & it's classification -Auditory prostheses using deep brain stimulation-Physiological Challenges-Single-multiple source systems-Real time programming.	9
2	Design of artificial organs: Introduction to heart mechanism- Heart valves- Pumping system-Kidney and it's artificial replacement- Anatomy of skin - Design of artificial skin- Liver and it's artificial replacement- ethical issues of implants.	9
3	Micro electrode technologies for Deep Brain Stimulators (DBS): Implant site -Stimulation patterns - Closed loop control of DBS- Introduction to bionic micro stimulation-Clinical applications-Brain control and sensing of artificial limbs-Stimulation of peripheral nerve interfaces-Construction of implant-wireless communication-prosthetic master control unit.	9
4	Fundamentals of Tissue Engineering: Engineering biomaterials for tissue	9

	engineering- Nano biomaterials for tissue engineering – Drug delivery & Gene therapy mechanism- Vascularization of Engineered Tissues-Tissue engineering of nervous system & kidney (basic level). Fundamentals of 3 D printing and its applications in Biomedical engineering: Introduction of 3D printing -3D printing process- Materials for 3D printing - Biomedical applications of 3 D printing technology- maxillofacial surgery & upper limb prosthetics.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand design aspects of neural prostheses.	K2
CO2	Explore different kind of artificial organs.	K2
CO3	Analyse design aspects of deep brain stimulators.	K4
CO4	Review applications of tissue engineering on various organs.	K4
CO5	Evaluate the use of 3 D printing technology in Biomedical Engineering.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-	1	-	-	-	1	-	3
CO2	3	3	2	1	-	1	-	-	-	1	-	3
CO3	3	3	2	1	-	1	-	-	-	1	-	3
CO4	3	3	2	1	-	1	-	-	-	1	-	3
CO5	3	3	3	1	-	1	-	-	-	1	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of artificial human joints & organs	Pal S.	Boston, MA: Springer US	2014
2	3D Printing in Biomedical Engineering	Singh S, Prakash C, Singh R	Springer	2020
3	Tissue engineering: principles and practices	Fisher JP, Mikos AG, Bronzino JD, Peterson DR	CRC Press	2012
4	Implantable neural prostheses	Zhou, David D., and Elias S. Greenbaum	Springer Science+ Business Media	2009
5	Principles of Tissue engineering	Robert Lanza, Anthony Atala	Textile eBook	2018
6	3D Printing for Tissue Engineering & Regenerative Medicine	Murat Guvendiren, Vahid Serpooshan	MDPI Books	2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced 3D-printed systems and nanosystems for drug delivery and tissue engineering	Du Toit L, Kumar P, Choonara Y, Pillay V	Elsevier	2020
2	Principles and Applications in Engineering Series: Tissue Engineering	Bernhard Palsson, Jeffrey A. Hubbell, Robert Plonsey, Joseph D. Bronzino	CRC Press	2003
3	Biomaterials, artificial organs and tissue engineering	Larry L. Hench, Julian R. Jones	Woodhead Publishing Limited	2005
4	Principles of Heart Valve Engineering	Kheradvar, Arash	Academic Press	2019
5	Engineering Neural Tissue from Stem cells	Dr. Stephanie Willerth	CCRM	2017
6	Textbook of Radiology for Cochlear Implants	Mohnish Grover, Gaurav Gupta, C. Preetam	Thieme	2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me130
2	https://onlinecourses.nptel.ac.in/noc21_mg60
3	https://onlinecourses.nptel.ac.in/noc22_ce87
4	https://onlinecourses.nptel.ac.in/noc22_bt61

SEMESTER 8

**ELECTRONICS AND BIOMEDICAL
ENGINEERING**

SEMESTER S8
TELEMEDICINE

Course Code	PEEBT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide an in-depth understanding of telemedicine systems, their components, and their evolution in healthcare.
2. To analyze the communication technologies used in telemedicine, develop mHealth applications, and address data security, ethical issues, and implementation strategies, while exploring the broad applications of telemedicine in healthcare.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction of telemedicine: History of Telemedicine, Block diagram of telemedicine system, Definition of telemedicine, Tele health, Tele care, Origin & development of telemedicine, Scope, benefits and limitation of telemedicine. Global and Indian scenario. Telecommunication based biomedical systems Tele-radiology system, Tele-pathology, Tele-cardiology, Tele-oncology, Tele-surgery, Tele dermatology, Telemedicine in neurosciences Tele- education and Tele-Monitoring.	9
2	Types of Communication & networks: PSTN, POTS, ATN, ISDN, Internet and Wireless. Data communication networks-protocols and standards-network models- the OSI model- layers in the OSI model- TCP/IP protocol suite- addressing- digital transmission- digital-to-digital conversion- analog-to-digital conversion- transmission modes- switching- circuit-switched networks- virtual-circuit networks- structure of a switch- cryptography- symmetric-key cryptography- asymmetric-key cryptography- security services- message confidentiality-message integrity- Communication infrastructure for telemedicine – LAN and WAN technology. Data exchange Network Configuration, Circuit and packet switching, H.320 series (Video	11

	phone-based ISBN) T.120, H.324 (Video phone based PSTN), Video Conferencing.	
3	Introduction to mHealth Types of Health Apps Advantages and Benefits of Delivering Interventions via Health Apps Digital health interventions (DHIs) Application Development and Testing, Sensors and the Internet of Things (IoT). Smartphone and Microcontroller Communication – Mobile Computing for Remote Monitoring - Limitations of the Mobile and IoT Environment.	9
4	Data security and standards& ethical and legal aspects of telemedicine Data security and standards: Standards to follow DICOM, HL 7, Ethical and legal aspects of telemedicine: Confidentiality and law, Patient rights and consent, access to medical records, consent treatment, intellectual property rights, jurisdictional issue.	7

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Review major components of a telemedicine system and its evolution in healthcare and explore the applications of telemedicine in healthcare	K4
CO2	Analyse communication technologies used in telemedicine	K3
CO3	Develop an application in mHealth.	K2
CO4	Explain data security, ethical issues and implementation strategies related to telemedicine	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				2		2				2
CO2	3	2				2		2				2
CO3	3	2				2		2				2
CO4	3	2				2		2				2
CO5	3	2				2		2				2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Essentials of Telemedicine and Telecare, John Wiley & Sons, 2002	A.C Norris	Katson Books	2009
2	Introduction to Telemedicine	R. Wootton& Victor Patterson	RSM Press	2006
3	Developing Medical Apps and mHealth Interventions.	Davies, Alan, and Julia Mueller.	Springer: Cham, Switzerland	2020
4	Advances in Telemedicine for Health Monitoring: Technologies, design and applications	Tarik A. Rashid, Chinmay Chakraborty, Kym Fraser	The Institution of Engineering and Technology	1 st edition 2020
5	Data communications and networking	Behrouz A Forouzan	TMH	
6	Data and computer communications	William stallings	PHI	
7	Fundamentals of Telemedicine and Telehealth	Shashi Bhushan Gogia (Editor)	Academic Press	2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Olga Ferrer-Roca & M. Sosa ludicissa, Handbook of Telemedicine, IOS Press 2002	Olga Ferrer-Roca & M. Sosa ludicissa	IOS Press	2002
2	Telemedicine and Telehealth: Principles, Policies, Performance and Pitfalls	A. Darkins & M. Cary,	Tata McGraw Hill	1 st edition Edition 2000
3	Current Principles and Practices of Telemedicine and e-Health	R. Latifi,	Ios press	edition, 2008
4	Medical Informatics-A Primer,	Mohan Bansal	Tata McGraw-Hill	2003
5	Computer networks	Andrew s.Tanenbaum	PHI	2003
6	Network Security: Private communication in a public world	CharlieKaufman, Radia Perlmon and mike speciner	PHI	2 nd edition
7	Internet working with TCP/IP: principles, protocols and architecture	Dougles E commer	PHI	3 rd edition
8	Telemedicine Technology and Applications (mHealth, TeleHealth and EHealth)	R. S. Khandpur	PHI	1 st edition, 2017
9	Telemedicine: The Computer Transformation of Healthcare	Editors: Ajay Rana, Avita Katal, Jung-Sup Um, Marwan Al-Akaidi, Tanupriya Choudhury	Springer International Publishing	1 st edition, 2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://elearn.nptel.ac.in/shop/nptel/fundamentals-of-telehealth-and-virtual-care/?v=c86ee0d9d7ed
2	https://www.coursera.org/learn/telehealth
3	https://onlinecourses.nptel.ac.in/noc22_cs19/preview

SEMESTER S8

MEDICAL DEVICE REGULATIONS AND STANDARDS

Course Code	PEEBT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course aims to equip students with the knowledge and skills necessary to ensure that medical devices meet all relevant standards and regulatory requirements, ultimately enhancing patient safety and device performance.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Regulatory Landscape: Evolution of medical device regulation, General structure – benefits of regulation, Medical Device Nomenclature (GMDN / EMDN & Alternate Options), Global Harmonization – GHTF & IMDRF, RAPS.HTA overview– Clinical investigation of Medical Devices - Clinical investigation plan for Medical Devices; Good clinical practice for clinical investigation of medical devices (ISO 14155:2011); DHF Overview, GMP, Adverse event reporting of medical device. Medical Device Approval: Regulatory Bodies & Notified bodies, leading organizations. WHO Medical Devices: Global efforts, Ensuring availability, affordability and accessibility (Technical specifications and regulatory efforts). Overview: List of basic documents of National or State level importance for regulatory efforts / importance for Hospital Management, Summary Technical Document (STED). Indian regulatory organizational structure; Purpose and Functions; Working Groups	9
2	Basic Regulatory Framework: Patient safety and risk management, Sterile / Non-Sterile devices & packaging, Basics of classification of In vitro diagnostics & Medical Devices (active / non active). Regulatory approval process for Medical Devices: Comparison of US FDA, CE and Indian Medical device Rule; Device risk classification, Registration, Clinical trials,	9

	Premarket Notification, Pre-Market Approval (PMA), Investigational Device Exemption (IDE) and In-vitro Diagnostics.	
3	Advanced Regulatory Framework: Quality System Requirements: Quality management of Medical Devices - International Organization for Standardization, ISO 9001 - QMP Overview & Process Approach, ISO 13485 (deeper understanding), ICMED 13485, ISO 14971 -Risk Management of Medical Devices, Unique Device Identification (UDI), Labelling requirements and Post market surveillance.	9
4	Quality: Reliability, Product testing and Technical Standards: Reliability: HFE, MTBF, FMEA, FTA, Black box testing and white box testing, Environmental Stress Screening / Testing, Accelerated Life Testing Product/Materials testing: Biocompatibility, overview of ASTM, ISO Technical standards, Indian National standards other international standards. System / Sub System & Software testing standards: International Electrotechnical Commission (IEC-60601(1-2), Overview of 61010 & IEC 62304 Compliances: RoHS, REACH, National level Safety testing's for ionized radiations (for devices & Hospital installations).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the need for regulatory control in medical device development	K2
CO2	Describe the global medical device regulatory landscape.	K2
CO3	Describe the structure of country specific medical device regulatory organizations and approval processes.	K2
CO4	Explain the requirements for quality systems and risk management of medical devices.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		2	2			2	2			2		3
CO2		2	2			2	2			2		3
CO3		2	2			2	2			2		3
CO4		2	2			2	2			2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical Device Quality Assurance and Regulatory Compliance	Richard C. Fries	CRC Press	1 st edition 2019
2	Medical Device Design: Innovation from Concept to Market	Peter J. Ogrodnik,	Academic Press	2nd edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Biomedical Devices and Systems	Paul H. King, Richard C. Fries, Arthur T. Johnson	CRC Press	4 th edition, 2018
2	Regulation of medical devices A step-by-step guide	World Health Organization	WHO Regional Publications Eastern Mediterranean Series	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ge14/preview

SEMESTER S8
MECHATRONICS

Course Code	PEEBT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min
Prerequisites (if any)	GYEST104, GBEST213	Course Type	Theory

Course Objectives:

1. The objective of this course is to get a basic idea about mechatronics – the working of differently powered systems as one unit and also to familiarize the mechanical actuators / mechanism and its electrical and electronic supporting systems and along with the application of mechatronic systems in industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mechatronics Design System and Interfacing: Introduction to Mechatronics-Key elements of mechatronic systems. Functions of mechatronic systems. -Design approach- Properties of conventional and mechatronic design systems, Ways of integration. Concurrent design procedure for mechatronic systems. Examples of mechatronic systems. Mechatronic system interfacing: Input signals and output signals of a mechatronic System	9
2	Actuation Systems: Pneumatic & hydraulic systems: process control valves, Directional and pressure control valves, Linear and rotary actuators. Mechanical Actuation Systems: Translational and Rotational motions, Kinematic Chains, Cams, Gear Trains, Ratchet and Pawl, Belt and Chain drives, Bearings.	9
3	Electrical Actuation Systems, Performance characteristics and System modelling: Mechanical and solid-state relays, Solenoids, DC & AC motors, Servo & stepper motors. Sensors and transducers- Static and dynamic performance characteristics, Internal sensors, External sensors and microsenors. System modelling - Basic building blocks of general mechanical, electrical, fluid and thermal systems.	9

4	Mechatronic System Controllers: Mechatronic System Controllers (basics only): ON/OFF, P, I, D, PI and PID Controllers, Digital controllers, Intelligent Controllers in Mechatronics. Fundamentals of numerical control - advantages of NC systems- classification of NC systems - point to point and contouring systems - NC and CNC - incremental and absolute systems -open loop and closed loop systems. Basics of Programmable Logic Controller and Industrial Robotics Programmable Logic Controller- Basic PLC structure, I/O processing. Industrial robotics - basic concepts - robot anatomy - robotics and automation - specification of robots - resolution - repeatability and accuracy of manipulator - classification of robots.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze various mechatronic instruments and tools	K4
CO2	Understand actuators and its electrical and electronic supporting systems	K2
CO3	Apply the principle of various controllers and numerical control machines for system design.	K3
CO4	Design and implement basic robotic solutions and analyze its specifications	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	-	-	-	-	2	-	3
CO2	3	3	2	-	-	-	-	-	-	2	-	3
CO3	3	3	2	-	-	-	-	-	-	2	-	3
CO4	3	3	2	-	-	-	-	-	-	2	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechatronics - Electronic Control systems in Mechanical and Electrical Engineering	Bolton. N,	Pearson Education	4/e, 2008
2	Mechatronics-An Introduction	Robert H Bishop	CRC press	1/e, 2005
3	Introduction to Mechatronics	K. K. Appukuttan	Oxford University Press	1/e, 2007
4	Mechatronics - Principles, Concepts and Applications	Nitaigour P Premchand	TMH	11/e, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechatronics-Integrated Technologies for Intelligent Machines	A.Smaili, F.Mrad	Oxford	1/e, 2009
2	Introduction to Mechatronics and Measurement Systems	David G Alciatore, Micheal	TMH	3/e, 2007
3	Mechatronics – Electronics in Products & Processes	Dradly. D.A, Dawson.D, Burd N.C and Loader A.J	Springer	1/e, 1993
4	Electromechanics- Principles concept and Devices	James Harter	Prentice Hall	2/e, 2002
5	Numerical Control of Machine Tools	Yoram Koren & Ben Yuri,	Khanna Publishers	1/e, 1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107298/ https://archive.nptel.ac.in/courses/112/101/112101304/
2	https://archive.nptel.ac.in/courses/112/107/112107298/ https://archive.nptel.ac.in/courses/112/101/112101304/
3	https://archive.nptel.ac.in/courses/112/107/112107298/ https://archive.nptel.ac.in/courses/112/101/112101304/
4	https://archive.nptel.ac.in/courses/112/107/112107298/ https://archive.nptel.ac.in/courses/112/101/112101304/

SEMESTER S8

REHABILITATION ENGINEERING

Course Code	PEEBT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Students will acquire knowledge in basic principles of orthoses and assistive devices and design of features of nerve stimulators.
2. Students will be able to understand features and principles of rehabilitation methods and its applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Orthoses and assistive devices: Introduction, Participatory Action Design, Policy, Human, Activity, assistance, Technology, and environment (PHAATE) Model, Telerehabilitation, Service Delivery Models, Principles of service delivery - Orthoses for spinal dysfunction-three column concept function-cervicothoracic and thoracic orthoses-thoracolumbar-lumbosacral-scoliosis-kinds of loading calculation of bending moments-treatment methods- Powered wheel chairs- multi functional wheel chairs- regulations and standards-micro controller based wheel chairs -design features-smart wheel chairs.	9
2	Electrical Stimulation (ES) Devices: Clinical Considerations of ES- Neuromuscular stimulation (NMS) of atrophied muscles, Foot drop and Wrist drop, Electrodes- Transcutaneous, Subcutaneous and Percutaneous electrodes, Spinal Cord Stimulation (SCS) and Deep Brain Stimulation (DBS). Clinical Applications- ES for Foot Drop, Ultralow power Electronics for SCS & DBS-Biomimetic design of neural prostheses-tissue device interface-neuro modulation through stimulation-sensory restoration motor	9

	restoration-integrated neural prostheses.	
3	Auditory and Visual Aid Devices : Types of hearing impairment, Hearing function assessment, Hearing assistance technology solutions, Surgical approaches- Middle-ear implants (MEIs) and bone anchored hearing aids (BAHAs), Non-surgical - Hearing Aids Assistive Listening Solutions, Voice synthesizer and speech trainer-Types of visual impairments, Visual function assessment, Assistive Technology- Medical/ Surgical Techniques, Cortical Implants, Retinal Implants, Optic-Nerve Stimulation, Auditory and tactile information display, Blind Mobility Aids, GPS Technology, Audible and Remotely Readable Signs	9
4	Neural Rehabilitation: Neural repair and rehabilitation-outcome measurement and diagnostic technology- Electromyography - neuro imaging –rehabilitation in chronic pain-loss of somatic sensation- gait disorders and rehabilitation-balance, vestibular and oculomotor dysfunction-rehabilitation of stroke- brain response to neural prostheses. Robotic assistive Technologies: Use of robots in rehabilitation-international safety Standards-Challenges- Human robot interaction for rehabilitation robots-types of information exchange-HRI matics-physical & social HRI- Assistive Robotic Manipulators (ARM)-Functions-Requirements-Upper limb and lower limb robotic prostheses-challenges and benefits-components-in corporation with human body.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Review the principles of some orthoses and assistive devices.	K2
CO2	Understand the design features of nerve stimulators	K2
CO3	Explore technical features of audio and visual rehabilitation measures	K4
CO4	Understand the Principles of neural rehabilitation methods and applications	K2
CO5	Appreciate the possible use of robots in assistive technologies	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	2	1	-	2	-	-	-	2
CO2	3	3	3	-	2	1	-	2	1	1	-	2
CO3	3	3	3	-	2	1	-	2	1	1	-	2
CO4	3	3	3	-	2	1	-	2	1	1	-	2
CO5	3	3	3	-	2	1	-	2	1	1	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An introduction to rehabilitation engineering.	Cooper, Rory A.; Ohnabe, Hisaichi; Hobson, Douglas A.	CRC Press	2006
2	Intelligent systems and technologies in rehabilitation Engineering	Horia-Nicolai Teodorescu, L.C. Jain	CRC Press	2000
3	Textbook of Rehabilitation	Sunder S.	Jaypee	2012
4	Textbook of Neural Repair and Rehabilitation	Michael Selzer, Stephanie Clarke, Leonardo Cohen, Gert Kwakkel, Robert Miller	Cambridge University Press	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical Instrumentation: Application and Design	John G. Webster	John Wiley & Sons	2009
2	Introduction to Biomedical Equipment Technology	Joseph J. Carr, John M. Brown	Pearson Education (Singapore) Pvt. Ltd.	2001
3	Principles of Applied Biomedical Instrumentation	Geddes & Baker	Wiley	1989
4	Orthotics and Prosthetics in Rehabilitation E-Book	Kevin C. Chui, Milagros Jorge, Sheng-Che Yen, and Michelle M. Lusardi	Elsevier Health Sciences	2019
5	Textbook of Neural Repair and Rehabilitation: Volume 1, Neural Repair and Plasticity	Michael Selzer, Stephanie Clarke, Leonardo Cohen, Gert Kwakkel, and Robert Miller	Cambridge University Press	2014
6	Robotic Assistive Technologies: Principles and Practice	Pedro Encarnação and Albert Cook		2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/aic19_ge01/preview
2	https://archive.nptel.ac.in/courses/108/108/108108188/

SEMESTER S8
MEDICAL INFORMATICS

Course Code	PEEBT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs.30 Min
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course is designed to offer an in-depth understanding of health informatics, blending theoretical frameworks, models, and practical applications within healthcare systems.
2. Students will explore key concepts including electronic health records (EHRs), telehealth, health information exchange, and healthcare analytics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of health informatics: Definition of health informatics, Theories & models underlying health informatics-Systems Theory, Chaos's Theory, Complexity Theory, Information Theory, Diffusion of innovation theory, Learning Theory, Change Theory, System Life cycle models, Staggers & Nelson system Life cycle model, Evidence based practice & Informatics-Evidence based practice models, ACE star model of knowledge transformation.	9
2	Information system in healthcare delivery: Electronic health Records– Definition, Electronic medical Record versus Electronic Health Record, EHR: components, Functions & Attributes, EHR applications used in clinical settings, benefits, Barriers to EHR adoption. Telehealth & Applications for	9

	delivering care at a distance–Technologies, Clinical Practice considerations for healthcare professionals, Operational & Organizational success factors & barriers, Telehealth challenges, Clinical Decision support system in Health care.	
3	Health information exchange & health care analytics: Health information exchange-Basic definitions, The Nationwide Health Information Network (NHIN) and, NHIN Exchange & NHIN Direct, Health Information Organizations (HIOs), Federal Gateway Overview, Web Services and Service Oriented Architecture, Healthcare Analytics-Introduction, Components of healthcare analytics, Tools: SQL (basics)-SQLite, Python Features-Lists and tuples, Dictionaries, Sets, Python libraries- pandas and scikit-learn- Linear regression model, Logistic Regression	9
4	Healthcare security: Patient medical Information-Privacy, Confidentiality, Security-Definition, Legal & historical context, Principles, laws & Regulations, Use of ICT in healthcare. Bioinformatics & mobile computing in health care: Basics for Bioinformatics- Cells, DNA and Chromosome, the Central Dogma, Genes and the Genome, DNA Sequencing, Transcriptomics and DNA Microarrays, Proteomics and Mass Spectrometer, ChIP-Chip and ChIP-Seq, Basic Principles of Genetics-Mendel and Morgan’s Legacy, Disease Gene Mapping in the Genomic Era.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the concepts of health information systems	K4
CO2	Identify the benefits and role of electronic health records.	K2
CO3	Assess the challenges of health information exchanges and healthcare analytics.	K3
CO4	Appraise the security and privacy risks of healthcare data. & familiarize the fundamentals of bioinformatics	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1	1	1		1				
CO2	2	1	1	2	1	1		1				
CO3	2	2	2	1	3	2		1				1
CO4	2	2	2	1	1	2	1	2				1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Health Informatics: An Interprofessional Approach	Ramona Nelson, Nancy Staggers	Elsevier	2 nd Edition, 2013
2	“Biomedical Informatics- Computer Applications in Health Care and Biomedicine”	Edward H. Shortliffe, James J. Cimino	Springer,	4 th edition, 2013
3	“An Introduction to Information Systems and Software in Medicine and Health	David J. Lubliner	“CRC Press, New Jersey Institute of Technology, Newark, USA,	2015
4	“Health care informatics	C Willam Hanson III MD	Medical Publishing Division McGraw-Hill Education / Medical;	1 st edition, 2006
5	“Healthcare Analytics Made Simple”	Vikas (Vik) Kumar	Packt Publishing,	2 nd edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Health Informatics: Digital Health Service Delivery - The Future is Now	K. Schaper	IOS Press	2013
2	“Ethical, Legal, and Social Issues in Medical Informatics”	Penny Duquenoy, Carlisle George, Kai Kimppa	Medical Information Science Reference, New York	2008
3	Clinical Decision Support Systems Theory and Practice	Eta S. Berner	Springer	2007
4	Basics of Bioinformatics	Rui Jiang • Xuegong Zhang • Michael Q. Zhan,	Springer	2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=VhhNcmPLKgo
2	https://elearn.nptel.ac.in/shop/iit-workshops/completed/healthcare-applications-role-relevance-of-ehrs-in-healthcare/?v=c86ee0d9d7ed
3	https://onlinecourses.nptel.ac.in/noc22_hs40/preview
4	https://nptel.ac.in/courses/102106065

SEMESTER S8

MODELLING OF PHYSIOLOGICAL SYSTEMS

Course Code	PEEBT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Quantitative approaches to understand various physiological systems is introduced.
2. Enable the student to understand the basic concepts of mathematical analysis of transport mechanisms across cell membranes, propagation of impulse, processes in respiratory systems, cardiac electrophysiology, and computational aspects of neuronal models.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Modeling approaches in physiological systems: Physiological complexity and need for models, Feedback and control in physiological systems. Modeling process-steps, Different approaches in system modeling-parametric and nonparametric modeling, linear and nonlinear, compartmental modeling. (Examples of each type in physiological system).	9
2	Mathematical analysis of cardiac physiology: Basic Cardiac Anatomy and Electro cardiology Conduction System: The Electrocardiogram (ECG) Cardiac Action Potential Models, Biophysical models Noble-Noble models, reduced cell models. Mathematical models of Cardiac Cells Arrangements: The Bidomain and monodomain models.	9
3	Mathematical models of computational neuroscience: Electrophysiology of neurons, Ionic currents and conductance, Equivalent circuit representation of a patch of cell membrane, resting membrane potential, voltage-gated channels, Hodgkin–Huxley model formulation for action potential, Compartmental models, Simplified models of neurons Integrate-and-fire neurons Spike-response model neurons Rate-based models	9
4	Dynamical systems in physiology: Dynamical system analysis- Types of fixed points, Phase plane analysis - nullclines - Hopf bifurcation and limit cycles. van der Pol oscillator, Fitzhugh Nagumo model Phase-plane analysis,	9

	showing excitability, bistability and oscillations. Simulation studies of simplified models (Eg. Izhikevich neuron models)	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

- 1. Problem Definition (K4 - 5 points)**
 - a. Clearly defines the problem.
 - b. Examine and identifies relevant factors.
- 2. Problem Analysis (K4 - 5 points)**
 - a. Break-down and presents a well-reasoned solution approach.
- 3. Evaluate (K5 - 5 points)**
 - a. Thoroughly evaluate the proposed solutions.
- 4. Conclusion (K3- 3 points, K5 – 2 points)**
 - a. Summarizes findings and insights. State which solution is most appropriate for the problem

Scoring:

- 1. Accomplished (4 points):** Exceptional analysis, and depth of understanding.
- 2. Competent (3 points):** Solid performance with minor areas for improvement.
- 3. Developing (2 points):** Adequate effort but lacks depth or clarity.
- 4. Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine the mathematical approaches of modeling physiological systems	K5
CO2	Analyze the mathematical principles of respiratory function	K4
CO3	Interpret models of cardiac electrophysiology	K4
CO4	Examine models of computational neuroscience	K5
CO5	Analyze the dynamical system properties of neuronal models	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2	2					2		2
CO2	3	1	3	2	2					2		2
CO3	3	3	2	2	3					2		2
CO4	3	3	2	1	3					2		2
CO5	3	3	3	3	3					2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Modeling in physiology and medicine.	Cobelli, Claudio, and Ewart Carson. .	Academic Press,	2019.
2	Mathematical cardiac electrophysiology	Franzone, Piero Colli, Luca Franco Pavarino, and Simone Scacchi.	Springer	. Vol. 13., 2014.
3	Principles of Computational Modeling in neuroscience	Sterratt, David, Bruce Graham, Andrew Gillies, and David Willshaw	Cambridge University Press	2011
4	Dynamical systems in neuroscience	Izhikevich, Eugene M.	MIT Press	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theoretical neuroscience: computational and mathematical modeling of neural systems.	Dayan, Peter, and Laurence F. Abbott.	. Computational Neuroscience Series	2001.
2	Applied mathematical models in human physiology.	Ottesen, Johnny T., Mette S. Olufsen, and Jesper K. Larsen.	Society for Industrial and Applied Mathematics.	2004.
3	An introduction to mathematical physiology and biology.	Mazumdar, Jagannath	Cambridge University Press,	1999.
4	Partial differential equation analysis in biomedical engineering: case studies with MATLAB.	Schiesser, William E.	. Cambridge University Press,	'2012
5	Mathematical biology: I. An introduction.	Murray, James D.	Springer	2007.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/102103056
1	https://archive.nptel.ac.in/courses/102/103/102103056/
1	https://nptel.ac.in/courses/111107113
2	https://nptel.ac.in/courses/112107214
2	https://nptel.ac.in/courses/111107118
2	https://nptel.ac.in/courses/102106023
3&4	https://nptel.ac.in/courses/102106066
4	https://nptel.ac.in/courses/102105100

SEMESTER S8

IOT & BIOMEDICAL APPLICATIONS

Course Code	OEEBT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of Internet of Things (IoT) concepts, including key protocols, architectural principles, and sensor technologies, with a focus on their theoretical applications and implications.
2. Examine the various IoT protocols and architectures, understanding their roles and interactions within IoT ecosystems, and evaluating their effectiveness in different contexts.
3. Explore the theoretical aspects of wearable health monitoring systems, including their design, functionality, and integration with data analytics and cloud technologies to address healthcare challenges.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT and Enabling Technologies: Introduction to Internet of Things – Definition and Characteristics of IoT, Physical Design of IoT – IoT Protocols, Logical Design of IoT - IoT communication models, IoT Communication APIs. IoT enabled Technologies - Wireless Sensor Networks, Cloud Computing, Embedded Systems. Sensor Interfaces – ECG, EEG, PPG, Pulse Oximeter, Temperature Sensors, and Pressure Sensors. Challenges of using IoT in Healthcare.	9
2	IoT Protocols, Architecture, and Standards: IoT Protocols & Architecture: Protocols - Protocol Standardization for IoT – Efforts – M2M and WSN Protocols – SCADA and RFID Protocols – Issues with IoT Standardization – Unified Data Standards – IEEE802.15.4 – BACNet Protocol – Modbus – KNX – Zigbee. Network and Security Architecture. IoT Open Source Architecture (OIC) - OIC Architecture & Design	9

	Principles. IoT Devices and Deployment Models - IoTivity: An Open-source IoT Stack - Overview - IoTivity Stack Architecture - Resource Model and Abstraction.	
3	Wearable Health Monitoring Systems and IoT Implementation: Wearable Health Monitoring System (WHMS): Basics of WHMS, Need for Wearable Systems, Applications of WHMS - Home Healthcare, Sports/Fitness. Design of Wearables: Introduction, Architecture, and Attributes. Design of Smart Textiles and Wearable Electronics. Issues of WHMS & Factors Inhibiting Growth. Building IoT with Arduino & Raspberry Pi: Programming – Python Program with Raspberry Pi, Controlling Hardware - Controlling AC Power Devices with Relays, Controlling Servo Motor, DC Motor. Sensors - Light Sensor, Motion Detection Sensors, Wireless Bluetooth Sensors.	9
4	Data Handling, Analytics, and Cloud for IoT: Data Handling & Analytics: Introduction, Big Data, Types of Data, Characteristics of Big Data, Data Handling Technologies, Data Acquisition, Data Storage. Introduction to Data Analytics, Types of Data Analytics, Local Analytics, Cloud Analytics, and Applications. IoT Physical Servers and Cloud Offerings – Introduction to Cloud Storage Models and Communication APIs. Webserver – Web Server for IoT, Cloud for IoT, Python Web Application Framework, Designing a RESTful Web API.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a clear understanding of various IoT protocols and architectural frameworks, including their roles and functionalities within IoT ecosystems.	K2
CO2	Apply theoretical designs for IoT systems using Raspberry Pi, focusing on system integration and hardware control concepts.	K3
CO3	Apply knowledge of wearable health monitoring systems to design and evaluate their components and functionalities in healthcare settings.	K3
CO4	Analyze data management practices in IoT, including data acquisition, storage, and analytics, to understand their impact on system performance and effectiveness.	K4
CO5	Apply concepts to deploy IoT applications and integrate them with cloud platforms for effective data management and analysis.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	3					2		2
CO2	3	2	2	1	3					2		2
CO3	3	2	2	1	3					2		2
CO4	3	2	2	2	3					2		2
CO5	2	2	3	1	3					2		2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things – A Hands-on Approach	Arshdeep Bahga, Vijay Madiseti	Universities Press	2015
2	From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence	Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle	Academic Press	1st Edition, 2014
3	Getting Started with Raspberry Pi	Matt Richardson, Shawn Wallace	O'Reilly Media, Inc.	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rethinking the Internet of Things: A Scalable Approach to Connecting Everything	Francis daCosta	Apress Publications	1st Edition, 2013
2	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, Jerome Henry	Cisco Press	2017
3	Internet of Things and Personalized Healthcare Systems	Venkata Krishna, Sasikumar Gurumoorthy, Mohammad S. Obaidat	Springer Briefs in Applied Sciences and Technology, Forensic and Medical Bioinformatics	2019
4	Architecting the Internet of Things	Dieter Uckelmann, Mark Harrison, Michahelles Florian (Eds)	Springer	2011
5	Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects	Michael Margolis	O'Reilly Media	2nd Edition, 2011
6	Principles of Internet of Things (IoT) Ecosystem: Insight Paradigm	Sheng-Lung Peng, Souvik Pal, Lianfen Huang (Eds)	Springer International Publishing	2020
7	Internet of Things (IoT) for Automated and Smart Applications	Yasser Ismail (Ed)	BoD–Books on Demand	2019
8	Internet of Things: For Things, and by Things	Abhik Chaudhuri	CRC Press	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105166/
2	https://onlinecourses.swayam2.ac.in/arp20_ap03/preview
3	https://onlinecourses.swayam2.ac.in/arp20_ap03/preview
4	https://onlinecourses.nptel.ac.in/noc22_cs53/preview

SEMESTER S8

HUMAN FACTORS IN ENGINEERING AND DESIGN

Course Code	OEEBT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the basic concepts and techniques employed in broad area of human factors engineering focusing on anthropometric principles in design of work places and to optimize the interaction between human and systems.
2. Integrate human factor considerations in the design of medical devices and health care tools for improved user experience and safety.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Human factors Engineering – Definition, Human Factors & Systems - Aspects of Human-machine systems. A model of Human information processing. Process of seeing- Visual capabilities-Accommodation, Visual acuity and Contrast sensitivity. Factors affecting visual capabilities. Human factor aspects of hard copy text and computer screen text. Graphic representations of Text & Data. Human factors in designing Symbols & Codes	9
2	Visual Displays - Qualitative visual displays, Quantitative visual displays, Signals & Warning lights, Representational Displays-Aircraft Bank Angle displays-Principles of aircraft position displays. Auditory Displays - Process of hearing. Principles of auditory displays, Auditory Displays for Specific Purposes. Cutaneous Senses- Tactual Displays- , Substitute for Hearing, seeing, Tactual graph. Compatibility – Types of Compatibility-Conceptual, movement, spatial and Modality Compatibility. Movement compatibility- Rotary controls and rotary displays in same plane and different plane.	9

3	Muscle Physiology- Work Physiology. Measure of physiological strain, Physical Workload, Strength & Endurance. Manual Material Handling (MMH), Bio-mechanical recommended limits of MMH. Reducing the risk of MMH overexertion. General principles of Biomechanical Design. Anthropometry -Anthropometric design principles. Static and Dynamic Anthropometry. Case study of Anthropometric Design- Viewing Angle determination of Cardiac Output Monitor. Work place and Seating Design- Work space envelope, Factors in design of workspace surfaces, Principles of seat design.	9
4	Classification of human errors-Dealing with human errors. Accidents - Theories of accident causation-CFAC model. Occupational Safety and Health - Industrial Safety and Health, Hazards and Control Measures, Warnings and Safety Programs. Human factors for Mobile medical device design - General and special considerations, Design principles for home healthcare; Consideration related to Hand tool design in health care. Case studies exemplifying design for device mobility and use in motion: Infusion pumps, Portable X-Ray machine, Portable Ultrasound Device, Transport ventilator.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the human factor aspects of visual capabilities and their implications for the design.	K2
CO2	Design and evaluate auditory, tactual displays based on human sensory processes and compatibility principles to enhance user interaction and performance in various applications.	K3
CO3	Identify the role of anthropometry and Ergonomics in product design	K3
CO4	Analyse and Apply principles of muscle physiology, biomechanics to design ergonomic work spaces and optimize the occupational settings.	K4
CO5	Classify and manage human errors, apply theories of accident causation, implement occupational safety and incorporate human factors principles in the design of mobile medical devices and health care tools.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3							2		2
CO2	3	2	3							2		2
CO3	3	2	3							2		2
CO4	3	2	3							2		2
CO5	3	2	3							2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Factors in Engineering and Design	Mark S. Sanders & Ernest J. McCormic	McGraw Hill	International Edition, 1993
2	Handbook of Human Factors in Medical Device Design	Matthew Bret Weinger, Michael E. Wiklund, Daryle Jean Gardner-Bonneau	CRC Press	2011
3	Introduction to Human Factors and Ergonomics for Engineers,	Mark Lehto, Steven J. Landry	CRC Press	2 nd Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Factors Engineering in System Design	Terence S. Andre, Aaron W. Schopper	British Columbia Teacher	1997
2	Human Factors Design Handbook	Wesley E. Woodson,	McGraw-Hill Professional	2 nd Edition, 1992
3	A Guide to Human factors and Ergonomics	Martin Helander	Taylor and Francis Group	2 nd Edition, 2005
4	Human Factors Engineering and Ergonomics - A Systems Approach,	Guastello, Stephen J.-	CRC Press	2 nd Edition, 2013
5	Human Factors Methods – A Practical Guide for Engineering and Design,	Neville Stanton, Paul Salmon, Guy Walker, Christopher Baber, Dan Jenkins	Ashgate Publishing Limited	2 nd Edition, 2015
6	Handbook of Cognitive Task Design,	Erik Hollnagel	Lawrence Erlbaum Publishers	1 st Edition 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/110105162
2	https://nptel.ac.in/courses/110105162
3	https://nptel.ac.in/courses/110105162
4	https://nptel.ac.in/courses/110105162

SEMESTER S8

MEDICAL IMAGE PROCESSING

Course Code	OEEBT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide an overview of the nature of human visual perception, the elements of an image-processing system which includes sampling, quantisation and two dimensional transforms.
2. To make the students familiar with image enhancement, restoration, segmentation, classification, machine learning and deep learning techniques for image analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Image perception -MTF of the visual system –monochrome vision models - color vision model. Image sampling and quantization: Two-dimensional sampling theory, Image quantization - uniform quantizer. Image transforms - Two dimensional orthogonal and unitary transforms - properties of unitary transforms, 2D DFT, DCT, Hadamard, Haar, KL transforms.	9
2	Image enhancement: contrast enhancement, clipping and thresholding - digital negative, intensity level slicing - bit extraction. Histogram processing, Magnification and interpolation. Image Enhancement in Spatial domain: Smoothing and sharpening operations in time domain, Filtering in frequency domain: Smoothing and sharpening frequency domain filters, Homomorphic filtering.	9
3	Image restoration: Image degradation, Noise, Blurring, Modelling image degradation, Inverse filtering, Wiener filtering. Image segmentation- Thresholding- optimal, adaptive, Region-based methods- region growing and region merging, Boundary-based methods- edge detection and linking,	9

	boundary tracking.	
4	Feature recognition and classification: Features - describing the contents and the shape of objects. Classification- Supervised and unsupervised learning. Discriminant analysis, k-nearest-Neighbor (kNN) classifier. Unsupervised- k-means clustering. Deep Learning for Medical Image Analysis, Convolutional Neural Networks. Image fusion for hybrid imaging, Super resolution imaging.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Infer the nature of processes involved in human visual image perception and also in image-processing system which includes sampling, quantisation and two dimensional transforms.	K2
CO2	Apply methods of image enhancement both in time and transform domains.	K3
CO3	Analyse the techniques for feature extraction, image segmentation and image restoration.	K4
CO4	Analyse machine and deep learning algorithms for medical image analysis.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										1
CO2	3	2			1							1
CO3	3	2	2	2	1							2
CO4	2	2	2	2	1							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Digital Image Processing	Jain Anil K	Prentice Hall of India	1989
2	Digital Image Processing for Medical Applications.	Geoff Dougherty	Cambridge University Press	2007
3	Digital Image Processing	Gonzalez Rafael C	Fourth Edition	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning for Medical Image Analysis	Edited by S. Kevin Zhou Hayit Greenspan Ding gang Shen	Academic Press, Elsevier	Ist Edition, 2017
2	Computer Vision Algorithms and Applications	Richard Szeliski	Springer	2011
3	Hands-On One-shot Learning with Python	Shruti Jadon Ankush Garg	Packt Publishing	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/7Pzq9IwscbE?si=4yMyNurWD3dfbQF5, https://youtu.be/nfEaPWZ_dRQ?si=r-Ith_KIUfLDbo52, https://youtu.be/7LXi7_lSJn0?si=dxNRVXnfr3t9yd7m,
2	https://youtu.be/ArKe6zMkXnk?si=v21e56QYZWgr5h5W,
3	https://youtu.be/ljCZM8mpj8I?si=yXQIEOrtXMwOP1lx,
4	https://youtu.be/Y_zQ9B4rWMY?si=DSVaLNRVDsVPaFRu, https://youtu.be/11WBbnlMkwg?si=etryD6j0WGYZRD, https://youtu.be/ljCZM8mpj8I?si=yXQIEOrtXMwOP1lx,

SEMESTER S8

REHABILITATION ENGINEERING

Course Code	OEEBT834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Students will be able to review the principles and design features of orthoses, assistive devices, nerve stimulators, and technical aspects of audio and visual rehabilitation measures.
2. Students can acquire knowledge of neural rehabilitation methods and explore the applications of robots in assistive technologies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Orthoses and assistive devices: Introduction, Participatory Action Design, Policy, Human, Activity, assistance, Technology, and environment (PHAATE) Model, Telerehabilitation, Service Delivery Models, Principles of service delivery - Orthoses for spinal dysfunction-three column concept function-cervicothoracic and thoracic orthoses-thoracolumbar-lumbosacral-scoliosis-kinds of loading calculation of bending moments-treatment methods- Powered wheel chairs- multi functional wheel chairs- regulations and standards-micro controller based wheel chairs -design features-smart wheel chairs.	9
2	Electrical Stimulation (ES) Devices: Clinical Considerations of ES- Neuromuscular stimulation (NMS) of atrophied muscles, Foot drop and Wrist drop, Electrodes- Transcutaneous, Subcutaneous and Percutaneous electrodes, Spinal Cord Stimulation (SCS) and Deep Brain Stimulation (DBS). Clinical Applications- ES for Foot Drop, Ultralow power Electronics for SCS & DBS-Biomimetic design of neural prostheses-tissue device	9

	interface-neuro modulation through stimulation-sensory restorationmotor restoration-integrated neural prostheses.	
3	Auditory and Visual Aid Devices : Types of hearing impairment, Hearing function assessment, Hearing assistance technology solutions, Surgical approaches- Middle-ear implants (MEIs) and bone anchored hearing aids (BAHAs), Non-surgical - Hearing Aids Assistive Listening Solutions, Voice synthesizer and speech trainer-Types of visual impairments, Visual function assessment, Assistive Technology- Medical/ Surgical Techniques, Cortical Implants, Retinal Implants, Optic-Nerve Stimulation, Auditory and tactile information display, Blind Mobility Aids, GPS Technology, Audible and Remotely Readable Signs	9
4	Neural Rehabilitation: Neural repair and rehabilitation-outcome measurement and diagnostic technology- Electromyography - neuro imaging –rehabilitation in chronic pain-loss of somatic sensation- gait disorders and rehabilitation-balance, vestibular and oculomotor dysfunction-rehabilitation of stroke- brain response to neural prostheses. Robotic assistive Technologies: Use of robots in rehabilitation-international safety Standards- Challenges- Human robot interaction for rehabilitation robots-types of information exchange-HRI matics-physical & social HRI-Assistive Robotic Manipulators (ARM)-Functions-Requirements-Upper limb and lower limb robotic robotic prostheses-challenges and benefits-components-in corporation with human body.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Review the principles of some orthoses and assistive devices.	K2
CO2	Understand the design features of nerve stimulators	K2
CO3	Explore technical features of audio and visual rehabilitation measures	K3
CO4	Understand the principles of neural rehabilitation methods and applications of robots in assistive technologies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	2	1	-	2	-	-	-	2
CO2	3	3	3	-	2	1	-	2	1	1	-	2
CO3	3	3	3	-	2	1	-	2	1	1	-	2
CO4	3	3	3	-	2	1	-	2	1	1	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An introduction to rehabilitation engineering.	Cooper, Rory A.; Ohnabe, Hisaichi; Hobson, Douglas A.	CRC Press	2006
2	Intelligent systems and technologies in rehabilitation Engineering	Horia-Nicolai Teodorescu, L.C. Jain	CRC Press	2000
3	Textbook of Rehabilitation	Sunder S.	Jaypee	2012
4	Textbook of Neural Repair and Rehabilitation	Michael Selzer, Stephanie Clarke, Leonardo Cohen, Gert Kwakkel, Robert Miller	Cambridge University Press	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical Instrumentation: Application and Design	John G. Webster	John Wiley & Sons	2009
2	Introduction to Biomedical Equipment Technology	Joseph J. Carr, John M. Brown	Pearson Education (Singapore) Pvt. Ltd.	2001
3	Principles of Applied Biomedical Instrumentation	Geddes & Baker	Wiley	1989
4	Orthotics and Prosthetics in Rehabilitation E-Book	Kevin C. Chui, Milagros Jorge, Sheng-Che Yen, and Michelle M. Lusardi	Elsevier Health Sciences	2019
5	Textbook of Neural Repair and Rehabilitation: Volume 1, Neural Repair and Plasticity	Michael Selzer, Stephanie Clarke, Leonardo Cohen, Gert Kwakkel, and Robert Miller	Cambridge University Press	2014
6	Robotic Assistive Technologies: Principles and Practice	Pedro Encarnação and Albert Cook		2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/1atOc5Jvr-A
2	https://youtu.be/wWjuZvFlh5Q